

BIC ISO Interface Manual

BASE24[®]



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What's New

The following table highlights the major changes that have been made in the most recent update to the *BASE24 BIC ISO Interface Manual*. The first column of the table lists the sections and appendixes in which major changes have been made. The second column of the table describes the major changes for each section or appendix. These changes update the manual to release 6.0, version 10.

November 2011

Section/ Appendix	Major Changes
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- | | |
|---|---|
| 1 | Clarifies processing of LCONF param ENHANCED-CRNCY-CONV |
| 2 | Clarifies definition of LCONF param ENHANCED-CRNCY-CONV |

December 2009

Section/ Appendix	Major Changes
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- | | |
|---|---|
| 1 | Updates the “Enhanced EMV Response Code Mapping” topic to refer to the <i>BASE24 BIC ISO Standards Manual</i> for a complete listing of response codes. |
| 2 | <p>Adds the following LCONF params: POS-B24-TO-POS-ACCT-TYP-SUPPRS, PRE-VERIFIED-PIN-PAD-CHAR, PSEUDO-3DES-CBC-MAC-TYPE, RACAL-ANSI-METHOD, REVERSAL-BAL-INQ, RSM-TERM-MASTER-KEY, SEND-ETX, and UPDT-ILF-ON-LATE-RESP.</p> <p>Updates the SETTLEMENT TIMER and ITF UPDATE INTERVAL fields on Enhanced Interchange Configuration File (ICFE) screen 12 to better describe default value processing.</p> <p>Updates the procedures for using the External Message File (EMF) to correct the value placed in the INTERF TYP field on an EMF screen for the BIC ISO Interface.</p> |
| A | Adds the following LCONF params to list of assigns and params read during initialization: ATM-LN-CUTOVER, POS-LN-CUTOVER, POS-B24-TO-POS-ACCT-TYP-SUPPRS, PRE-VERIFIED-PIN-PAD-CHAR, and PSEUDO-3DES-CBC-MAC-TYPE. |

Preface

The ***BASE24 BIC ISO Interface Manual*** describes the BIC ISO Interface process, which is used to establish a communications link between a BASE24 network and a co-network. This co-network may be another BASE24 network, but may also be any transaction processing network which conforms to the standards set forth in the ***BASE24 BIC ISO Standards Manual*** for message formats and processing.

This manual describes the BIC ISO Interface process, which uses the International Organization for Standardization (ISO) 8583:1987 message formats.

Audience

This manual is intended as a reference for those persons responsible for the interface between BASE24 and a co-network. It provides a comprehensive description of all the functions performed by the BIC ISO Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Prerequisites

The structure and content of the BIC external message is patterned after the BASE24 external message. The BASE24 external message is similar in nature to the standard external message developed by the International Organization for Standardization (ISO)—described in the ISO 8583 (1987) standard, ***Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions***. Because of the similarity between the BASE24 external message and the ISO standard, familiarity with the above document, commonly referred to as “ISO 8583,” is recommended.

However, there are several differences between the BASE24 external message and the ISO standard that may be of interest to readers familiar with the ISO standards. These differences are described throughout this manual. The standards for message formats and processing to which BIC conforms are set forth in the ***BASE24 BIC ISO Standards Manual***.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring the BIC ISO Interface process.
- The ***BASE24 BIC ISO Standards Manual*** discusses in detail the specifications for the BIC ISO External Message. It includes a general overview of the message, explanations of the external message types used by the BIC ISO Interface process, a set of message defaults for each BASE24 product, a description of each data element contained in the message, and descriptions of the message flows between BASE24 and a co-network.
- The ***BASE24 External Message Manual*** discusses in detail the BASE24 external message on which the BIC ISO Interface process is based.
- The ***BASE24 Event Message Reference Manual*** discusses in detail how to electronically access event message documentation generated by the BIC ISO Interface process.
- The ***BASE24 Text Command Reference Manual*** describes each text command sent using the network control facility that can be issued for the BIC ISO Interface process. This manual includes the corresponding command syntax used with each command.
- The ***BASE24 Base Files Maintenance Manual*** documents the BASE24 data entry screens for the External Message File (EMF), the Key File (KEYF), the Key 6 File (KEY6), the Enhanced Interchange Configuration File (ICFE), the Acquirer Processing Code File (APCF), the Issuer Processing Code File (IPCF), the Card Prefix File (CPF), and the Surcharge File (SURF). The ICFE is used to configure the parameters described in this manual for the BIC ISO Interface process. The APCF and IPCF are used to configure the transactions allowed to be sent and received by the BIC ISO Interface process. The CPF and SURF are used to configure surcharging. The EMF and the KEYF or KEY6 are used to configure the messages and parameters described in this manual.

- The ***BASE24 Files and Record Structures Reference Manual*** documents how to access the file and record structures for the Interchange Log File (ILF).
- The ***BASE24-pos Address Verification Manual*** documents the BASE24-pos Address Verification add-on product.
- The ***BASE24 Tokens Manual*** documents token processing. It includes information on how to configure the tokens that get logged to the Interchange Log File (ILF), sent in BIC ISO messages, and extracted from the ILF.
- The ***BASE24-atm Transaction Processing Manual*** documents the BASE24-atm standard internal message format (STM).
- The ***BASE24-atm Files Maintenance Manual*** documents the data entry screens for the BASE24-atm Terminal Data files (ATD), which is used to link the SURF with a card prefix or terminal group.
- The ***BASE24-pos Transaction Processing Manual*** documents the BASE24-pos standard internal message format (PSTM).
- The device-specific BASE24-atm Self-Service Banking manuals document the BASE24-atm Self-Service Banking add-on product.
- The ***BASE24 Transaction Security Manual*** discusses in detail Personal Identification Numbers (PINs), message authentication codes (MACs), and dynamic key management used to set up transaction security for the BIC ISO Interface process.

This manual also references the following NET24 publications:

- The ***NET24-XPNET Network Control Command Reference Manual*** contains details on process, station, line, and device attributes used in configuring the BIC ISO Interface process.
- The ***NET24 Performance Monitoring Manual*** provides detail on the NET24 Performance Monitoring subsystem.

In addition to the BASE24 and NET24 publications, this manual contains references to the ISO 4217 standard, ***Codes for the Representation of Currencies and Funds***.

Software Release

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The *BASE24 BIC ISO Interface Manual* is divided into the following sections:

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides basic information about the BIC ISO Interface process and provides an overview of BIC ISO Interface functionality.

Section 2, “Configuration and Control,” describes the configuration and control of the BIC ISO Interface process.

Section 3, “Network Management and Operations,” describes the network management messages supported by the BIC ISO Interface process and provides message flows that illustrate the processing that occurs when these messages are sent.

Section 4, “Store-and-Forward Processing,” describes how the BIC ISO Interface process performs store-and-forward processing. It includes information about what messages can be placed in the SAF and when messages from the SAF are sent. It also provides message flows that illustrate store-and-forward processing.

Section 5, “Transaction Handling,” describes how the BIC ISO Interface process handles various types of transactions.

Section 6, “Cutover and Reporting,” describes cutover processing and report generation. Message flows describing how the BIC ISO Interface process handles various types of cutover transactions are also included in this section.

Appendix A, “Startup and Control,” provides information about the configuration of the BIC ISO Interface process. It also describes the processing for initialization and network control.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, SW-DS004-01 for the ***BASE24 BIC ISO Interface Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Nov-2011 for November, 2011) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

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Conventions Used in This Manual

This section explains how field descriptions are documented in this manual.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the Enhanced Interchange Configuration File (ICFE).

Example

ACQ STAND-IN AUTH — A flag indicating whether the BASE24 system, when acting as the acquirer, is allowed to stand in on behalf of the co-network to authorize transactions. This field is verified at logon to make sure the co-network is set up correctly. Valid values are as follows:

Y = Stand-in is allowed
N = Stand-in is not allowed

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: N
Data Name: ICFEBICI.ACQ-STAND-IN

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is <i>System protected</i>.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

1: Introduction

This section introduces the BASE24 Interchange (BIC) ISO Interface and provides an overview of BIC ISO Interface process functionality. It also identifies the BASE24 files used by the BIC ISO Interface process, and the message types supported.

BIC ISO Interface Process Overview

The BIC ISO Interface process, shown in the illustration on the next page, is responsible for the online message transfer between BASE24 and its co-networks. This process performs the following functions:

- Facilitates message exchange between the BASE24 system and the co-network.

When the BASE24 system receives a transaction from the co-network, the BIC ISO Interface translates the message into the appropriate internal message format and sends the message to a BASE24 Authorization process. When the BASE24 system receives a transaction for the co-network to authorize, the BIC ISO Interface process translates the message from the internal message format to the external message format and forwards the message to the co-network.

- Provides store-and-forward capabilities.

When the link between the BASE24 system and the co-network is down, the BIC ISO Interface process stores certain messages intended for the co-network in its Store-and-Forward File (SAF). When the link is restored, these messages are retrieved from the file and sent to the co-network.

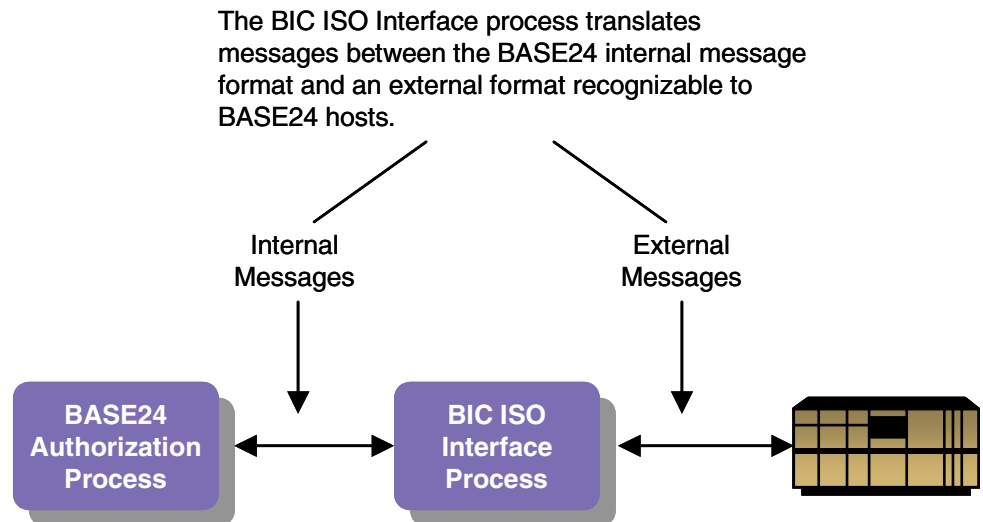
- Facilitates settlement between the BASE24 system and the co-network.

The BIC ISO Interface process produces a set of interchange reports to assist in reconciling transactions between the BASE24 system and the co-network. These reports allow the BASE24 participants to reconcile the settlement positions of the co-network and the BASE24 system. The BIC ISO Interface process also handles online exchange of reconciliation totals between the BASE24 system and the co-network.

- Manages communications between the BASE24 system and the co-network.

The BIC ISO Interface process sends and receives network management messages to determine the status of the link between the BASE24 system and the co-network. Network management messages include logon, logoff, echo test, dynamic key management, and cutover messages.

BIC ISO Interface Process



The BIC ISO Interface process supports communication with co-networks using the International Organization for Standardization (ISO) message format.

Co-Networks

The BASE24 Interchange, or BIC, connects a BASE24-atm or BASE24-pos network with a *co-network*—another BASE24 network or another transaction processing network that conforms to BIC ISO standards. These standards include sending and accepting the appropriate form of the BIC ISO external message and providing the proper message flows to effect processing.

BIC serves as a link between two networks. It allows a cardholder of one network to initiate a transaction at a terminal belonging to the other network. Although the co-network can be another BASE24 network, for the purposes of this manual the co-network is considered to be unknown.

Stations

Stations provide the means by which communications takes place between the BASE24 system and a co-network. All messages sent to a co-network are actually sent to the stations defined for the co-network. Likewise, all messages received from a co-network are actually received from stations.

Stations and their lines must be defined to the XPNET system. Stations and lines are tied to their respective co-networks through the Enhanced Interchange Configuration File (ICFE). At least one station is required for each co-network, and a maximum of 20 stations are allowed per co-network. For information on defining stations and lines to BASE24, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

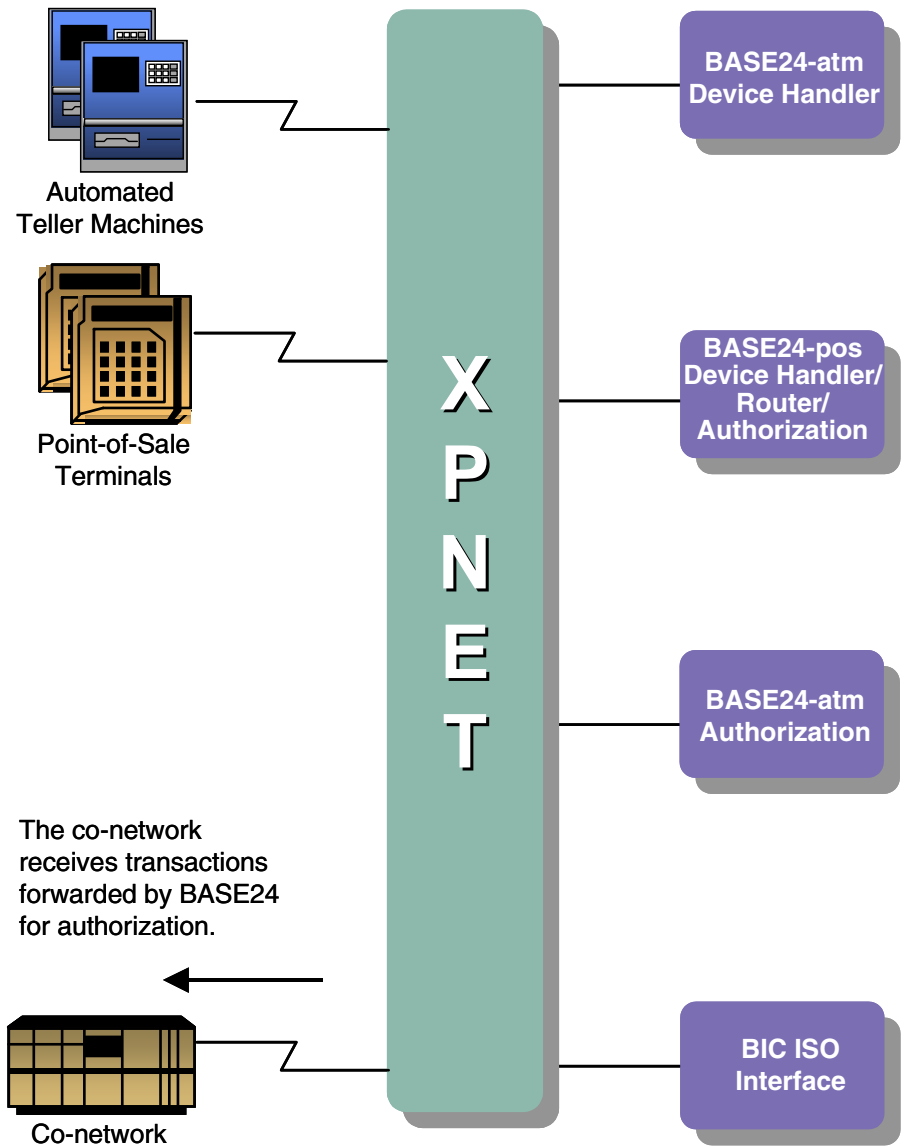
Note: All protocol-level functions between BASE24 and the stations are actually handled by the XPNET process using HP's NonStop communications control software. The BIC ISO Interface process sends and receives messages using the XPNET process. The XPNET process maintains communications links with the stations according to the protocols under which the stations are defined.

Issuer and Acquirer Co-Networks

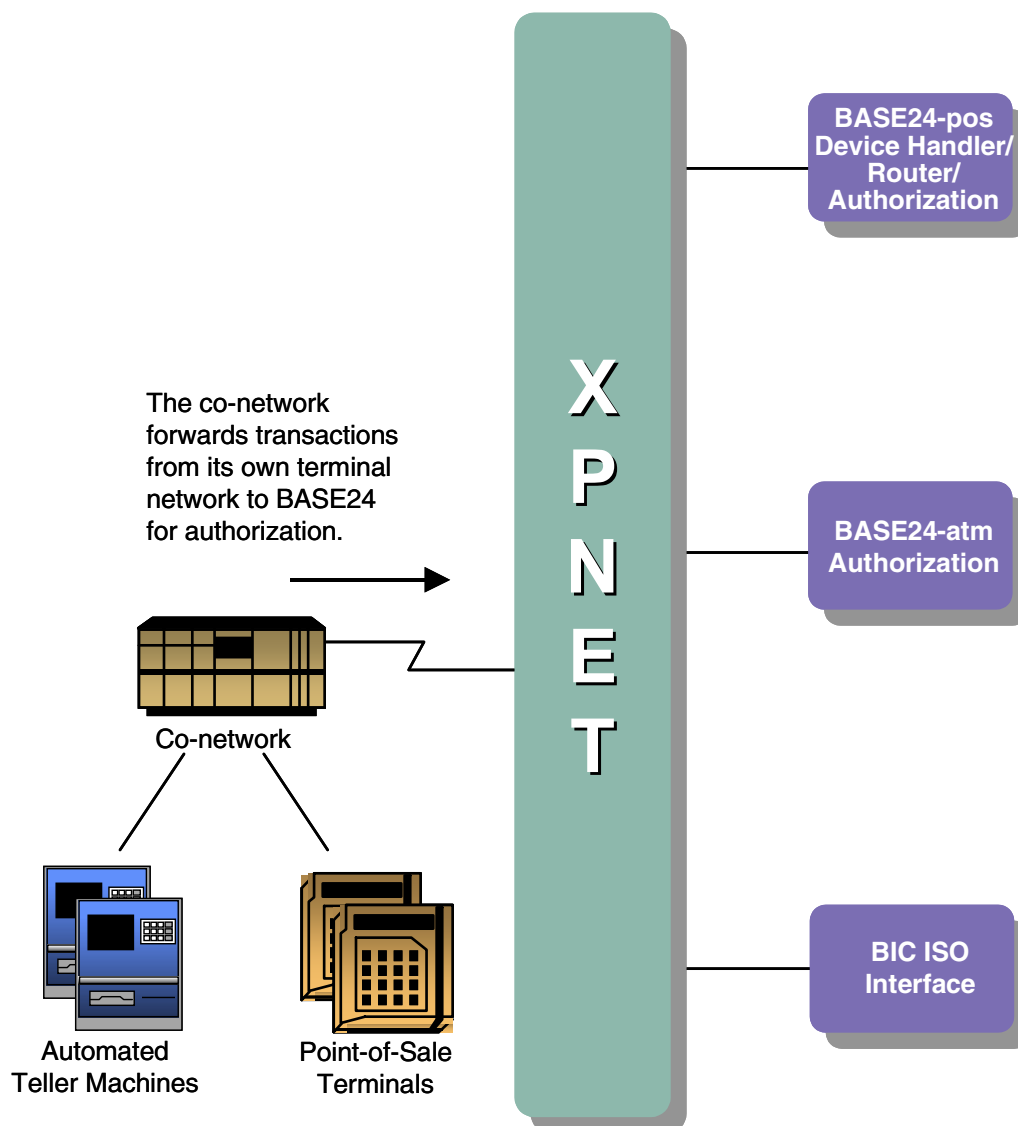
In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorize and process transactions.

Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorizing cardholder transactions and keeping track of cardholder accounts. *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

Co-networks acting on behalf of an issuer in a BASE24 message exchange are referred to as *issuer co-networks*. Issuer co-networks authorize or post transactions sent to them by the BASE24 system, as shown in the following diagram.



Co-networks acting on behalf of an acquirer in a BASE24 message exchange are referred to as *acquirer co-networks*. Acquirer co-networks support ATM or POS networks external to the BASE24 system and forward transactions to BASE24 for authorization, as shown in the following diagram.



Note: Because of their general position in the transaction path, issuer co-networks are also referred to as back-end co-networks and acquirer hosts are also referred to as *front-end co-networks*.

Issuer and Acquirer Co-Network Support

The BIC ISO Interface process can function in the role of an issuer or an acquirer. When the BIC ISO Interface process sends financial transactions to an issuer co-network for authorization, the BIC ISO Interface process is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the BIC ISO Interface process for authorization from an acquirer co-network, the BIC ISO process is representing the card issuer, even though the BIC ISO process may only forward the transaction to an issuer co-network for authorization.

Co-networks can also function on behalf of both card issuers (or account owners) and transaction acquirers, as would be the case when an institution issues cards and supports its own network of ATM or POS devices.

These options are controlled using the ACQUIRER and ISSUER fields on ICFE screen 3.

Internal Message Format

The internal message format is used whenever information is routed between the BIC ISO Interface process and the rest of the BASE24 system. Depending on the BASE24 product receiving the message, the internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

For more information on the STM or PSTM, see the BASE24-atm or BASE24-pos transaction processing manual.

BIC ISO External Message Format

The structure and content of the external message used by the BIC ISO Interface processes to communicate with their co-networks is patterned after the BASE24 external message. The BASE24 external message is based on the standard external message developed by the International Organization for Standardization (ISO). It is a variable-length, variable-content message that can be configured differently, based on the type of message being sent. Specifications for how the BIC ISO external message differs from the BASE24 external message can be found in the *BASE24 BIC ISO Standards Manual*.

Store-and-Forward Function

A store-and-forward function is incorporated into the BIC ISO Interface process to collect transaction messages during periods when a co-network is unavailable.

When a co-network is unavailable, messages required as notification to the co-network are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored. For more information on store-and-forward processing, see section 4.

Multiple Product Support

The same BIC ISO Interface process can be set up to process co-network messages for multiple BASE24 products. If an institution purchases multiple BASE24 products, the appropriate product code is bound into the BIC ISO Interface process to allow it to recognize and process transaction messages for each product.

Message Processing

Due to the variances in processing that must occur in handling messages for different products, the BIC ISO Interface processes messages differently based on the message type and product ID for the message. For more information on how messages flow back and forth between the co-network and the BIC ISO Interface process, please refer to the following sections, based on the type of message:

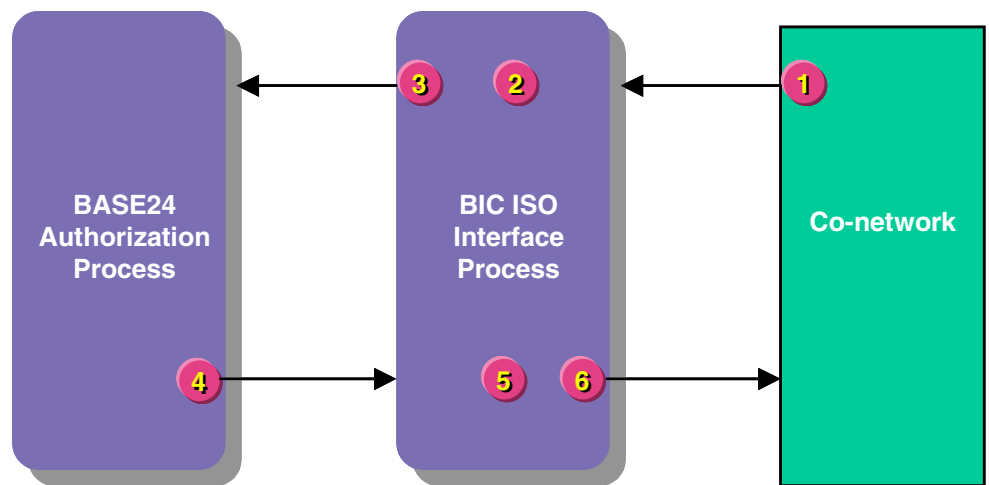
- Network management message flows, including those for dynamic key management, are documented in section 3.
- Store-and-forward message flows are documented in section 4.
- BASE24 transaction flows are documented in section 5.
- Cutover message flows are documented in section 6.

The remainder of this subsection provides an overview of transaction flows for normal and link-down message processing.

Normal Message Flow

When a message from the co-network is received by the BIC ISO Interface process, it is translated from the external message into the appropriate internal message format. The internal message is then sent to the appropriate BASE24 Authorization process. When the message has been processed, the authorizing process sends an internal response back to the BIC ISO Interface process. Using information from the internal message, the BIC ISO Interface process formats an external response message and sends it back to the co-network.

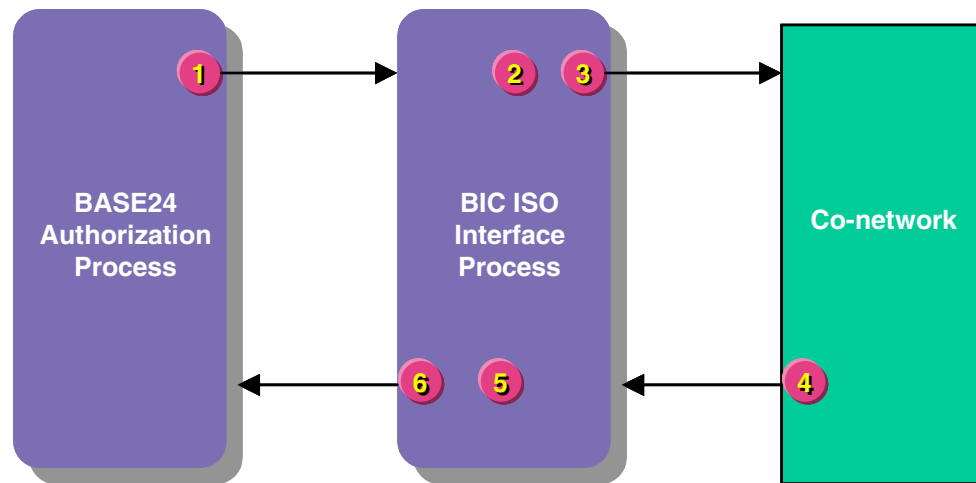
The following diagram illustrates this incoming message flow. Processing steps are provided following the illustration.



1. The co-network sends an external request message to the BASE24 system.
2. The BIC ISO Interface process translates the external request message into an internal request message.
3. The BIC ISO Interface process sends the internal request message to the BASE24 Authorization process for authorization.
4. The BASE24 Authorization process authorizes the request and sends an internal response message to the BIC ISO Interface process.
5. The BIC ISO Interface process translates the internal response message into an external response message.
6. The BIC ISO Interface process sends the external response message to the co-network.

When a message from a BASE24 Authorization process is received by the BIC ISO Interface process, it is translated from the internal message into the external message format. The external message is then sent to the co-network to be authorized. When the message has been processed, the co-network sends an external response back to the BIC ISO Interface process. Using information from the external message, the BIC ISO Interface process formats an internal response message and sends it back to the BASE24 Authorization process.

The following diagram illustrates this outgoing message flow. Processing steps are provided following the illustration

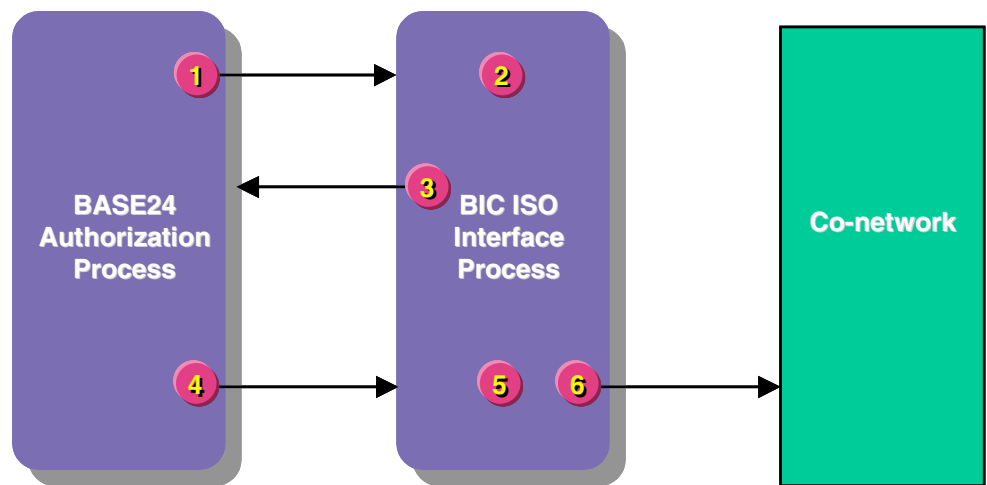


1. The BASE24 Authorization process sends an internal request message to the BIC ISO Interface process.
2. The BIC ISO Interface process translates the internal request message into an external request message.
3. The BIC ISO Interface process sends the external request message to the co-network for authorization.
4. The co-network authorizes the request and sends an external response message to the BIC ISO Interface process.
5. The BIC ISO Interface process translates the external response message into an internal response message.
6. The BIC ISO Interface process sends the internal response message to the BASE24 Authorization process.

Link-Down Message Flow

If stand-in authorization is allowed, the acquirer network has the option of authorizing requests when the link to the issuer network is down. The BIC ISO Interface process returns the request message to the BASE24 Authorization process as a failed request needing offline authorization. If the transaction is approved, a force post message is forwarded to the issuer network when the link is reestablished. In the illustration below, BASE24 is the acquirer and the co-network is the issuer.

The following diagram illustrates this outgoing message flow. Processing steps are provided following the illustration.



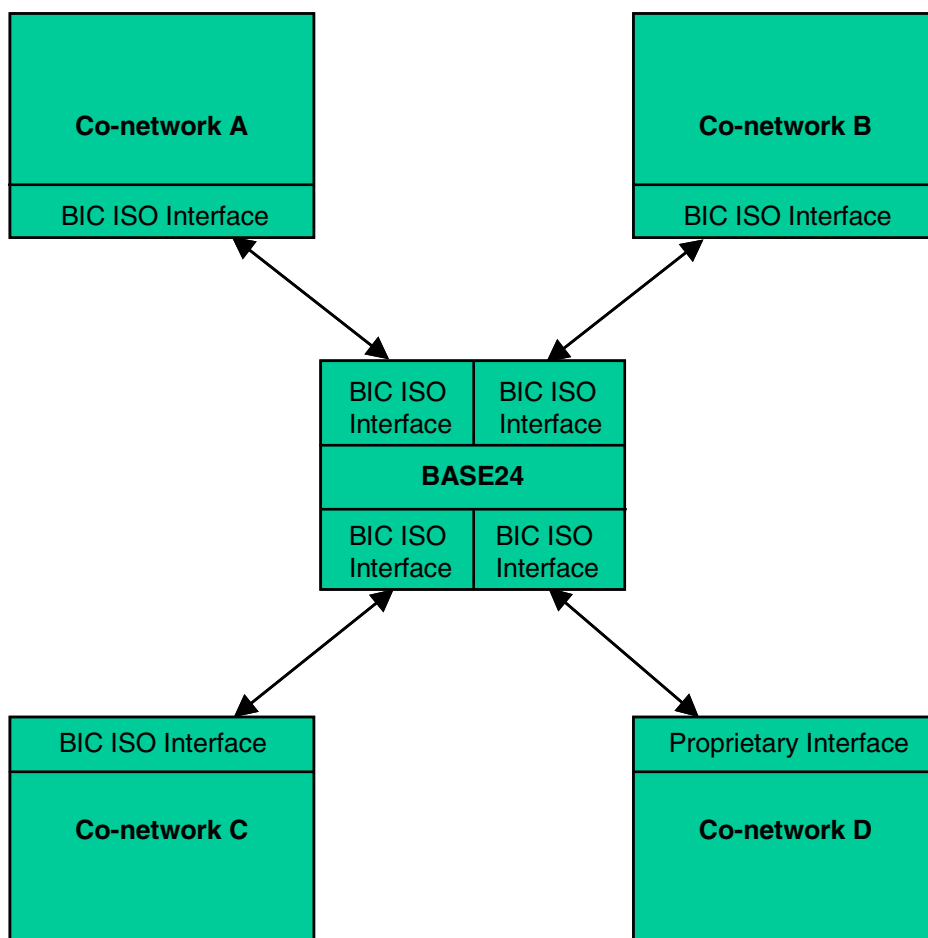
1. The BASE24 Authorization process sends an internal request message to the BIC ISO Interface process.
2. The BIC ISO Interface process determines that the link to the co-network is down.
3. The BIC ISO Interface process returns the internal request message to the BASE24 Authorization process.
4. The BASE24 Authorization process approves the transaction and sends an internal force post message to the BIC ISO Interface process.
5. The BIC ISO Interface process translates the message into an external force post message and stores the message until the link to the co-network is restored.
6. The BIC ISO Interface process sends the external force post message to the co-network.

If stand-in authorization is not allowed and an alternate destination is available, the link-down situation is handled by using alternate routing to send the transaction to the alternate destination. However, if stand-in authorization is not allowed and an alternate destination is not available, the BIC ISO Interface process responds to a link-down situation by denying all transactions until the link is reestablished.

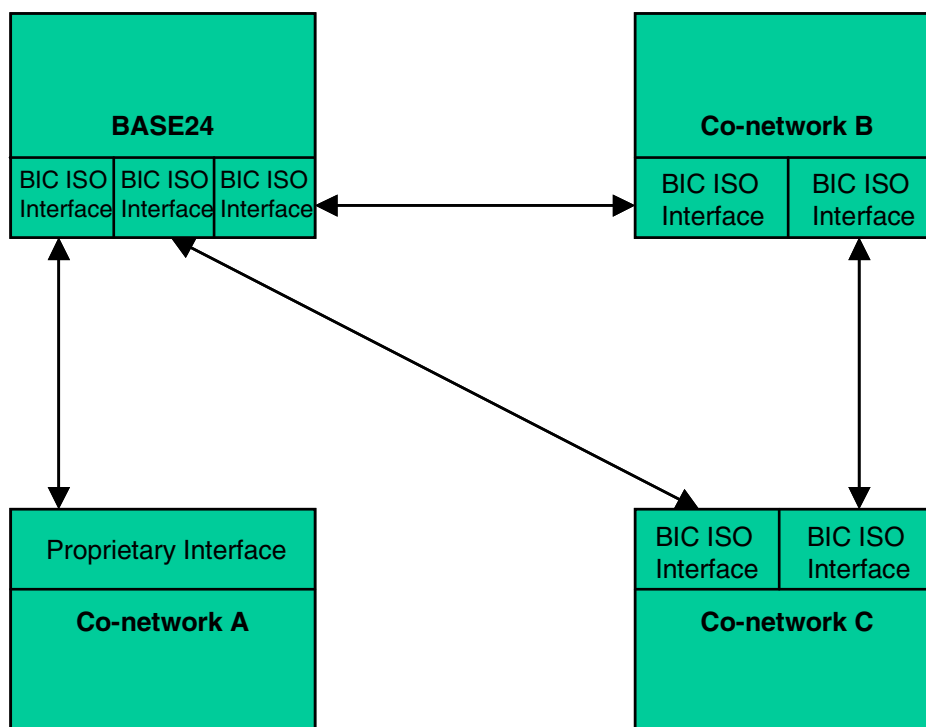
BIC ISO Networks

The main purpose of this manual is to describe the connection between a BASE24 system and a single co-network. However, it is possible to connect a BASE24 system to multiple co-networks. This type of a system requires a separate BIC ISO Interface process for each co-network to which the BASE24 system is connected.

A BIC ISO network can be configured to require that all transactions be routed through a single member of the BIC ISO network. The example illustrated below has a primary BASE24 system configured as the focal point for exchanging transactions within the entire BIC ISO network. In this type of arrangement, the primary BASE24 system has a separate BIC ISO Interface process for each co-network to which it is linked. Co-networks A, B, and C are other BASE24 systems and would each require one BIC ISO Interface process. Co-network D is a non-BASE24 system and would require a proprietary interface that conforms to BIC ISO standards for message formats and processing.



A BIC ISO network can also be configured to allow for certain transactions to be exchanged directly between members of the BIC ISO network. An example of this is illustrated below. In this illustration, the configuration consists of a primary BASE24 system and three co-networks, two of which are other BASE24 systems. The primary BASE24 system can exchange transactions with each of the three co-networks; co-networks B and C, which are also BASE24 systems, are configured to exchange transactions directly with each other. Co-network A is a non-BASE24 system. In this type of arrangement, the primary BASE24 network would require three BIC ISO Interface processes, co-networks B and C would each require two BIC ISO Interface processes, and co-network A would require a proprietary interface that conforms to BIC ISO standards for message formats and processing.



Supported Transactions

The BIC ISO Interface process provides support for both BASE24-atm and BASE24-pos transactions.

Different transaction sets are supported by the BIC ISO Interface process depending on the BASE24 products installed.

Supported BASE24-atm Transactions

If BASE24-atm is installed, the following transactions are supported by the BIC ISO Interface process. These transactions are supported for checking, savings, and credit accounts. The BIC ISO Interface process also supports transactions for which no account type has been specified.

- Balance inquiries
- Deposits
- Deposits with cash back
- EMV PIN unblock
- Log-only transactions
- Messages to financial institutions
- Payments enclosed
- Payments from/to
- PIN change
- Split deposits
- Transfers
- Withdrawals

Supported BASE24-pos Transactions

If BASE24-pos is installed, the following transactions are supported by the BIC ISO Interface process. These transactions are supported for checking, savings, and credit accounts. The BIC ISO Interface process also supports transactions for which no account type has been specified.

- Balance inquiries
- Card verifications
- Cash advances
- Cash advance adjustments
- Check guarantee
- Check verification
- Mail or telephone orders
- Merchandise returns
- Merchandise return adjustments
- Normal purchases
- Preauthorizations
- Preauthorization completions
- Purchase adjustments
- Purchases with cash back
- Purchase with cash back adjustments

File Usage

The BIC ISO Interface process uses a number of BASE24 files. These files can be divided into control files and transaction files, and are identified below.

Control Files

Control files contain configuration information and parameters that determine how the BIC ISO Interface process operates. The control files that contain configuration information for the BIC ISO Interface process include the following:

- Acquirer Processing Code File (APCF)
- Check Authorization Routing File (CART)
- Enhanced Interchange Configuration File (ICFE)
- External Message File (EMF)
- Extract Configuration File (ECF)
- Issuer Processing Code File (IPCF)
- Key File (KEYF) or Key 6 File (KEY6)
- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Token File (TKN)

For information on how these files are used to configure the BIC ISO Interface process, see section 2.

Transaction Files

Transaction files are used to store information about individual transactions processed through the BIC ISO Interface process. The transaction files used by the BIC ISO Interface process include the following:

- Interchange Log File (ILF)
- Interchange Totals File (ITF)
- Store-and-Forward File (SAF)

For information on what the ILF contains when it is used by the BIC ISO Interface process, refer to the ILF Data Definition Libraries (DDLs). For information on the SAF, refer to the SAF DDLs. The ILF and SAF DDLs can be accessed using the DDLDOC utility. For information on the DDLDOC utility, refer to the ***BASE24 Files and Record Structures Reference Manual***. For information on the ITF, refer to section 6.

Supported Message Types

The BIC ISO Interface process supports the following messages between the BASE24 system and the co-network. Notations appear for each message type to indicate its use or availability by BASE24 product.

Type	Description	BASE24-atm	BASE24-pos
0100	Authorization Request		✓
0110	Authorization Request Response		✓
0120	Authorization Advice		✓
0121	Authorization Advice Repeat		✓
0130	Authorization Advice Response		✓
0200	Financial Transaction Request	✓	✓
0210	Financial Transaction Request Response	✓	✓
0220	Financial Transaction Advice	✓	✓
0221	Financial Transaction Advice Repeat	✓	✓
0230	Financial Transaction Advice Response	✓	✓
0402	Card Issuer Reversal Request		✓
0412	Card Issuer Reversal Request Response		✓
0420	Acquirer Reversal Advice	✓	✓
0421	Acquirer Reversal Advice Repeat	✓	✓
0430	Reversal Advice Response	✓	✓
0500	Acquirer Reconciliation Request	✓	✓
0502	Issuer Reconciliation Request	✓	✓
0510	Acquirer Reconciliation Request Response	✓	✓
0512	Issuer Reconciliation Request Response	✓	✓
0520	Acquirer Reconciliation Advice	✓	✓

Type	Description	BASE24-atm	BASE24-pos
0521	Acquirer Reconciliation Advice Repeat	✓	✓
0522	Issuer Reconciliation Advice	✓	✓
0523	Issuer Reconciliation Advice Repeat	✓	✓
0530	Acquirer Reconciliation Advice Response	✓	✓
0532	Issuer Reconciliation Advice Response	✓	✓
0800	Network/Key Management Request	✓	✓
0810	Network/Key Management Request Response	✓	✓
0820	Network Management Advice	✓	✓
0821	Network Management Advice Repeat	✓	✓
0830	Network Management Advice Response	✓	✓

Interchange Reporting

The BIC ISO Interface process produces two reports: the SETL01-BIS and the STAT09-BIC. The purpose of these reports is to facilitate settlement between the BASE24 system and the co-network. For information on these reports, refer to section 6.

Address Verification

The BIC ISO Interface process supports BASE24-pos address verification processing. Merchants can use address information supplied by the customer to verify whether the customer is entitled to use an account when it is not possible to physically examine the card and the customer's signature.

Address verification data is added to data element P-63 as a token and is passed along with any other tokens to the co-network in the BIC ISO external message for BASE24-pos message types. This token contains address information and verification results. The presence of data element P-63 in the BIC ISO external message is determined by the setting of this data element in the External Message File (EMF). In order for P-63 to be included in the BIC ISO external message, it must be defined as mandatory or conditional in the EMF.

For more information on BASE24-pos Address Verification add-on product, refer to the ***BASE24-pos Address Verification Manual***.

Self-Service Banking (SSB)

The BIC ISO Interface process supports the split deposit transaction option available through BASE24-atm self-service banking (SSB). Split deposit transactions can be performed only if the customer has purchased the BASE24-atm self-service banking Base Application.

Split deposit transactions allow BASE24-atm customers to deposit money into two separate accounts by entering a single transaction. The form of deposit can consist of an envelope or a check. The following example describes how a split deposit transaction works. A customer selects a split deposit transaction from an ATM. The customer is then prompted to enter the split deposit type. The split deposit type identifies the two account types into which the deposit is to be made. These accounts can be two checking accounts, two savings accounts, or one checking account and one savings account. Next, the customer is prompted to enter the full deposit amount followed by the deposit amount for the first account. BASE24 then computes the amount deposited into the second account. A transaction request is initiated for the deposit and the request is sent to the authorizing entity.

Data element P-54 can be included in the BIC ISO external message for BASE24-atm message types. This data element contains the deposit credit amount established for split deposit transactions. The presence of P-54 in the BIC ISO external message is determined by the setting of this data element in the EMF. In order for this data element to be included in the BIC ISO external message, it must be defined as mandatory or conditional in the EMF.

For more information on the split deposit transaction and on the BASE24-atm self-service banking add-on product in general, refer to the device-specific BASE24-atm SSB manual.

Dynamic Key Management

The BIC ISO Interface uses working keys to protect transaction data moving between networks. Working keys are used to maintain the proper security to protect personal identification numbers (PINs) and verify message authentication codes (MACs). Working keys should be changed regularly to ensure security. The monitoring and changing of these keys is known as key management and can be performed manually using text commands or automatically using dynamic key management.

Dynamic key management allows co-networks to automatically change working keys based on thresholds set for time, number of transactions, and number of single or consecutive errors. One or more working keys can be changed when any one of the defined thresholds is surpassed.

Dynamic key management requires the use of hardware security modules to perform PIN encryption, PIN verification, and message authentication. It is not supported when these functions are performed in software.

Fields in the Key File (KEYF) or Key 6 File (KEY6) and the External Message File (EMF) must be set to specific values to allow for dynamic key management. See section 2 for a listing of KEYF, KEY6 and EMF settings.

For more information about dynamic key management, refer to the ***BASE24 Transaction Security Manual***.

Five-day Settlement

The BIC ISO Interface process can support 5-day settlement. An interface using 5-day settlement creates an ILF for Monday when it cuts over on Friday instead of creating one for Saturday. All transactions taking place after this cutover until the next cutover on Monday are logged to Monday's ILF. The totals calculations are also processed this way.

The cutover date can also be configured to allow for holiday dates. These dates are treated like weekends: when the interface cuts over on the day before the holiday, it creates the ILF for the day after the holiday. For example, when the interface cuts over on December 31, it creates an ILF for January 2. All transactions taking place on January 1 and January 2 are logged to the ILF for January 2.

Five-day settlement is configured on screen 2 of the Enhanced Interchange Configuration File (ICFE). See section 2 for more information about the ICFE screen and its field settings.

Note: Because the ILF for Monday will contain transactions for Saturday, Sunday, and Monday, the file extents for the ILF should be large enough to allow for an extra two days' volume.

Currency Conversion Support

The BIC ISO Interface process supports the conversion of currency between two institutions that do not share the same currency. Institutions have the option of using the same currency, two different currencies, or different currencies for acquirer, issuer, and settlement.

The co-networks must first decide on a currency in which to settle transactions. If the settlement currency is different than the currency of the acquiring bank, the BIC ISO Interface process converts the local currency to the settlement currency. If the settlement currency is different from the local currency of the issuer, the BIC ISO Interface process converts the settlement currency into the local currency.

The following illustrations show transactions taking place between a BASE24 network in the United States and a co-network based in the United Kingdom. The co-networks have decided to settle transactions in pounds. The first illustration shows the BASE24 network as the acquirer.



In this example, a customer of the U.K. institution requests a transaction at a device owned by the U.S. institution. The BASE24 Authorization process sends the transaction to the BIC ISO Interface process. The interface converts it to pounds and sends it to the U.K. institution's co-network. The co-network sends the response to the BIC ISO Interface process in pounds. The BIC ISO Interface process converts the amount to dollars and sends the response to the BASE24 Authorization process. It also logs the transaction in dollars to its Interchange Log File (ILF).

The next illustration shows a transaction requested by a customer of the U.S. institution at a device owned by the U.K. institution. The BASE24 network is the issuer in this example.



The U.K. co-network sends the transaction to the BIC ISO Interface process in pounds. The BIC ISO Interface process converts the amount to dollars and forwards the transaction to the BASE24 Authorization process. The Authorization process formats a response message and sends it to the BIC ISO Interface process. The BIC ISO interface converts the currency to pounds and sends the response to the co-network. The BIC ISO Interface process logs the transaction to its ILF in dollars.

Amounts passed in the internal message within BASE24 are expressed in the local currency. Amounts logged to the Interchange Totals File (ITF) are expressed in the settlement currency. Amounts in the SETL01 and STAT09 reports are expressed in both the local and settlement currencies.

The BIC ISO Interface process does not perform currency conversion if the BASE24 system supports Multiple Currency processing, as described in the “Multiple Currency Support” topic.

Configuration Considerations

Currency conversion requires changes to the following:

- COBNAMES file on BAxxSRC subvolume, where *xx* is the number of the current release. The CURRENCY-CODE-TABLE needs to contain shared settlement currency code information. Instructions for adding a currency code are included in that file.
- Enhanced Interchange Configuration File (ICFE). The CURRENCY CODE field on ICFE screen 2 needs to be set to the settlement currency. The NETWORK MANAGEMENT MESSAGE ENABLED field on ICFE screen 3 needs to be set to a value of Y so the settlement currency code can be exchanged in logon messages. Two fields on ICFE screen 12, BUY RATE and SELL RATE, need to contain values for issuers and acquirers. For more information about the ICFE, see section 2.

- External Message File (EMF). Previously unused fields in the Standard External Message (SEM) message need to be turned on. See section 2 for more information about configuring the EMF.
- Logical Network Configuration File (LCONF). See section 2 for more information about LCONF parameters.
 - The BIC ISO Interface process uses the CURRENCY-CODE param to indicate the local currency code.
 - The BIC ISO Interface process uses the ENHANCED-CRNCY-CONV param to determine whether the amounts in fields P-54, S-95, or S-123 are specified in a single currency (either the transaction currency or the settlement currency), or in two currencies (both the transaction currency and the settlement currency). The BIC ISO Interface process on an acquirer system checks this param whenever it converts the transaction amounts to the settlement currency. The BIC ISO Interface process on an issuer system checks this param whenever it receives a request or advice message that contains the Settlement Currency Code (data element P-50). The presence of this data element indicates the co-network has converted the transaction amounts to the settlement currency.

Note: The amount fields in the SEM are limited to 12 digits. Network managers should keep this in mind when choosing the shared settlement currency. The working fields used in the calculation of the amount to be sent can accommodate 19 digits, but when the amount is converted to ASCII, more than 12 digits will cause the transaction formatting to fail. To prevent this possibility, some test amounts of varying sizes, including representative maximum amounts, should be calculated on paper before modifying the ICFE conversion rates.

Original Currency Release 6.0 Token

The BIC ISO Interface process on an issuer system adds the Original Currency Release 6.0 token (token ID BE) if the transaction currency code (data element P-49) does not match the local currency code (the CURRENCY-CODE param in the LCONF).

When the Original Currency 6.0 token is added, data is moved from the BIC ISO external message to the token, as shown in the following table. The Original Currency Release 6.0 token can then be logged to the Interchange Log File (ILF), Transaction Log File (TLF), and POS Transaction Log File (PTLF) for use in settlement processing with the co-network.

BIC ISO External Message Field	Original Currency Release 6.0 Token Field
TRAN-AMT	AMT-1
ADD-AMTS.B24-DEF.AMT	AMT-2
CRNCY-CDE	CRNCY-CDE
SETL-CONV-RAT	CONV-RATE
CONV-DAT	CONV-DAT

Multiple Currency Support

The BASE24-atm and BASE24-pos Multiple Currency add-on products allow institutions to offer accounts in different currencies, with the ability to define each account with its own currency. They also allow cardholders to perform transactions in different currencies at different ATMs and point-of-sale (POS) devices. This add-on product is available with the BASE24-atm NCR 5XXX and Diebold 10XX/478X Device Handler processes and the BASE24-pos Standard POS Device Handler (SPDH) and APACS 30/40 Device Handler modules.

The BIC ISO Interface process allows multiple currency transactions to pass unimpeded to and from the co-network.

Customers who use the Multiple Currency add-on products should not configure the currency conversion processing described in the “Currency Conversion” topic. If a BASE24 system is set up with a Multiple Currency add-on product and transactions flow through the BASE24 system in more than one transaction currency, all BIC ISO Interface processes within that system should be configured to support the Multiple Currency add-on products.

Note: When using a Multiple Currency add-on product, the BIC ISO Interface process bypasses Interchange Totals File (ITF) processing because it is not possible to accumulate totals in different currencies without currency conversion.

You can configure Multiple Currency add-on product support by setting the MULTI CURRENCY field on Enhanced Interchange Configuration File (ICFE) screen 3 to a value of Y. The value Y indicates that transactions can pass through the BIC ISO Interface process in more than one currency, and single currency conversion processing is not used (i.e., the rates set with the BUY RATE and SELL RATE fields on ICFE screen 12). If the MULTI CURRENCY field on ICFE screen 3 is set to a value of Y, the co-network must also support multiple currencies.

Refer to the *BASE24 Base Files Maintenance Manual* for more information about ICFE screen 3. For more information about the BASE24-atm Multiple Currency and BASE24-pos Multiple Currency add-on products, refer to the *BASE24-atm Multiple Currency Support Manual* and the *BASE24-pos Multiple Currency Support Manual*, respectively.

Performance Monitoring Support

If you have purchased the NET24 Performance Monitoring subsystem, the BIC ISO Interface process supports the performance monitoring of transactions, in real time, between the BASE24 system and interchanges for both the BASE24-atm and BASE24-pos products. The standard application counters for interchanges are defined below. For more information on the NET24 Performance Monitoring subsystem, refer to the *NET24 Performance Monitoring Manual*.

Counter	Description
Issuer Approval Count	The total number of requests to the issuer that resulted in an approved response during the retrieval interval.
Issuer Denial Count	The total number of requests to the issuer that resulted in a declined response during the retrieval interval.
Issuer Force Post Count	The total number of requests to the issuer that resulted in an advice response during the retrieval interval.
Issuer Reversal Count	The total number of reversals during the retrieval interval.
Issuer Bad PIN Count	The total number of requests to the issuer that resulted in a declined response during the retrieval interval because of a bad PIN.
Issuer Key Sync Count	The total number of requests to the issuer that resulted in a key synchronization error during the retrieval interval.
Inbound Request Time-out Count	The total number of requests to the issuer that timed out during the retrieval interval.

Counter	Description
Inbound Response Time	The average response time of all issuer response transactions during the retrieval interval. This counter consists of two counters with the same name. One counter is a message rate counter that is the total number of requests to the issuer during the retrieval interval. The second counter is a response time counter that displays the average response time. The average response time is determined by dividing the total response time by the total number of requests to the issuer. The total response time is measured from the time that the request enters the BASE24 system until the response exits the BASE24 system.
Acquirer Approval Count	The total number of acquirer requests that resulted in an approved response during the retrieval interval.
Acquirer Denial Count	The total number of acquirer requests that resulted in a declined response during the retrieval interval.
Acquirer Force Post Count	The total number of acquirer requests that resulted in an advice response during the retrieval interval. Note that this counter is available for the BASE24-pos product only.
Acquirer Reversal Count	The total number of reversals during the retrieval interval.
Acquirer Bad PIN Count	The total number of acquirer requests that resulted in a declined response during the retrieval interval because of a bad PIN.
Acquirer Key Sync Count	The total number of acquirer requests that resulted in a key synchronization error during the retrieval interval.
Outbound Request Time-out Count	The total number of acquirer requests that timed out during the retrieval interval.

Counter	Description
Outbound Response Time	The average response time of all acquirer response transactions during the retrieval interval. This counter consists of two counters with the same name. One counter is a message rate counter that is the total number of acquirer requests received during the retrieval interval. The second counter is a response time counter that displays the average response time. The average response time is determined by dividing the total response time by the total number of acquirer requests. The total response time is measured from the time that the request enters the BASE24 system until the response exits the BASE24 system.

Surcharging Support

The BIC ISO Interface process supports the capability of surcharging. An institution can configure the BASE24-atm system to assess acquirer fees (surcharges) on transactions initiated at selected ATMs. Surcharges can be based on attributes such as terminal type (for example, the location of the ATM), card prefix or transaction routing, card type, and transaction type. In addition, surcharges can be based on a flat fee per transaction or on a percentage of the transaction amount. A negative flat fee (rebate) value is supported.

Surcharging is configured in the Surcharge File (SURF). Fields in the Card Prefix File (CPF) and the BASE24-atm Terminal Data files (ATD) link the SURF with a card prefix or terminal group. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about configuring the SURF and CPF for surcharging. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about configuring the ATD.

The surcharge amount is carried in data element P-28 of the BIC ISO external message. The presence of data element P-28 in the BIC ISO external message is determined by the setting of this data element in the External Message File (EMF). In order for this data element to be included in the BIC ISO external message, it must be defined as mandatory or conditional in the EMF. Refer to the ***BASE24 External Message Manual*** for more information about this data element.

In addition, the Surcharge Data token is appended to the message and passed along with any other tokens to the co-network. This token is used to store surcharge assessment information defined in the BASE24-atm Terminal Data files (ATD) and Surcharge File (SURF) and other information filled in at the entry point and by the Authorization process. Refer to the ***BASE24 Tokens Manual*** for more information about the Surcharge Data token.

Enhanced EMV Response Code Mapping

BASE24 maps response codes carried internally with ISO response codes used in the BASE24 external message. Two types of mapping are supported: standard mapping and enhanced EMV mapping.

The difference between standard mapping and enhanced EMV mapping is how EMV-related response codes are handled. Using standard mapping, EMV-related response codes are mapped to more general response codes which can obscure the reason an EMV transaction is declined. Using enhanced EMV mapping, however, unique ISO response codes are used that allow EMV-related response codes to retain their EMV characteristics. For example, an outbound BASE24-atm EMV response code of 084 would map to 05 under standard mapping and U4 under enhanced EMV mapping as shown in the table below. Where the former does not retain the specific reason for the response (due to an ATC check failure), the latter does.

BASE24-atm	Standard Mapping	Enhanced EMV Mapping
084 ATC check failure	05 Do not honor	U4 ATC check failure

For a complete list of BASE24-atm response code mappings, refer to the ***BASE24 BIC ISO Standards Manual***.

The type of mapping used for BASE24-atm and BASE24-pos EMV messages is controlled by the ENHANCED-EMV-RC-MAPPING param in the LCONF. Param values control the type of mapping as follows:

Param Value	Response Code Mapping	
	BASE24-atm	BASE24-pos
0	Standard Mapping	Standard Mapping
1	Enhanced EMV Mapping	Standard Mapping
2	Standard Mapping	Enhanced EMV Mapping
3	Enhanced EMV Mapping	Enhanced EMV Mapping

2: Configuration and Control

This section describes the configuration and control of the BIC ISO Interface process and includes the following:

- Descriptions of files used by the BIC ISO Interface process
- Process configuration information
- Descriptions of institution- and co-network-level controls
- A description of how stations are defined and used by the BIC ISO Interface process
- Descriptions of the timers used by the BIC ISO Interface process and the processing performed when each timer expires

LCONF Assigns and Params

The BIC ISO Interface process uses a number of Logical Network Configuration File (LCONF) assigns and params in its processing. These assigns and params are identified below. For more information on configuring the BIC ISO Interface process using these assigns and params, refer to the *BASE24 Logical Network Configuration File Manual*.

Assigns

The following assigns identify the names of files that contain BIC ISO Interface process configuration and control information, or that the BIC ISO Interface process uses during processing. The BIC ISO Interface process reads these assigns regardless of the products supported.

APCFEMT — Identifies the name of the Acquirer Processing Code File (APCF) extended memory table (APCFEMT), which contains APCF records.

EMF — Identifies the name of the External Message File (EMF).

ICF — Identifies the name of the Interchange Configuration File (ICF). This assign is required if the BASE24-REL param is not present or does not contain the value of 60.

ICFE — Identifies the name of the Enhanced Interchange Configuration File (ICFE). This assign is required if the BASE24-REL param contains the value of 60.

ILF — Identifies the name of the Interchange Log File (ILF).

IPCFEMT — Identifies the name of the Issuer Processing Code File (IPCF) extended memory table (IPCFEMT), which contains IPCF records.

ITF — Identifies the name of the Interchange Totals File (ITF).

KEY6 — Identifies the name of the Key 6 File (KEY6).

KEYF — Identifies the name of the Key File (KEYF).

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

RPT — Identifies the name of the Report Perusal File (RPT). This assign is required if reports are to be run.

RPT-ERROR-FILE — Identifies the name of the spooler location to receive report error messages for interchange reports. This assign is required if reports are to be generated.

SAF — Identifies the name of the Store-and-Forward File (SAF).

SW-RPT-OBEY-FILE — Identifies the name of the obey file used to start the interchange reports. This entry is required if reports are to be generated.

SW-RPT01-PROG-NAME

SW-RPT09-PROG-NAME — Identify the names of the interchange report programs. These entries are required if these reports are to be generated.

TKN — Identifies the name of the Token File (TKN).

Params

Several params are used by the BIC ISO Interface process during processing. These params are as follows.

ATM-DISCR-DATA-STM-SUPPRS — Specifies whether discretionary data in the BASE24-atm Standard Internal Message (STM) should be suppressed. Valid values are Y (suppress) and N (do not suppress), with a default value of N.

ATM-LN-CUTOVER — The time of day all BASE24-atm processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This param is required for BASE24-atm.

CURRENCY-CODE — A code indicating the primary type of currency used by the logical network. Each code consists of a three-digit value. For the currency code values, refer to the ISO 4217 standard *Codes for the Representation of Currencies and Funds*.

DATE-DISPLAY-TYPE — A flag indicating the format of the month and day portion of the dates on reports. This param does not affect dates which include the year. Valid values are as follows:

- 1 = DDMM
- 2 = MMDD (default)

DES-TRIPLE-SINGLE — A code indicating whether an Interchange Interface process translates PINs from encryption under a double length key using the Triple Data Encryption algorithm (T-DEA) to encryption under a single length key using the Data Encryption algorithm (T-DEA) and vice versa. Valid values are as follows:

- Y = Yes, translate PINs from T-DEA encryption to DEA encryption and vice versa.
- N = No, do not translate PINs from T-DEA encryption to DEA encryption and vice versa.

If this param is set to a value of Y, the BIC ISO Interface process uses the Key 6 file (KEY6). Otherwise, it uses the Key File (KEYF).

ACCT-NUM-TYP — A flag indicating whether account numbers are stored in binary or decimal form. The binary form allows the account number to be up to 28 digits in length. The decimal form allows the account number to be up to 19 digits in length. Valid values are as follows:

- D = Decimal format (default)
- B = Binary format

ENHANCED-CRNCY-CONV — A code indicating whether enhanced currency conversion processing is performed. If data elements P-54, S-95, or S-123 are present in the ISO-based external message for messages acquired by the co-network or BASE24, this param indicates whether the amounts in these fields are specified in a single currency or in two currencies. Enhanced currency conversion is performed if the amounts are specified in two currencies. Valid values are as follows:

- 0 = Do not perform enhanced currency conversion processing. (This is the default value.)

- 1 = Perform enhanced currency conversion processing for ATM transactions only.
- 2 = Perform enhanced currency conversion processing for POS transactions only.
- 3 = Perform enhanced currency conversion processing for ATM and POS transactions.

This param is not used if the system supports Multiple Currency processing (the MULT CURRENCY field on ICFE screen 3 is set to Y).

ENHANCED-EMV-RC-MAPPING — A code indicating whether enhanced EMV response code mapping is used. Certain BASE24-atm and BASE24-pos response codes are used to identify EMV-specific failure conditions. Without enhanced EMV response code mapping, these response codes are mapped to a generic ISO response code (e.g., 05—do not honor), and the acquiring system often cannot identify the reason for a declined EMV transaction. With enhanced EMV response code mapping, unique ISO response codes are added to identify EMV-specific failure conditions. Valid values are as follows:

- 0 = Do not perform enhanced EMV response code mapping. (This is the default value.)
- 1 = Perform enhanced EMV response code mapping for ATM transactions only.
- 2 = Perform enhanced EMV response code mapping for POS transactions only.
- 3 = Perform enhanced EMV response code mapping for ATM and POS transactions.

ILF-NOTIFY-INTERVAL — The number of writes made to the ILF between calculating the percentage of file full. Valid values are 10–32766. The default value is 50.

ILF-NOTIFY-PERCENTAGE — The percentage of file full at which the BIC ISO Interface process logs a message indicating the ILF is becoming full. The BIC ISO Interface process continues to log messages at the interval indicated in the ILF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10–99. The default value is 70.

OVRRD-INST-ID-FLDS — A code indicating whether the institution ID can be overridden with values from fields in the Enhanced Interchange Configuration File (ICFE) record. The ICFE fields that can be overridden are the SWITCH ID and DEFAULT ACQUIRER ID NUM fields on ICFE screen 1. Valid values are as follows:

- 0 = Do not override institution ID values. (This is the default value.)
- 1 = Override acquiring institution ID only.
- 2 = Override forwarding institution ID only.
- 3 = Override acquiring and forwarding institution ID.
- 4 = Override receiving institution ID.
- 5 = Override acquiring and receiving institution ID.
- 6 = Override forwarding and receiving institution ID.
- 7 = Override acquiring, forwarding, and receiving institution ID.

POS-B24-TO-POS-ACCT-TYP-SUPPRS — Specifies whether the account type in the BASE24-pos Standard Internal Message (PSTM) should be suppressed when the PSTM is mapped to the external message. The BIC ISO Interface process suppresses the account type from the PSTM by using an account type of zeroes in the external message. Valid values are Y (suppress) and N (do not suppress), with a default value of N.

POS-DISCR-DATA-PSTM-SUPPRS — Specifies whether discretionary data in the BASE24-pos Standard Internal Message (PSTM) should be suppressed. Valid values are Y (suppress) and N (do not suppress), with a default value of N.

POS-LN-CUTOVER — The time of day all BASE24-pos processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This param is required for BASE24-pos.

PRE-VERIFIED-PIN-PAD-CHAR — Specifies the value to be repeated in a PIN block to identify the PIN as pre-verified. The default value is F, but it is possible for an actual encrypted PIN to result in a PIN block value of all Fs. If used, this value should be set to a non-hexadecimal character (i.e., a character other than 0-9 or A-F) because the use of any repeated hexadecimal character could have the same issue as the F character. If this value is set, the host must also be modified to recognize the new pre-verified PIN block character.

BASE24-pos Note: BASE24-pos transactions may or may not contain a PIN, and authorizing systems must be able to differentiate between transactions with a pre-verified PIN and transactions that do not contain a PIN (i.e., that were not entered

with a PIN). Therefore, for incoming and outgoing BASE24-pos transactions, a PIN block containing this LCONF pad character is always used to indicate a PIN has been pre-verified. The absence of the P-52 (Personal Identification Number Data) data element in the external message is used to indicate the transaction did not contain a PIN (for incoming BASE24-pos messages, a space-filled P-52 data element is also acceptable to indicate the transaction did not contain a PIN).

BASE24-atm Note: BASE24-atm transactions always require a PIN. Therefore, for BASE24-atm transactions, a pre-verified PIN can be accurately identified by the absence of the P-52 (Personal Identification Number Data) data element in the external message. For outgoing messages, BASE24-atm always identifies a pre-verified PIN by omitting the P-52 data element. For incoming messages, the host has several options—BASE24-atm assumes a PIN has been pre-verified if the P-52 data element is omitted, contains spaces, or is filled by the pad character specified in this param.

PSEUDO-3DES-CBC-MAC-TYPE — Specifies whether to use the Pseudo 3DES Cipher Block Chaining method of MACing instead of the default 3DES Cipher Block Chaining method used by the ISO Host Interface process.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

RACAL-ANSI-METHOD — Specifies whether to use the ANSI X9.17 Encryption Method for key exchanges instead of the default Variant Encryption Method used by the ISO Host Interface process. The ANSI X9.17 Encryption Method can be used with a Thales (formerly Racal) Hardware security module.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

REVERSAL-BAL-INQ — Specifies whether balance inquiry transactions can be reversed. This param is only read if BASE24-atm or BASE24-pos support is bound into the BIC ISO Interface process.

RSM-TERM-MASTER-KEY — Specifies an encrypted terminal master key used during hardware message authentication by Thales (formerly Racal) security devices.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

SAF-NOTIFY-INTERVAL — The number of writes made to the SAF between calculating the percentage of file full. Valid values are 10–32766. The default value is 50.

SAF-NOTIFY-PERCENTAGE — The percentage of file full at which the BIC ISO Interface process logs a message indicating the SAF is becoming full. The BIC ISO Interface process continues to log messages at the interval indicated in the SAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10–99. The default value is 70.

SEND-ETX — A flag indicating whether an interchange interface process sends the end-of-text (ETX) control character. Valid values are as follows:

- Y = Yes, send the ETX character with external messages (default value).
- N = No, do not send the ETX character with external messages.

SPAN-FLAG — A value indicating whether totals are written to the Span Clearings File. Valid values are as follows:

- Y = Yes, write totals to the Span Clearings File.
- N = No, do not write totals to the Span Clearings File.

SWI-ACQ-AVS-DATA-SUPPRS — A flag indicating whether logging of address verification data for acquirer transactions to the ILF should be suppressed. Valid values are as follows:

- Y = Yes, suppress logging of address verification data.
- N = No, do not suppress logging of address verification data.

SWI-ACQ-CVD2-DATA-SUPPRS — A flag indicating whether logging of card verification data (CVD2) for acquirer transactions to the ILF should be suppressed. Valid values are as follows:

- Y = Yes, suppress logging of card verification data (CVD2).
- N = No, do not suppress logging of card verification data (CVD2).

SWI-ACQ-DISCR-DATA-SUPPRS — A flag indicating whether logging of discretionary data from Track 1 or Track 2 for acquirer transactions to the ILF should be suppressed. Valid values are as follows:

Y = Yes, suppress logging of discretionary data from Track 1 or Track 2.

N = No, do not suppress logging of discretionary data from Track 1 or Track 2.

SWI-ISS-AVS-DATA-SUPPRS — A flag indicating whether logging of address verification data for issuer transactions to the ILF should be suppressed. Valid values are as follows:

Y = Yes, suppress logging of address verification data.

N = No, do not suppress logging of address verification data.

SWI-ISS-CVD2-DATA-SUPPRS — A flag indicating whether logging of card verification data (CVD2) for issuer transactions to the ILF should be suppressed. Valid values are as follows:

Y = Yes, suppress logging of card verification data (CVD2).

N = No, do not suppress logging of card verification data (CVD2).

SWI-ISS-DISCR-DATA-SUPPRS — A flag indicating whether logging of discretionary data from Track 1 or Track 2 for issuer transactions to the ILF should be suppressed. Valid values are as follows:

Y = Yes, suppress logging of discretionary data from Track 1 or Track 2.

N = No, do not suppress logging of discretionary data from Track 1 or Track 2.

SW-ILF-FILE-RETENTION — The number of days an ILF is to remain on the system. After the specified number of days, the ILF is purged from the system by the BIC ISO Interface process. Valid values are 3–99. The default value is 7.

SW-RPT-STARTUP — The time the BIC ISO Interface process runs its report obey file. Valid values, in military time, are from 00:00 to 23:59.

UPDT-ILF-ON-LATE-RESP — A flag indicating whether the BIC ISO Interface should retrieve the ILF record upon receipt of a late response and update it to reflect that a reversal was sent out. Valid values are as follows:

- Y = Yes, retrieve the ILF record upon receipt of a late response and update it to reflect that a reversal was sent out (default value).
- N = No, do not retrieve the ILF record upon receipt of a late response and update it to reflect that a reversal was sent out.

Configuring a Co-Network

Each co-network with which BIC ISO Interface process communicates requires a record in the Enhanced Interchange Configuration File (ICFE). Co-networks are assigned stations and tied to the BIC ISO Interface processes with which they communicate using the ICFE.

A number of processing options are controlled at the co-network level, which enable institutions to set up co-networks with individual processing requirements. The majority of processing options available at the co-network level are stored in the ICFE. Other options are specified in the KEYF or KEY6. The processing options controlled at the co-network level are described below.

Specifying Timer Lengths

The BIC ISO Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICFE, KEYF, or LCONF. Some timer intervals are specified once at the interface level, while others are specified at the interface and product level (e.g., you can specify one time limit for inbound BASE24-atm messages and a different time limit for inbound BASE24-pos messages).

The table below identifies each timer used by the BIC ISO Interface process, along with the level at which it is defined and the file in which it is defined. Each of these timers are discussed later in this section.

Timer	Interface Level	Interface/Product Level
Network Management Message timer	ICFE	
Extended Network Management Message timer	ICFE	
Store-and-Forward Message timer	ICFE	
Wait-for-Traffic timer	ICFE	
Performance timer	ICFE	
Outbound Request timer		ICFE

Timer	Interface Level	Interface/Product Level
Inbound Request timer		ICFE
Completion Acknowledgment timer		ICFE
PIN Key Exchange timer	KEYF or KEY6	
MAC Key Exchange timer	KEYF or KEY6	
Clear Old Key timer	KEYF or KEY6	
Report Startup timer	LCONF	
Settlement Interval timer	ICFE	
ITF Update timer	ICFE	

Interspersing Store-and-Forward Messages

Store-and-forward messages are interspersed with request messages. Interspersing store-and-forward messages allows other transactions to be processed during store-and-forward activity, while receiving all store-and-forward messages prior to further requests may cause requests to be processed offline or be declined because of an inability to send them to the co-network.

In addition, messages sent to the SAF are *single-threaded*. Single-threaded means that only one message of any kind can be outstanding to a station at a time. In this case, the BIC ISO Interface process must receive a response or the message must time out before another message can be sent to the station.

The BIC ISO Interface process sends the following messages types directly to the SAF instead of to the co-network:

- Reversal messages
- Force post messages
- Adjustment messages

- Chargeback messages
- Cutover messages
- Administrative messages

Sending Text-Level Acknowledgments to a Co-Network

Acquiring institutions must specify whether the co-network expects text-level acknowledgments from the BIC ISO Interface process. If text-level acknowledgments are required, BASE24 returns a 0130 message for each 0120 and 0121 message; a 0230 message for each 0220 or 0221 message; a 0430 message for each 0420 or 0421 message; and a 0530 message for each 0520 or 0521 message.

Institutions indicate whether text-level acknowledgments are required using the ACK TO SWITCH field on ICFE screen 1.

Expecting Text-Level Acknowledgments from a Co-Network

Issuer institutions must indicate whether the BIC ISO Interface process should expect text-level acknowledgments. If text-level acknowledgments are required, the BASE24 system expects a 0130 message for each 0120 and 0121 message; a 0230 message for each 0220 or 0221 message; a 0430 message for each 0420 or 0421 message; and a 0530 message for each 0520 or 0521 message.

Institutions indicate whether they send text-level acknowledgments using the ACK FROM SWITCH field on ICFE screen 1.

Note: The BIC ISO Interface process always requires acknowledgments for its store-and-forward messages. If text-level acknowledgments are not to be sent, the BIC ISO Interface process requests a *logical acknowledgment* from the XPNET process. A logical acknowledgment notifies the BIC ISO Interface process that the message has been sent successfully to the co-network.

If message authentication is being performed, the setting for this field should be carefully considered. Some message verification errors can be retried (for example if the co-network's security device is down at the time it receives the message to be authenticated). If this field indicates that text-level acknowledgments are not required and the message was a store-and-forward message, the original message will not be available to resend to the co-network. This is because the message would have been deleted when the XPNET process returned a logical acknowledgment that the message was sent successfully.

Specifying Maximum Timeouts

Institutions must indicate the number of consecutive timeouts of non-network management and non-store-and-forward messages that can occur at a station before the station is marked down. For example, if 3 is specified, the BIC ISO Interface process marks the station down upon the third consecutive timeout. However, if only two consecutive timeouts have occurred, the station is not marked down.

Institutions control when a station is marked down because of timeouts using the MAXIMUM TIMEOUTS field on ICFE screen 1.

Initiating Network Management Messages

Institutions specify whether the BIC ISO Interface process is allowed to initiate network management messages (that is, 0800 messages for new key, change key, verify key, or repeat key).

Institutions indicate whether the BIC ISO Interface process can initiate network management messages using the NETWORK MANAGEMENT MESSAGE ENABLED field on ICFE screen 3.

Note: This option only controls whether the BIC ISO Interface process can initiate the network management actions; the BIC ISO Interface process always responds to the network management messages it receives.

Enabling Dynamic Key Management

Institutions specify whether the BIC ISO Interface process is allowed to initiate network management messages for dynamic key management (that is, 0800 messages for new key, change key, verify key, or repeat key commands). This option limits the ability of the BIC ISO Interface process to perform or solicit key management requests.

Network management messages for dynamic key management are controlled by the timers and thresholds on KEYF screen 3 or KEY6 screen 4. If all of the timers and thresholds on KEYF screen 3 or KEY6 screen 4 are set to the value 0, the BIC ISO Interface process does not initiate network management messages for dynamic key management. If any of the timers or thresholds contains a nonzero value, the BIC ISO Interface process initiates a key management message when the timer expires or when the threshold is passed.

Specifying Extract File Format

The BIC ISO Interface process supports Format 2 files. The FILE FORMAT field on Extract Configuration File (ECF) screen 3 indicates whether the disk file is created as a Format 1 or Format 2 file.

Specifying Maximum Store-and-Forward Retries

Institutions set the maximum number of times the BIC ISO Interface process attempts to send a store-and-forward message to a co-network before sending that message to the log and deleting it from the SAF.

This option can be used to stop processing a message when, due to the number of timeouts on a store-and-forward message, it becomes apparent that the co-network cannot process the message. Institutions control this option using the MAX SAF RETRY field on ICFE screen 3.

Specifying Maximum Outbound Messages

The BIC ISO Interface process allows institutions to specify the maximum number of request messages (that is, 0100, 0200, 0300, 0402, and 0500 messages) or advice messages (that is, 0120, 0220, 0320, 0420, and 0520 messages) that can be outstanding to an issuer co-network before the BIC ISO Interface process stops sending additional messages. Outstanding messages are those messages that have not been received (for example, a 0210 for a 0200 or a 0230 for a 0220).

This option allows the BIC ISO Interface process to prevent processing when it becomes apparent that the co-network cannot process a message within the time limits required because of the number of messages already outstanding to the co-network.

Once the number of outstanding messages reaches this limit, the BIC ISO Interface process marks any new request messages destined for the co-network as failed and returns them to their originating process. The BIC ISO Interface process places any new advice messages destined for the co-network in the SAF. At this point, the co-network does not receive additional request or advice messages until the messages in the queue are either responded to or have timed out.

Institutions control the number of outstanding request and advice messages to a co-network using the OUTBOUND field on ICFE screen 3.

Specifying Maximum Inbound Requests

The BIC ISO Interface process also provides a parameter for limiting the number of outstanding 0100 and 0200 messages to a BASE24-atm or BASE24-pos Authorization process. This parameter controls requests being sent from acquirer co-networks. Once the number of outstanding 0100 and 0200 messages reaches this limit, the BIC ISO Interface process declines any new 0100 and 0200 messages coming from the co-network and returns them as 0110 and 0210 messages.

The number of outstanding requests to an Authorization process is controlled using the INBOUND field on ICFE screen 3.

Note: This feature only affects 0100 and 0200 messages. Other message types are not included in the count of outstanding messages, nor is their processing affected in any way.

Defining Data Prefix Characters

The BIC ISO Interface process allows institutions to define certain characters they want included in front of the messages they receive from the BIC ISO Interface process. An institution can specify up to nine characters to precede messages to their co-network system. Each time the BIC ISO Interface process creates an external message to a co-network, it checks the ICFE for data prefix characters. If the ICFE contains data prefix characters for the co-network, the BIC ISO Interface process places these characters at the front of the message.

Data prefix characters are placed in the ICFE using the hexadecimal equivalents for the ASCII versions of the characters to be sent. For example, suppose a co-network required the following three characters in front of the message:

ESC 1 B
(escape)

To place these three values in an ICFE record, a co-network would have to enter hexadecimal equivalents (six characters) for the ASCII representations of these values, which would be:

1B 31 42

These values would be stored in the ICFE as six characters (1B3142) and would be converted back to the three ASCII characters at the time the message is created.

Because the BIC ISO Interface process does not support transparent data communications, protocol characters cannot be used as data prefix characters. Some characters to avoid are:

Hex	ASCII Display	Hex	ASCII Display
01	SOH	02	STX
03	EXT	04	EOT
05	ENQ	06	ACK
10	DLE	15	NAK
16	SYN	17	ETB
1F	ITB		

Authorization Process

The BIC ISO Interface processes handling requests from acquirer networks determine the Authorization processes to which to route inbound transaction requests using the AUTH PROCESS field on ICFE screen 8 for the BASE24-atm product or ICFE screen 10 for the BASE24-pos product.

Setting Up Co-Network Stations

Stations serve as the links between BASE24 and a co-network. All messages sent to a co-network are actually sent to the stations defined for that co-network. Likewise, all messages received from a co-network are actually received from the stations defined to that co-network.

Up to 20 stations can be defined for each co-network, although a recommended maximum is 10 to 12. A station must first be defined as a co-network station in the Network Environment File (NEF). The station's symbolic name can then be placed in the ICFE to fully define the station.

The BIC ISO Interface process reads the ICFE at initialization and builds a table of the stations defined for each co-network.

Station Types

Co-networks can define two types of stations for use by the BIC ISO Interface process: send/receive stations and receive-only/no output stations. These stations are described below.

Station types are defined using the TYPE fields on ICFE screen 13.

Send/Receive Stations

Send/receive stations allow for both incoming and outgoing messages. The BIC ISO Interface process can send messages to and receive messages from the co-network using this type of station.

Receive-Only/No Output Stations

Receive-only/no output stations only allow for messages coming in to BASE24. BASE24 never uses receive only/no output stations for sending messages to the co-network. The receive-only/no output station is used to avoid line connection when using a point-to-point or SNA protocol. In this case, communications are received on one station and sent from a different station.

Setting Up the ICFE Station Table

Stations are numbered from left to right, top to bottom on the ICFE file maintenance screens. This numbering scheme is of particular importance to co-networks with multiple stations on their lines.

When selecting a station to which to send a message, the BIC ISO Interface process goes through its station table from start to finish until it finds a station available to receive the message.

If lines have multiple stations on them, the stations from each line should be alternated when setting up the ICFE station table, and not numbered sequentially. Alternating the stations balances the transaction load so that if one station is chosen from one line, the next station chosen is on a different line.

Selecting Stations for Sending Messages

To select the station to send a message to, the BIC ISO Interface process uses a technique called *round robin*. The BIC ISO Interface process begins with the first station in the ICFE station table. For each subsequent message that the interface needs to send, it increments the index in the ICFE station table to determine the next station. The interface evaluates the next station and determines the availability of the station based upon whether the station is a send or send and receive station and whether the station is marked up and logged on. The BIC ISO Interface compares the number of pending transactions for the next available station to the number of pending transactions for the current station and sends the message to the station that is least busy.

When the BIC ISO Interface process has sent a message to the last station in the ICFE station table, it begins its next search using the first station in the ICFE station table. The BIC ISO process never sends messages to receive only stations.

Configuring Data Suppression

The BIC ISO Interface process offers several options for suppressing data contained in response messages and logged to the Interchange Log File (ILF).

The BIC ISO Interface process can suppress discretionary data such as fields other than the primary account number (PAN) and the card expiration date in BASE24-atm and BASE24-pos response messages exchanged between co-hosts or that is logged to the ILF. It can also suppress card verification data (CVD2) and address verification data for transactions logged to the ILF.

The BIC ISO Interface process uses the following Logical Network Configuration File (LCONF) params to determine whether the data is logged to the ILF:

Response Messages

ATM-DISCR-DATA-STM-SUPPRS

POS-DISCR-DATA-PSTM-SUPPRS

ILF Logging When the Interface Acting as an Acquirer

SWI-ACQ-AVS-DATA-SUPPRS

SWI-ACQ-CVD2-DATA-SUPPRS

SWI-ACQ-DISCR-DATA-SUPPRS

ILF Logging When the Interface Acting as an Issuer

SWI-ISS-AVS-DATA-SUPPRS

SWI-ISS-CVD2-DATA-SUPPRS

SWI-ISS-DISCR-DATA-SUPPRS

If the corresponding param exists and the value indicates that the BIC ISO Interface process is to suppress discretionary, address verification, or CVD2 information, the data is suppressed. Refer to the “LCONF Assigns and Params” topic or the *BASE24 Logical Network Configuration Manual* for more information about these params.

Response message discretionary data suppression occurs in the internal message structure before it can be used to create the response message, while ILF data suppression occurs after transaction data has been moved to the ILF record structure and before it is written to disk.

The suppression of discretionary data can affect the content of messages exchanged between co-hosts. This is true whether the data has already been suppressed in response messages that BASE24 receives from an interchange or co-host, or the data is suppressed by a BASE24 authorization process. In particular, BASE24 may have to construct the Track 2 Data (P-35) data element, and the

default External Message File (EMF) records configure the Track 2 Data data element. As a result, users should consider configuring EMF records that include the Primary Account Number (P-2) data element and modify the default settings for the Track 2 (P-35) data element. Refer to the ***BASE24 ISO Standards Manual*** for additional information for modification suggestions.

Configuring Timers

The BIC ISO Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICFE.

Network Management Message Timer

A Network Management Message timer is set each time an 0800 message is sent to a co-network station. The Network Management Message timer limits the length of time that the BIC ISO Interface process waits for a response after transmitting an 0800 message. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on ICFE screen 3.

The actions taken when a Network Management Message timer expires depend on the type of message for which the timer was set. If the message was a logon, logoff, or echo test message, the BIC ISO Interface process performs as follows:

1. Marks the station down.
2. Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ISO Interface process marks the co-network down.
3. Sets an Extended Network Management Message timer for the station. If the 0800 message was a dynamic key management message, the BIC ISO Interface process performs as follows:
 - a. Adds one to the count of consecutive timeouts accumulated for the station.
 - b. Checks the number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICFE screen 1.
 - 1) If the current number of consecutive timeouts is less than the maximum allowed, processing continues with step c.
 - 2) If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ISO Interface process marks the station down and verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, so the BIC ISO Interface process marks the co-network down.

- c. Sets an Extended Network Management Message timer for the station.
- d. If the message that timed out was a new key or a change key message, sends a message to the network logging facility indicating that the message expired and that the message will automatically be retried.

Extended Network Management Message Timer

An Extended Network Management Message timer is set each time a co-network station is marked down and each time a new key or change key message times out. The Extended Network Management Message timer specifies the following intervals:

- The amount of time that the BIC ISO Interface process waits after a station is marked down before attempting to reestablish communications with the station.
- The amount of time the BIC ISO Interface process waits after a new key or change key message times out before attempting to resend the message.

The number of seconds to use for this timer is taken from the EXTENDED NETWORK field on ICFE screen 3.

If an Extended Network Management Message timer that was set when a station was marked down expires before any messages are received from the station, the BIC ISO Interface process performs the following:

1. Sends an echo test message to the station.
2. Sets a Network Management Message timer for the echo test message.

When an Extended Network Management Message timer that was set after a new key or change key message time out expires, the BIC ISO Interface process determines whether there is a station available to which to send the message. For more information on how the BIC ISO Interface process selects a station for sending a message, see the topic “Selecting Stations for Receiving Messages” earlier in this section. Depending on whether a station is available, the BIC ISO Interface process performs as follows:

- If a station is available, the BIC ISO Interface process sends the new key message or change key message to that station. The BIC ISO Interface process sets a Network Management Message timer to wait for the response to the new key or change key message.

- If no stations are available, the BIC ISO Interface process, sets an Extended Network Management Message timer. When this timer expires, the BIC ISO Interface process attempts to send the message again.

Store-and-Forward Message Timer

A Store-and-Forward Message timer is set each time a store-and-forward message is sent to a co-network. The Store-and-Forward Message timer limits the length of time that the BIC ISO Interface process waits for an acknowledgment after transmitting a store-and-forward message. The number of seconds to use for this timer is taken from the STORE AND FORWARD field on ICFE screen 8 for BASE24-atm and ICFE screen 10 for BASE24-pos.

The BIC ISO Interface process only expects text-level acknowledgments to store-and-forward messages if the ACK FROM SWITCH field on ICFE screen 3 contains the value Y. If text-level acknowledgments are not required from the co-network, the BIC ISO Interface process expects a logical acknowledgment from the XPNET process once the message has been sent successfully to the co-network. In this case, if the co-network rejects the message for any reason, the message is not resent because it will have already been deleted from the SAF.

If this timer expires before an appropriate acknowledgment is received in response, the BIC ISO Interface process performs as follows:

1. Reads the message from the SAF.
2. Adds one to the SAF retries count for that record. The SAF retries count indicates the number of times the SAF record has been sent to the co-network. A separate count is kept for each SAF record.
3. Checks the SAF retries count for the record against the maximum allowed by the MAX SAF RETRY field on ICFE screen 3.
 - a. If the SAF retries count is equal to the maximum allowed, the BIC ISO Interface process performs as follows:
 - 1) Writes the SAF record to its log.
 - 2) Deletes the record from the SAF.
 - b. If the SAF retries count is less than the maximum allowed, the BIC ISO Interface process continues with the next step.
4. Marks the SAF message to be resent.

5. Updates the message type, if necessary.

If the message type is 0120, 0220, 0320, 0420, or 0520, the BIC ISO Interface process reads the corresponding record in the SAF and updates the message type in place to 0121, 0221, 0321, 0421, or 0521 respectively. Message types 0402 and 0412 are not changed.

Wait-for-Traffic Timer

A Wait-for-Traffic timer is set each time a station is marked up and each time a previously set Wait-for-Traffic timer expires. The Wait-for-Traffic timer specifies the length of time that the BIC ISO Interface process waits for traffic on the line before sending an echo test message. The number of seconds to use for this timer is taken from the WAIT FOR TRAFFIC field on ICFE screen 3.

The BIC ISO Interface process sets Wait-for-Traffic timers for its send/receive stations only. Wait-for-Traffic timers are not set for receive-only/no output stations.

When this timer expires, the BIC ISO Interface process determines whether there has been any message traffic to or from the station since the timer was set. Depending on whether there has been traffic on the line, the BIC ISO Interface process performs as follows:

1. If there has been message traffic, the BIC ISO Interface process simply sets another Wait-for-Traffic timer.
2. If there has not been message traffic, the BIC ISO Interface process:
 - a. Sends an echo test message to the station.
 - b. Sets a Network Management Message timer for the echo test message.

Performance Timer

A performance timer is set for each co-network at the start of that co-network's new performance statistics accumulation period. The Performance timer specifies the interval the BIC ISO Interface process uses for gathering performance statistics. The number of minutes to use for this timer is taken from the PERFORMANCE PERIOD field on ICFE screen 3.

For each co-network, the BIC ISO Interface process tracks the number of requests sent to the co-network, the number and percentage of requests that time out, and the average response time. These statistics are tracked on an interval basis and are

available for the most current four intervals (interval zero through interval three with interval zero being the most current). The Performance timer triggers the end of one interval and the beginning of the next.

When a Performance timer expires, the BIC ISO Interface process performs as follows:

1. Moves the performance statistics from interval two to interval three.
2. Moves the performance statistics from interval one to interval two.
3. Moves the performance statistics from interval zero to interval one.
4. Zeros out the performance statistics in interval zero to begin accumulating statistics again.
5. Sets a Performance timer for the co-network.

Note: These statistics are accessed using the PERFORMANCE command entered from a network control facility. For information concerning the entry of the PERFORMANCE command and the display of the statistics, please refer to the description of the PERFORMANCE command in the ***BASE24 Text Command Reference Manual***.

Outbound Request Timer

An Outbound Request timer is set each time a 0100, 0200, or 0300 message is sent to a co-network. The Outbound Request timer limits the amount of time that the BIC ISO Interface process waits for a response from the co-network after transmitting a request message. The number of seconds to use for this timer is taken from the OUTBOUND field on the product-specific ICFE screen.

If this timer expires before a response is received to the message, the BIC ISO Interface process performs as follows:

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the BIC ISO Interface process performs as follows:
 - 1) Retrieves the original message from memory.

The BIC ISO Interface process retains a copy of the internal message for each message it sends to the co-network until it is notified that the message was received.

- 2) Sets the STM to indicate the destination was not available. This identifies the 0200 message as a failure.
- 3) Sets the STM to cause the Authorization process to generate a 0220 message if it subsequently stands in and authorizes the transaction.
- 4) Sends the STM to the process where the message originated.
 - a) If the message is a BASE24-pos message, the BIC ISO Interface process performs as follows:

- 1) Retrieves the 0200 message from memory (all 0100 messages are handled as 0200 messages internally).

The BIC ISO Interface process retains a copy of the PSTM for each 0100 and 0200 message it sends to the co-network until it is notified that the message was received.

- 2) Sets the PSTM to indicate the destination was not available. This identifies the 0200 message as failed.
 - 3) Sets the PSTM to cause the Authorization module to generate a 0200 message if it subsequently stands in and authorizes the transaction.
 - 4) Sends the PSTM to the process where the message originated.
2. Adds one to the count of consecutive timeouts accumulated for the station.
 3. Adds one to the number of request timeouts for the co-network.
- The BIC ISO Interface process tracks this information as part of its co-network performance statistics.
4. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICFE screen 3.
 - a. If the current number of consecutive timeouts is equal to the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ISO Interface process performs as follows:
 - 1) Marks the station down.
 - 2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ISO Interface process marks the co-network down.
 - 3) Sets an Extended Network Management Message timer for the station.

Inbound Request Timer

An Inbound Request timer is set each time a 0100 or 0200 message is sent to a BASE24-atm or BASE24-pos Authorization process. The Inbound Request timer limits the length of time that the BIC ISO Interface process waits for a response from a BASE24 Authorization process after transmitting a request message. The number of seconds to use for this timer is taken from the INBOUND field on the product-specific ICFE screen.

If an Inbound Request timer expires before a response message is received, the BIC ISO Interface process performs as follows:

1. Retrieves the 0200 message from memory.
The BIC ISO Interface process retains a copy of the external message for each 0200 message it sends to the Authorization process until a response is received (the BASE24-pos product handles all 0100 messages as 0200 messages internally).
2. Formats a 0110 or 0210 external message to send to the co-network using the information from the request. The BIC ISO Interface process:
 - a. Sets the Responder Code in the BASE24 External Message Header to a value of 6, indicating that the request is being responded to by the BIC ISO Interface process.
 - b. Sets the Response Code (P-39) field in the message to a value of 05 (denied).
 - c. Sets the Card Issuer FIID and Logical Network (P-61) field in the message to zeros.
3. Sends the external message to the co-network.
4. Adds one to the number of request timeouts for the BIC ISO Interface process.

Completion Acknowledgment Timer

A Completion Acknowledgment timer is set each time a 0210, 0220, 0320, 0402, 0412, 0420, or 0520 message is sent to a co-network. The Completion Acknowledgment timer limits the amount of time that the BIC ISO Interface process waits for an acknowledgment from the co-network after transmitting an advice message. The number of seconds to use for this timer is taken from the COMPLETION ACK field on the product-specific ICFE screen.

The BIC ISO Interface process only expects text-level acknowledgments to completion messages if the ACK FROM SWITCH field on ICFE screen 3 contains the value Y. If text-level acknowledgments are not required from the co-network, the BIC ISO Interface process expects an acknowledgment from the XPNET process instead. In this case, the XPNET process returns an acknowledgment once the message has been sent successfully to the co-network.

If a Completion Acknowledgment timer expires before an appropriate acknowledgment is received, the BIC ISO Interface process performs as follows:

1. Retrieves a copy of the external message from memory.

The BIC ISO Interface process retains a copy of the external message for each message it sends to the co-network until it is notified that the message was received.

2. Changes the message type from 0120, 0220, 0320, 0420, or 0520 to 0121, 0221, 0321, 0421, or 0521 respectively. Message types 0402 and 0412 are not changed.
3. Places the external message in the SAF.
4. Adds one to the count of consecutive timeouts accumulated for the station.
5. Adds one to the number of request timeouts for the co-network.
6. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICFE screen 3.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, the BIC ISO Interface process takes no further action.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ISO Interface process:
 - 1) Marks down the station.
 - 2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ISO Interface process marks the co-network down.
 - 3) Sets an Extended Network Management Message timer for the station.

Key Exchange Timer

Key Exchange timers are used to automatically initiate PIN and MAC key exchanges. If the BASE24 product is configured for dynamic key management, one Key Exchange timer is set for each PIN and MAC key being used by the BIC ISO Interface process. Key Exchange timers are set or reset at the following times:

- When the BIC ISO Interface process is initialized or warmbooted
- When the key for which the timer has been set is exchanged for any reason

The BIC ISO Interface process sets and uses Key Exchange timers for PIN keys when the PIN KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 contains a nonzero value. PIN Key Exchange timers limit the length of time that the BIC ISO Interface process uses a particular PIN key to encrypt and decrypt PINs sent between the BIC ISO Interface process and the co-network. The length of time to use for these timers is taken from the PIN KEY TIMER INTERVAL and PIN KEY TIMER VALUE fields on KEYF screen 3 or KEY6 screen 4.

The BIC ISO Interface process sets and uses Key Exchange timers for MAC Keys when the MAC KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 contains a nonzero value. MAC Key Exchange timers limit the amount of time that the BIC ISO Interface process uses a particular MAC key to encrypt and decrypt MACs sent between the BIC ISO Interface process and the co-network. The amount of time to use for these timers is taken from the MAC KEY TIMER INTERVAL and MAC KEY TIMER VALUE fields on KEYF screen 3 or KEY6 screen 4.

If a Key Exchange timer expires, the BIC ISO Interface process performs as follows:

1. Determines the key that needs to be exchanged.
2. Changes the status of the co-network to indicate that a key exchange is pending.
3. Determines the type of message to send based on the value in the KEY PROCESSING TYPE field on KEYF screen 3 or KEY 6 screen 4 and the key direction, as described below:
 - a. If the KEY PROCESSING TYPE field contains the value N (no key processing), then no further steps are taken.
 - b. If the KEY PROCESSING TYPE field contains the value M (main processor), the message type is a new key message.

- c. If the KEY PROCESSING TYPE field contains the value S (secondary processor), the message type is a change key message.
 - d. If the KEY PROCESSING TYPE field is contains the value C (co-network processor) and the key direction is outbound, the message type is a new key message.
 - e. If the KEY PROCESSING TYPE field is contains the value C (co-network processor) and the key direction is inbound, the message type is a new key message.
4. Determines whether a key exchange is already in progress for the specific key. If a key exchange is already in progress for the specific key, no further steps are taken. If a key exchange is not in progress for the key, processing continues with the next step (step 5).
5. Determines whether there is a station available to which to send the message. Depending on whether a station is available, the BIC ISO Interface process performs as follows:
 - a. If no stations are available, the BIC ISO Interface process sets an Extended Network Management Message timer. When this timer expires, the BIC ISO Interface process attempts to send the message again. No further processing is performed at this time.
 - b. If a station is available, the BIC ISO Interface process performs according to the type of message, as described below:
 - 1) If the message is a new key message, the BIC ISO Interface process sends the message to the selected station. The BIC ISO Interface process sets a Network Management Message timer to wait for the response. The BIC ISO Interface process also sets a flag to indicate that a key exchange is in progress and deletes the Clear Old Key timer for the key being exchanged (if one is still set).
 - 2) If the message is a change key message, the BIC ISO Interface process sends the message to the selected station. The BIC ISO Interface process sets a Network Management Message timer to wait for the response.
6. Adds one to the count of messages outstanding to the station.

Note: Upon receiving a response from the co-network, the BIC ISO Interface process clears the flag that indicated a key exchange was pending and sets a new Key Exchange timer.

Clear Old Key Timer

The clear old key timer limits the amount of time that BIC ISO Interface process attempts to use the old key to verify a PIN or MAC after a key exchange has taken place.

The INBOUND PIN KEY, OUTBOUND PIN KEY, INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEYF screen 2 can each contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key is currently in use. At the time of a successful key exchange, the CURRENT INDEX is updated to point to the new key and the key that was in use becomes the old key.

When processing a transaction, the BIC ISO Interface process uses the new key for PIN processing or message authentication. If the security device cannot verify the PIN or the MAC because of a key error, the transaction is tried again using the old key, if one exists. This timer controls how long the old key exists. When the timer expires, the old key is cleared from the KEYF.

A clear old key timer is set under the following circumstances:

- When the BIC ISO Interface process receives a duplicate new key message from the co-network (carrying a key that was already received in a previous new key message and stored in the KEYF.)
- When the BIC ISO Interface process approves a new key message from the co-network.
- When the BIC ISO Interface process receives an approved new key response message from the co-network (i.e., BASE24 sent the host a new key, and the co-network has responded that the new key has been implemented successfully).

When a clear old key timer expires, the BIC ISO Interchange Interface processBIC ISO Interchange Interface process performs as follows.

Note: KEYF information is read into memory by the BIC ISO Interface process at initialization, and the BIC ISO Interface process performs the following steps on the KEYF information in memory and in the KEYF file.

1. Determines the type of key for which the timer was set.
2. Determines the key to clear using the value in the CURRENT INDEX field.
Since the value in the CURRENT INDEX field points to the newest key, the BIC ISO Interface process selects the key not being pointed to by the CURRENT INDEX field as the one to clear.
3. Sets the selected key value and its check digits in memory to ASCII zeros.
4. Sets the selected key value and its check digits in the KEYF or KEY6 to binary zeros.

Report Startup Timer

The BIC ISO Interface process uses a report startup timer to start the SETL01 and STAT09 reports. The BIC ISO Interface process starts the timer at initialization and at interchange cutover. The time for this timer is taken from the LCONF param SW-RPT-STARTUP.

When the report startup timer expires, the obey file containing the commands to run the report programs is executed.

Settlement Interval Timer

The settlement interval timer specifies how long to wait after the interface has cut over successfully before the reconciliation totals are sent to the co-network. The number of minutes used for this timer is taken from the SETTLEMENT TIMER field on ICFE screen 12.

The SETTLEMENT TIMER field has a default value of 0 when an ICFE record is added and the operator can modify this value to any value between 0 and 999. However, the interface process uses a value of 20 minutes in processing when the SETTLEMENT TIMER field is set to a value of 0.

The BIC ISO Interface process starts this timer when it receives a reply to the cutover message sent to the co-network. The BIC ISO Interface process waits for any transactions in progress for the previous BASE24 settlement day to finish processing. When the timer expires, the BIC ISO Interface process updates the reconciliation totals for the previous BASE24 settlement day for the last time and sends these totals to the co-network.

ITF Update Timer

The BIC ISO Interface process uses an ITF update timer to determine when the totals kept in the Interchange Totals File (ITF) should be updated. The number of minutes used for this timer is taken from the ITF UPDATE INTERVAL field on ICFE screen 12.

The ITF UPDATE INTERVAL field has a default value of 30 minutes when an ICFE record is added and the operator can modify this value to any value between 0 and 999. However, the interface process uses a value of 30 minutes in processing when the ITF UPDATE INTERVAL field is set to a value of 0.

The BIC ISO Interface process also uses a transaction counter to determine when ITF totals should be updated. The ITF UPDATE MAX TRANSACTIONS field on ICFE screen 12 indicates the maximum number of transactions that can be processed between ITF updates.

The BIC ISO Interface process performs the following steps when the ITF update timer expires or when the transaction count reaches the maximum:

1. Updates the transaction totals in the ITF.
2. Resets the ITF update timer.
3. Resets the transaction counter.

Configuring External Messages

The BIC ISO Interface process has a standard set of message structures, or defaults, that it uses for creating and interpreting external messages. These message defaults are hard-coded in the BIC ISO Interface process.

If the default message structures need to be modified, records can be created in the EMF specifying the new message structures. The BIC ISO Interface process allows incoming and outgoing messages to be configured individually by an institution, depending on the information the co-network chooses to send and receive.

Data elements can be defined in the EMF as mandatory, conditional, or not required.

- If a data element is defined as mandatory, it is always sent in outgoing external messages and must be present in incoming external messages.
- If a data element is defined as conditional, it is sent in outgoing messages under certain conditions. If the element contains data and the data is valid, the BIC ISO Interface process sends the element in its outgoing external messages. If the element contains blanks or zeros, it is not sent. Likewise, the data element may or may not be present in incoming messages.
- If a data element is defined as not required, the field in the EMF used to define a data element as mandatory or conditional is left blank. Data elements defined in this manner are not sent in outgoing external messages and are not expected in incoming external messages.

Message Type

The message type identifies the external message type being defined by the record. Any external message type allowed by the BIC ISO Interface process is valid in this field, except for repeat messages. A repeat message contains the same information as the original message. The value entered in this field can only be used with certain values entered in the PROD # field and IN-OUT-IND field. A table showing valid relationships for external messages to and from a co-network is provided as follows:

PROD #	Product	MSG TYP	IN-OUT-IND
00	Base	0800, 0810	I, O, B
		9000	O
01	ATM	0200, 0210, 0220, 0230, 0420, 0430	I, O, B
02	POS	0100, 0110, 0120, 0130, 0200, 0210, 0220, 0230, 0402, 0412, 0420, 0430	
		0510, 0530	I
		0500, 0520	O

Note: Message types in the 9000 range are used to denote rejected messages and message authentication errors. For example, a message type of 0200 would be changed to 9200 if it were rejected and a message type of 0420 would be changed to 9420 if it were rejected. Setting up a record in the EMF for a message type of 9000 is not to control the message structure, but rather to allow for assigning a single IMS/CICS transaction code equivalent to all rejected messages returned to the co-network. For further information on assigning an IMS/CICS transaction code equivalent to a 9000 message, refer to the description of EMF screen 3 in the *BASE24 Base Files Maintenance Manual*.

Changing the Defaults

When the default values for a particular message type are not appropriate, a separate EMF record can be created for each message type. To modify the default values for any message type, the following steps must be performed. These steps assume that you are logged on to the BASE24 CRT access system.

Warning: Special care must be used when adding or updating an EMF record. If IMS or CICS transaction code equivalents are being used, access to EMF screen 3 is required in order to perform file maintenance operations (in addition to any other EMF screen access that may be defined). EMF screen 3 is used to assign IMS or CICS transaction code equivalents to BASE24 transaction codes and can consist of up to five pages. The last page containing transaction codes must be displayed when the function key is pressed to add or update the record or transaction codes are lost. For example, if the function key is pressed while screen 1 or screen 2 is displayed, all transaction codes on screen 3 are not included in the new or updated record. If the function key is pressed while one of the screen pages is displayed, but it is not the last page containing transaction codes, the transaction codes on the remaining screen 3 pages are not included in the new or updated record.

1. Display EMF screen 1 by entering EMF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu press the **F1** key. From a file screen, press the **F16** key.
2. Read the EMF record being changed by completing the field information shown below and pressing the **F2** key. If no record is found, the BIC ISO Interface process uses EMF default values. The default values can be displayed by pressing the **F7** key.
 - Set the INTERF TYP to BIC
 - Set the PROCESS NAM field to the symbolic name of the BIC ISO Interface process
 - Set the PROD #, MSG TYP, and IN-OUT-IND fields using any combination of values from each row of the following table:

PROD #	Product	MSG TYP	IN-OUT-IND
00	Base	0800, 0810	I, O, B
		9000	O
01	ATM	0200, 0210, 0220, 0230, 0420, 0430	I, O, B

PROD #	Product	MSG TYP	IN-OUT-IND
02	POS	0100, 0110, 0120, 0130, 0200, 0210, 0220, 0230, 0402, 0412, 0420, 0430	
		0510, 0530	I
		0500, 0520	O

3. Review the values for the message type being displayed. If the value for a specific data element needs to be changed, enter an M (mandatory), a C (conditional), or a blank (not required) in the field directly to the right of the appropriate data element number.
4. Display EMF screen 2 by pressing the **F9** key. EMF screen 2 allows you to specify individual data elements to be used during message authentication.
5. Review the values for the message type being displayed. If the value for a specific data element needs to be changed, enter a Y (include in authentication) or a blank (do not include in authentication) in the field directly to the right of the appropriate data element number.
6. Display EMF screen 3 by pressing the **F9** key. EMF screen 3 allows you to assign CICS or IMS equivalents to BASE24 transaction codes.
7. Determine whether any BASE24 transaction codes require IMS or CICS transaction code equivalents, and perform as follows:
 - a. If no IMS or CICS transaction codes are required, or if all of the IMS or CICS transaction codes are displayed on the current page of screen 3, press the **F3** key to add the record. Continue with step 8.
 - b. If additional IMS or CICS transaction codes are required, add the IMS or CICS equivalents by making the appropriate entries in the TRANS, IMS TRAN, and L fields.

If you need more than 30 transaction codes, go to the next screen 3 page by pressing the **F8** key and add the necessary IMS or CICS equivalents. You must go to a new screen 3 page each time you add 30 transaction codes.

When you have finished adding the necessary IMS or CICS transaction code equivalents, add the record to the EMF by pressing the **F3** key.

Note: The last screen 3 page containing entries must be displayed when you add or update the record.

8. Return to EMF screen 1 by pressing the **F9** key. Repeat steps 2 through 7 for any remaining message types to be added to the EMF.
9. Reinitialize the BIC ISO Interface process in order to read the new values.

Configuring Tokens to be Sent in the External Message

A *token* is a collection of related data required to perform a specific function. As transactions are processed and additional information is required, BASE24 processes add tokens to the internal message. The Token File (TKN) is used to configure the following token processing:

- The tokens that get logged to the transaction log files. The transaction log files affected are the BASE24-atm Transaction Log File (TLF), POS Transaction Log File (PTLF), and the Interchange Log Files (ILFs).
- The tokens that get extracted from the transaction log files by the Super Extract process. Extracting allows the token information to be processed by the co-network.
- The tokens that get sent to the co-network with each transaction in the ISO-based external message.

For more information on configuring the tokens that get logged, extracted, and sent in the external message, please refer to the ***BASE24 Tokens Manual***.

Configuring Key Management

The BIC ISO Interface process reads the Key File (KEYF) for the information it needs to handle PIN encryption, PIN translation, message authentication, and dynamic key management. The KEYF contains one record for each BIC ISO Interface process. The ***BASE24 Transaction Security Manual*** describes in detail the configuration of PIN encryption, PIN translation, message authentication, and dynamic key management. The key management options available for the BIC ISO Interface process are described briefly in the paragraphs that follow.

Co-Network Encryption Type

The institution can specify the type of PIN encryption used for messages going to and coming from the co-network. There are three options:

- Clear PIN processing. All PINs handled by the co-network are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by the co-network are encrypted under the secure keys of a security module. The security modules that can be used include Atalla Security Modules and the Thales e-Security (Racal) Host Security Modules.
- Encrypted PINs—software DES PIN processing. All PINs handled by the co-network are encrypted in software, using the DES algorithm. No security module is in use.

This option is controlled using the ENCRYPT TYPE field on KEYF or KEY6 screen 1.

BASE24 Encryption Type

The BASE24 system and the co-network are not required to handle PIN encryption in the same manner. Each has its own PIN encryption options that are controlled separately. The BASE24 system can handle PIN encryption in two different ways:

- Clear PIN processing. All PINs handled by the BASE24 system are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by the BASE24 system are encrypted under the secure keys of a security module.

This option is controlled using the BASE24 ENCRYPT TYPE field on KEYF or KEY6 screen 1.

PIN Block Type

The institution can specify the type of PIN block used during encryption. PINs are not simply fed into the encryption routine as is. Rather, they are presented to the encryption routines in what is called a *PIN block*. A PIN block is a 16-digit series of hexadecimal characters, of which the PIN is a part.

The BIC ISO Interface process supports two types of PIN blocks—ANSI PIN blocks and PIN/PAD PIN blocks. The interchange can specify one of these two PIN block types or specify clear PIN, indicating that PINs are not encrypted.

This option is controlled using the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.

PAN Characters Used in PIN Block

If the ANSI PIN block is selected, the institution can specify which PAN characters to use in formatting the PIN block. There are three options:

- Use the 12 right-most digits of the PAN, excluding the check digit
- Use the 12 right-most digits of the PAN, including the check digit
- Use the 12 left-most digits of the PAN

This option is controlled using the ANSI PAN FORMAT field on KEYF or KEY6 screen 1.

PAD Character Used in PIN Block

If the PIN/PAD PIN block is selected, the institution must specify the PAD character to use in formatting the PIN block. The PAD character is a single hexadecimal character used to fill out the remaining positions of the PIN block.

The PAD character is specified using the PIN PAD CHARACTER field on KEYF or KEY6 screen 1.

Number of Keys

The number of keys indicates whether the inbound and outbound keys supported by the BIC ISO Interface process are the same or different. If the keys are the same, only one key is generated and used for both inbound and outbound keys. If the keys are different, a separate key is generated for inbound and outbound keys. This field is used to determine the number of keys being used for PINs and message authentication.

The number of different PIN or message authentication keys supported by the BIC ISO Interface process is specified using the NUMBER OF KEYS field on KEYF or KEY6 screen 1.

Key Length

The key length indicates the type of key management supported by the BIC ISO Interface process. This field identifies whether the BIC ISO Interface process supports single-length Key Encryption Keys (KEKs) or double-length KEKs. Single-length KEKs consist of one 64-bit key. Double-length KEKs consist of a key pair consisting of 64 bits each. For more information on single-length and double-length KEKs, refer to the *BASE24 Transaction Security Manual*.

The type of key management supported by the BIC ISO Interface process is specified using the KEY LENGTH field on KEYF or KEY6 screen 1.

PIN Key Variant

The PIN key variant indicates whether Key Exchange Keys (KEKs) or PIN Encryption Keys (PEKs) need to be generated for PIN key translations.

The PIN key variant option supported by the BIC ISO Interface process is specified using the PIN KEY VARIANT field on KEYF screen 3 or KEY6 screen 4.

Key Processing Type

The key processing type describes whether the BIC ISO Interface process is performing main, secondary, co-network, or no key processing. The following list describes each key processing type.

- Main — Indicates that this process controls all key generation.
- Co-network — Indicates that this process shares the control of key management processing with its co-network. Each co-network is only responsible for exchanging its inbound key.
- Secondary — Indicates that this process is not responsible for any key generation.
- None — No key processing is supported.

The key processing type supported by the BIC ISO Interface process is specified using the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4.

PIN Management Keys

There are four types of keys identified by the BASE24 system for use in managing PINs between the BIC ISO Interface process and an interchange: inbound encryption keys, outbound encryption keys, intermediate keys, and exchange keys. The institution must specify the values for these keys.

These options are controlled using KEYF or KEY6 screen 2.

Configuring Message Authentication

Message authentication options are configured using parameters set up in two files: the External Message File (EMF) and the Key File (KEYF) or Key 6 File (KEY6). The paragraphs that follow discuss the configuration options for using message authentication with a BIC ISO Interface process.

MAC Encryption Type

The message authentication encryption type indicates whether authentication is supported, and if so where the encryption is done. BASE24 and the co-network are required to handle the encryption of the message authentication code (MAC) in the same manner. Currently, BASE24 only supports message authentication using a security device.

This option is controlled using the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1.

MAC Data Type

The message authentication data type indicates the character set of the data being authenticated using MACs. The character set can be either ASCII or EBCDIC. This is the format of the data prior to performing the MAC calculation. The BASE24 system and the co-network are required to have the same message authentication data type format.

The message authentication data character format is specified using the MAC DATA TYPE field on KEYF or KEY6 screen 1.

Message Elements Encrypted

The BIC ISO Interface process has a standard set of defaults that it uses for determining which of the 128 data elements are to be included when authenticating each external message.

These defaults can be overridden using settings in the EMF . These settings allow the institution to include or exclude data elements when authenticating the message. The EMF allows a co-network to alter the combinations of data elements that are included when authenticating messages.

The BASE24 system also offers the options of authenticating all data elements in an individual message, or of authenticating all data elements in all messages. Full message authentication is specified in the EMF for individual messages. If an EMF record for the specific message type does not exist, the BIC ISO Interface process checks the KEYF or KEY6 to determine whether all of its messages should use full message authentication. If the FULL MESSAGE MAC field on KEYF or KEY6 screen 1 contains the value Y, the entire message is included in creating the MAC. If the FULL MESSAGE MAC field on KEYF or KEY6 screen 1 contains the value N, the BIC ISO Interface process uses its defaults to determine which elements should be included in creating and interpreting the MAC.

In addition to using the specified data elements in the MAC generation, the BIC ISO Interface process uses the ISO literal, BASE24 header, message type identifier, and primary bit map when it generates a MAC. These message components are always used in MAC generation, regardless of the EMF, KEYF or KEY6 settings.

MAC Key Variant

The MAC key variant indicates whether key exchange keys (KEKs) or MAC key exchange keys (KEKs) need to be generated for message authentication key translations.

The message authentication key variant option supported by the BIC ISO Interface process is specified using the MAC KEY VARIANT field on KEYF screen 3 or KEY6 screen 4.

Configuring Processing Options

The following options are used to determine how the BIC ISO Interface process is configured for cutover and reconciliation with the co-network, transactions allowed, stand-in authorization, definition of product indicators and stations, automatic logons and displaying customer balance information.

Cutover and Reconciliation Handling

The following options control how the BIC ISO Interface process handles cutover and reconciliation with the co-network.

Reconciliation Totals

The BIC ISO Interface process can be set up to specify whether it should maintain reconciliation totals to be used during cutover processing. This option is controlled using the PROCESSING MODE field on ICFE screen 3.

Cutover Relationship

The cutover relationship between BASE24 and the co-network can be defined as an equal partner or a main and secondary relationship. This option is controlled using the CUTOVER STATUS field on ICFE screen 12.

Updating the ITF

The BIC ISO Interface process can be set up to specify how often the Interchange Totals File (ITF) should be updated during the processing day. This option is controlled using the ITF UPDATE INTERVAL and ITF UPDATE MAX TRANSACTIONS fields on ICFE screen 12.

Sending a Reconciliation Message

The BIC ISO Interface process can be set up to specify how long it should wait after a successful cutover before it sends a reconciliation message to the co-network. This option is controlled using the SETTLEMENT TIMER field on ICFE screen 12.

Transactions Allowed

The BIC ISO Interface process ensures that inbound transactions from the co-network are allowed using the Acquirer Processing Code File (APCF) prior to forwarding the transaction to the authorization process. Similarly, the interface ensures that outbound transactions are allowed to be sent to the co-network using the Issuer Processing Code File (IPCF) prior to forwarding the transaction to the co-network.

The BIC ISO Interface process can be set up to specify the transactions that can be sent to or from the co-network for authorization. Transactions allowed are configured using the transaction profiles in the Enhanced Interchange Configuration File (ICFE). These profiles correspond to records in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

Outbound Transactions

Institutions define the transactions that can be sent to a particular co-network by specifying an issuer transaction profile value in the ISSUER TXN PROFILE field on screen 8 (ATM) and screen 10 (POS) of the Enhanced Interchange Configuration File (ICFE). The value of the issuer transaction profile is used to read the Issuer Processing Code File (IPCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an issuer transaction profile is not specified, the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the IPCF extended memory table.

Issuer Transaction Profile Processing. Once the issuer transaction profile value used to read the IPCF extended memory table has been determined, the BIC ISO Interface process attempts to read the IPCF extended memory table using the issuer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The BIC ISO Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the IPCF extended memory table.

Using the issuer transaction profile, message category, and ISO processing code, the BIC ISO Interface process steps through the records in the IPCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Issuer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the ON-US TRANSACTION ALLOWED field on IPCF screen 2 is set to a nonzero value, the transaction is allowed. If the transaction is allowed and the COMPLETION REQUIRED TO HOST field on IPCF screen 2 is set to a value of Y, the Router/Authorization module will generate a completion message indicating whether the transaction is approved or declined and send it to the host after the transaction has been authorized.

Inbound Transactions

The ACQUIRER TXN PROFILE field on ICFE screens 8 and 10 defines the transactions allowed from the co-network. The value of the acquirer transaction profile is used to read the Acquirer Processing Code File (APCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an acquirer transaction profile is not specified, the BIC ISO Interface process uses a value of all asterisks (wildcard characters) to read the APCF extended memory table. If BASE24 receives a transaction from an interchange that has not been defined as allowed, the BIC ISO Interface process denies the transaction.

Acquirer Transaction Profile Processing. Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Interchange Interface process attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The BIC ISO

Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the BIC ISO Interface process steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Acquirer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed.

The APCF also allows you to specify an authorization destination for a transaction other than the BASE24-atm Authorization process or BASE24-pos Device Handler/Router/Authorization process specified in the AUTH PROCESS field on ICFE screen 8 or 10, respectively. This mechanism allows you to route transactions received from an interchange to an application process. When a transaction is routed in this way, you can also specify whether the transaction response returned from the authorization destination is logged to the transaction log file.

Stand-in Authorization

The BIC ISO Interface process can be set up to specify whether the issuer or acquirer can stand in to authorize transactions if the link to the authorizing entity is down. These options are controlled using the ACQ STAND-IN AUTH and ISS STAND-IN AUTH fields on ICFE screen 12.

Customer Balance Display

The BIC ISO Interface process can be set up to specify whether customer balance information for incoming responses from the co-network should be displayed, printed, or both displayed and printed. This option is controlled using the CUSTOMER BALANCE DISPLAY field on ICFE screen 1. It is only used when a transaction is authorized by the co-network.

Automatic Logon

The BIC ISO Interface process can be set up to automatically log on to the co-network at startup. If this option is selected, no operator intervention is required to log on to the co-network. It is done automatically as part of the initialization of the BIC ISO Interface process. This option is controlled using the AUTO SIGNON START field on ICFE screen 3.

Product Indicators

The BIC ISO Interface process can be set up to specify which products it can support. The BIC ISO Interface process currently supports the BASE24-atm and BASE24-pos products. This option is controlled using the PRODUCT INDICATOR fields on ICFE screen 12.

Defining Stations

The BIC ISO Interface process can be connected to a maximum of 20 stations. These stations are defined using the STATION and TYPE fields on ICFE screen 13. The stations defined in the ICFE must also have a corresponding attribute in the XPNET system.

Configuring Institution IDs

Under certain conditions, the BIC ISO Interchange Interface process can override the values moved between the following data elements in ISO-based external messages and the corresponding fields in the Standard Internal Message (STM) or POS Standard Internal Message (PSTM), as shown in the following table.

Identifier	External Message	STM	PSTM
Acquiring	Acquiring Institution ID Code (P-32)	RQST.ACQ-INST-ID-NUM	ACQ-INST-ID-NUM
Forwarding	Forwarding Institution ID Code (P-33)	FRWD-INST-ID-NUM	FRWD-INST-ID-NUM
Receiving	Receiving Institution ID Code (S-100)	RCV-INST-ID-NUM	RCV-INST-ID-NUM

The SWITCH ID field on ICFE screen 1 can contain an override identifier for the institution that owns the BASE24 system. The DEFAULT ACQUIRER ID NUM field on ICFE screen 1 can contain an override identifier for the co-network institution.

One or more override institution identifiers can be used, and the OVRRD-INST-ID-FLDS param in the LCONF uses the values 0 through 7 to specify the desired combination of identifiers to be used.

- When a transaction containing institution identifiers is acquired by the institution that owns the BASE24 system, three values in the external message can be overridden with values from ICFE screen 1 before the message is sent to the co-network. The value in the SWITCH ID field can be used to override the acquiring ID, forwarding ID, or both, and the value in the DEFAULT ACQUIRER ID NUM field can be used to override the receiving ID.
- When a transaction is acquired by the co-network institution, the external message contains institution identifiers. One value in the external message can be overridden with a value from ICFE screen 1 before the message is sent to the co-network. The value in the SWITCH ID field can be used to override the receiving ID.
- When a transaction is acquired by the institution that owns the BASE24 system, one value from ICFE screen 1 can be moved to the STM or PSTM if the STM or PSTM field is empty and the corresponding field in the external

message received from the co-network is also empty. The value in the DEFAULT ACQUIRER ID NUM field can be used to override the receiving ID.

- When a transaction is acquired by the co-network institution, one value from ICFE screen 1 can be moved to the STM or PSTM if the corresponding fields in the external message received from the co-network are empty. The value in the DEFAULT ACQUIRER ID NUM field can be used to override the acquiring ID.

ICFE Screens

This section contains a listing of ICFE fields used by the BIC ISO Interface process. The ICFE is used by the institution to define institution and terminal interchange sharing holidays, transaction handling in offline and online situations, and settlement handling.

The following tables indicate the ICFE fields used by the BIC ISO Interface process and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields. The BIC ISO Interface process does not use all the fields on all the ICFE screens. These fields are noted with “not used” in the Values column.

For complete documentation on ICFE screens 1 through 11 and their corresponding field descriptions, refer to the ***BASE24 Base Files Maintenance Manual***. ICFE screens 12 and 13 and descriptions of their fields are provided later in this section.

Note: The information contained in the following tables does **not** indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

ICFE Screen 1

ICFE screen 1 contains fields establishing interchange identifiers, station names, default values for certain fields, and report title names to be printed on the SETL01-BIS and STAT09-BIC reports. The fields on ICFE screen 1 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 1	
Field	Value
INTERCHANGE FIID	Interchange FIID.
PROCESS	Process name as defined in the XPNET system.
SWITCH TYPE	“BICI”
INTERCHANGE LOGICAL NET	Logical network identifier.

ICFE Screen 1	
Field	Value
REPORTING NAME	BIC ISO Interface name as it appears on reports.
INSTITUTION ID	The institution identifier uniquely distinguishing this interchange.
SWITCH ID	Identifier for the institution that owns the BASE24 system.
STATION 1	Not used.
STATION 2	Not used.
SIC CODE	Not used.
CURRENCY CODE	Currency code. If implementing currency conversion, this value is the settlement currency.
DEFAULT TERM NUM	Not used.
DEFAULT ACQUIRER ID NUM	Identifier for the co-network institution.
CUSTOMER BALANCE DISPLAY	0, 1, 2, or 3.

ICFE Screen 2

ICFE screen 2 contains interchange settlement information and pertinent holiday dates. The fields on ICFE screen 2 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 2	
Field	Value
SETTLEMENT HOUR	00–23
SETTLEMENT MINUTE	00–59

ICFE Screen 2	
Field	Value
SETTLEMENT DAYS	0 for five-day settlement, 1 for seven-day settlement
REPORT PRIORITY	0–255
REPORT CPU	CPU number
SWITCH POSTING DATE	YYMMDD
HOLIDAY DATES	YYMMDD (used with five-day settlement only)

ICFE Screen 3

ICFE screen 3 is split up into two sections. The first section defines transaction message timer limits used by the BIC ISO Interface process. The second section defines the processing options used by the BIC ISO Interface process. The fields on ICFE screen 3 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 3	
Field	Value
NETWORK MANAGEMENT	0–9999 (seconds)
ACQUIRER	Y or N
EXTENDED NETWORK	0–9999 (seconds)
ISSUER	Y or N
WAIT FOR TRAFFIC	0–9999 (seconds)
PROCESSING MODE	0 if no totals are to be kept, 1–70 if totals are to be kept
PERFORMANCE PERIOD	0–9999 (minutes in performance monitoring period)
AUTO SIGNON START	Y or N

ICFE Screen 3	
Field	Value
MAXIMUM TIMEOUTS	0–9999 (timeouts allowed)
MAX SAF RETRY	0–9999 (SAF retries allowed)
OUTBOUND	0–9999 (outbound transactions allowed)
ACK TO SWITCH	Y or N
INBOUND	0–9999 (inbound transactions allowed)
ACK FROM SWITCH	Y or N
NETWORK MANAGEMENT MESSAGE ENABLED	Y or N. This field should be set to Y if implementing currency conversion.
TYPE OF INTERCHANGE REPORTS	Not used
ILF EXTRACT NUMBER	3–9 (ILFs to be extracted)

ICFE Screen 8

ICFE screen 8 contains BASE24-atm processing information for interchanges, including acquirer and issuer transaction profiles for the transactions allowed. The fields on ICFE screen 8 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 8	
Field	Value
AUTH PROCESS	Authorization process name
DEFAULT ROUTING NUMBER	Terminal routing group number
ACQUIRER TXN PROFILE	Acquirer transaction profile for BASE24-atm processing

ICFE Screen 8	
Field	Value
ISSUER TXN PROFILE	Issuer transaction profile for BASE24-atm processing
DEFAULT MERCHANT TYPE	Default merchant type
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	Not used
COMPLETION ACK	Not used
SHARING GROUPS	Sharing group codes

ICFE Screen 10

ICFE screen 10 defines BASE24-pos processing information for interchanges, including acquirer and issuer transaction profiles for the transactions allowed. The fields on ICFE screen 10 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 10	
Field	Value
AUTH PROCESS	Authorization process name
REFERRAL PHONE NUMBER	Telephone number
RETAILER ID DEFAULT	Not used
TIMEOUT ACTION	0, 1, or 2
SETTLE ENTITY	0 or 1
ACQUIRER TXN PROFILE	Acquirer transaction profile for BASE24-pos processing

ICFE Screen 10	
Field	Value
ISSUER TXN PROFILE	Issuer transaction profile for BASE24-pos processing
ADJUSTMENT FLAG	Y or N
CHARGEBACK FLAG	Y or N
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	Not used
COMPLETION ACK	Not used

ICFE Screen 11

ICFE screen 11 defines POS preauthorization default time and amount, approval code length, and services allowed to the co-network. The fields on ICFE screen 11 used by the BIC ISO Interface process and their expected values are indicated in the following table.

ICFE Screen 11	
Field	Value
DEFAULT PRE-AUTH AMOUNT	Not used
APPROVAL CODE LENGTH	2–6
PRE-AUTH HOLD INCREMENT	0, 1, or 2
PRE-AUTH HOLD TIME	00–99
ALLOWED SERVICES	Service code

ICFE Screen 12

ICFE screen 12 is an interchange-specific screen used exclusively by the BIC ISO Interface process. It defines options used at logon to determine processing requirements for the BIC ISO Interface process. All fields on ICFE screen 12 can be used by the BIC ISO Interface process. ICFE screen 12 is shown below followed by descriptions of its fields.

```

BASE24-ATM   ICFE FILE MAINTENANCE   LLLL      YY/MM/DD  HH:MM  12 OF 13
INTERCHANGE FIID:                        PROCESS:

                                     BASE24 INTERFACE OPTIONS

ACQ STAND-IN AUTH: N                     ISS STAND-IN AUTH: N
CUTOVER STATUS: 0                         VERSION: 00
FIXED LENGTH FORMAT: 0

ITF UPDATE INTERVAL: 030 (MINUTES)        ITF UPDATE MAX TRANSACTIONS: 0200
SETTLEMENT TIMER: 000 (MINUTES)           PROTOCOL TYPE: 00
SELL RATE: 00000000                      BUY RATE: 00000000

PRODUCT INDICATOR (Y/N): N (ATM)   N (POS)

                                     DATA PREFIX CHARACTERS
1.      2.      3.      4.      5.      6.      7.      8.      9.
VALID ENTRIES FOR DATA PREFIX CHARACTERS ARE (0 THRU 9) AND (A THRU F)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                                     F12-HELP

```

BASE24 INTERFACE OPTIONS

The following fields contain information relevant to the functionality for the BIC ISO Interface process.

ACQ STAND-IN AUTH — A flag indicating whether BASE24, when acting as the acquirer, is allowed to stand in on behalf of the co-network to authorize transactions. This field is verified at logon to make sure the co-network is set up correctly. Valid values are as follows:

Y = Stand-in is allowed
N = Stand-in is not allowed

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: N
Data Name: ICFEBICI.ACQ-STAND-IN

ISS STAND-IN AUTH — A flag indicating whether the co-network, when acting as the acquirer, is allowed to stand in for BASE24 to authorize transactions when BASE24 is the issuer. This field is verified at logon to make sure the co-network is set up correctly. Valid values are as follows:

Y = Stand-in is allowed
N = Stand-in is not allowed

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: N
Data Name: ICFEBICI.ISS-STAND-IN

CUTOVER STATUS — A flag specifying BASE24's relationship to the co-network for cutover and reconciliation processing. Valid values are as follows:

0 = Equal
1 = Main
2 = Secondary

The value in this field must correspond with the value defined for the co-network when the relationship is main/secondary or equal/equal. If it is main/secondary, the main network controls when cutover should occur and is responsible for sending totals for reconciliation. If it is equal/equal, each partner is responsible for its own cutover and each must send totals at cutover.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: 0
Data Name: ICFEBICI.CTVR-STAT

VERSION — The current release of the external message format to be used. This field is verified at logon to make sure the co-network is set up correctly.

In a release 4.0 BASE24 network, the value of this field must be set to 00. In a release 5.x BASE24 network, the value must be set to 00 or 01. The value cannot be set to 60. In a release 6.0 BASE24 network, the value can be set to 00, 01 or 60. The BASE24 co-networks must use the value of the message format of the oldest release of BASE24 used by the two co-networks.

Valid values are as follows:

00 = Release 4.0 format
 01 = Release 5.x format
 60 = Release 6.0 format

Field Length: 2 numeric characters
 Required Field: Yes
 Default Value: 00
 Data Name: ICFEBICI.VERSION-NUM

FIXED LENGTH FORMAT — A flag indicating whether certain variable-length data elements in the BIC ISO external message should be sent as fixed-length fields. Valid values are as follows:

0 = Send variable-length fields
 1 = Send fixed-length fields

Field Length: 1 numeric character
 Required Field: Yes
 Default Value: 0
 Data Name: ICFEBICI.FIXED-LGTH-FRMT

ITF UPDATE INTERVAL — The length of time in minutes, the BIC ISO Interface process waits between updates to the Interchange Totals File (ITF). The ITF is always updated at the time interval specified in this field. The ITF may also be updated based on the value specified in the ITF UPDATE MAX TRANSACTIONS field.

This field has a default value of 30 minutes when an ICFE record is added, and the operator can modify this value to any value between 0 and 999. However, the interface process uses a value of 30 minutes in processing when the ITF UPDATE INTERVAL field is set to a value of 0.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 30
Data Name: ICFEBICI.ITF-UPDT-INTERVAL

ITF UPDATE MAX TRANSACTIONS — The maximum number of transactions that may occur before the ITF is updated on disk throughout the processing day. When the number of transactions that have occurred is equal to the value in this field, the ITF is updated, regardless of the setting in the ITF UPDATE INTERVAL field. Valid values are 0–9999.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0200
Data Name: ICFEBICI.ITF-UPDT-MAX-TRANS

SETTLEMENT TIMER — The number of minutes after the interface has cut over successfully that the reconciliation message is created.

This field has a default value of 0 when an ICFE record is added, and the operator can modify this value to any value between 0 and 999. However, the interface process uses a value of 20 minutes in processing when this field is set to a value of 0.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000
Data Name: ICFEBICI.SETL

PROTOCOL TYPE — A code indicating the type of protocol message header processing to be performed. Valid values are as follows:

00 = None
01 = SNA/CICS Protocol
02 = Bisync Multipoint Protocol
03 = X.25 Protocol
04–99 = Reserved for future use

If this field is set to 00, no special protocol processing is required. Protocol processing is performed by the XPNET process.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: ICFEBICI.PROTO-TYP

SELL RATE — The rate used to calculate the settlement amount in the shared currency when the BASE24 network is acting as the issuer. This rate is used when the shared currency is different from the local currency of the issuer.

The first digit of this field is a scale factor indicating where the decimal should be placed from the end of the field. The remaining seven characters represent the conversion factors used to multiply the amounts to convert to settlement or local amounts.

Field Length: 8 numeric characters
Required Field: Required only when supporting currency conversion
Default Value: 00000000
Data Name: ICFEBICI.SELL-RATE

BUY RATE — The rate used to calculate the settlement amount in the shared currency when the BASE24 network is acting as the acquirer. This rate is used when the shared currency is different from the local currency of the acquirer.

The first digit of this field is a scale factor indicating where the decimal should be placed from the end of the field. The remaining seven characters represent the conversion factors used to multiply the amounts to convert to settlement or local amounts.

Field Length: 8 numeric characters
Required Field: Required only when supporting currency conversion
Default Value: 00000000
Data Name: ICFEBICI.BUY-RATE

PRODUCT INDICATOR

The following fields indicate the products this interface supports.

ATM — A flag indicating whether BASE24-atm transactions are processed. Valid values are as follows:

Y = BASE24-atm transactions are processed

N = BASE24-atm transactions are not processed

Field Length: 1 alphanumeric character

Required Field: Yes

Default Value: N

Data Name: ICFEBICI.PROD-IND

POS — A flag indicating whether BASE24-pos transactions are processed. Valid values are as follows:

Y = BASE24-pos transactions are processed

N = BASE24-pos transactions are not processed

Field Length: 1 alphanumeric character

Required Field: Yes

Default Value: N

Data Name: ICFEBICI.PROD-IND

DATA PREFIX CHARACTERS — Alphanumeric characters that the BIC ISO Interface process must include at the start of any message being sent to the co-network. Up to 9 bytes are allowed and they are stored in hex character display format. These are the ASCII (not EBCDIC) codes. Valid characters are 0–9 and A–F.

Field Length: 2 alphanumeric characters

Occurs: 9 times

Required Field: No

Default Value: No default value

Data Name: ICFEBICI.DATA-PREFIX-CHARS

ICFE Screen 13

ICFE screen 13 is an interchange-specific screen used exclusively by the BIC ISO Interface process. It describes the stations associated with the BIC ISO Interface process. All fields on ICFE screen 13 can be used by the BIC ISO Interface process. ICFE screen 13 is shown below, followed by a description of its fields.

```

BASE24-ATM  ICFE FILE MAINTENANCE  LLLL      YY/MM/DD  HH:MM  13 OF 13
INTERCHANGE FIID:                      PROCESS:

                                     BASE24 INTERFACE STATIONS

      STATION      TYPE  DESCRIPTION      STATION      TYPE  DESCRIPTION

*****
NEW PAGE:          FILE DESTINATION:  BASE24 *****
                                NEW LOGICAL NETWORK ID:
                                F12-HELP

```

BASE24 INTERFACE STATIONS

The following fields contain the names of up to 20 stations and the type of each station with which the BIC ISO Interface process is associated. The stations defined in the ICFE must also be defined in the XPNET system.

STATION — The symbolic name of the stations to be used for transaction requests and responses to and from the co-network. At least one station must be defined.

Field Length:	16 alphanumeric characters
Occurs:	Up to 20 times
Required Field:	No
Default Value:	No default value
Data Name:	ICFEBICI.STA.NAM

TYPE — A flag indicating the type of station the interface can use. The type of station must be either send/receive or receive only. Valid values are as follows:

0 = Send/receive
1 = Receive only

Field Length: 1 numeric character
Occurs: Up to 20 times
Required Field: Required only when the station name is present
Default Value: No default value
Data Name: ICFEBICI.STA.TYP

DESCRIPTION — The description of the value in the type field.

Field Length: System protected
Occurs: Up to 20 times
Data Name ICFEBICI.STA.TYP

KEYF and KEY6 Screens

This section contains a listing of KEYF fields used by the BIC ISO Interface process. The KEY6 contains the same fields as the KEYF, except that PIN working key fields are on KEY6 screen 2, MAC working key fields are on KEY6 screen 3, and the dynamic key management fields are on KEY6 screen 4. In addition, the key fields on KEY6 screens allow for double-length keys (i.e., 32 hexadecimal characters instead of 16 hexadecimal characters).

The following tables indicate the KEYF fields used by the BIC ISO Interface process and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields. The BIC ISO Interface process does not use all the fields on all the KEYF screens. These fields are noted with *Not used* in the Values column.

For documentation on the KEYF and KEY6 screens and their corresponding field descriptions, refer to the ***BASE24 Base Files Maintenance Manual***. For information on BASE24 security, refer to the ***BASE24 Transaction Security Manual***.

Note: The information contained in the following tables does **not** indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

KEYF Screen 1

KEYF screen 1 defines the number of PIN and MAC keys currently being used by the interface, the length of the key exchange keys (KEKs) used by the interface, and the PIN and MAC exchange keys used by the interface. In addition, screen 1 contains security information used to set up the PIN and MAC parameters associated with the BIC ISO Interface process. The fields on KEYF screen 1 used by the BIC ISO Interface process and their expected values are indicated in the following table.

KEYF Screen 1	
Field	Value
DPC/FIID	Host DPC number or financial institution identifier
INTERFACE PROCESS	Interface process name

KEYF Screen 1	
Field	Value
ENCRYPT TYPE	0 for clear PINs, 1 for security module PIN management, 2 for software DES PIN management. This field must contain the value of 1 if using dynamic PIN key management.
BASE24 ENCRYPT TYPE	0 for clear PINs, 1 for security module PIN management
MAC ENCRYPT TYPE	0 for no MAC processing, 1 for security module MAC processing, 2 for software MAC processing. This field must contain a 1 if using dynamic MAC key management.
ANSI PAN FORMAT	0 for the last 12 digits of the PAN, excluding the check digit; 1 for the last 12 digits of the PAN, including the check digit; or 2 for the first 12 digits of the PAN.
PIN BLOCK FORMAT	0 for clear PINS, 1 for ANSI (PIN/PAD) PIN block, 3 for PIN/PAD PIN block. This field must contain a 1 or 3 if using dynamic PIN key management.
PIN PAD CHARACTER	A–F
MAC DATA TYPE	0 or 1
NUMBER OF KEYS	1 if the same key is used for inbound and outbound messages, or 2 if different keys are used for inbound and outbound messages or if the KEY PROCESSING TYPE field on KEYF screen 3 is set to a value of C.

KEYF Screen 1	
Field	Value
KEY LENGTH	1 for single-length key exchange keys (KEKs), or 2 for double-length KEKs
FULL MESSAGE MAC	N for authentication of selected fields, or Y for full message authentication.
INTERMEDIATE KEYS: CLEAR	Clear version of intermediate key
INTERMEDIATE KEYS: ENCRYPTED	Encrypted version of intermediate key
INTERMEDIATE KEYS: CHECK DIGITS	0–9, A–Z
EXCHANGE KEYS: PIN KEY	Two sets of hexadecimal characters
EXCHANGE KEYS: CHECK DIGITS	0–9, A–Z
EXCHANGE KEYS: MAC KEY	Two sets of hexadecimal characters
EXCHANGE KEYS: CHECK DIGITS	0–9, A–Z

KEYF Screen 2

KEYF screen 2 contains inbound and outbound working key information for PINs and MACs. The fields on KEYF screen 2 used by the BIC ISO Interface process and their expected values are indicated in the following table.

KEYF Screen 2	
Field	Value
OUTBOUND KEYS: PIN KEY1	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
OUTBOUND KEYS: CURRENT INDEX 1	1 for the value in the PIN KEY1 field, or 2 for the value in the PIN KEY2 field.
OUTBOUND KEYS: PIN KEY2	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
OUTBOUND KEYS: KEY COUNTER	6 hexadecimal characters
INBOUND KEYS: PIN KEY1	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
INBOUND KEYS: CURRENT INDEX 1	1 for the value in the PIN KEY1 field, or 2 for the value in the PIN KEY2 field.
INBOUND KEYS: PIN KEY2	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
INBOUND KEYS: KEY COUNTER	6 hexadecimal characters
OUTBOUND KEYS: MAC KEY1	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
OUTBOUND KEYS: CURRENT INDEX 1	1 for the value in the MAC KEY1 field, or 2 for the value in the MAC KEY2 field.
OUTBOUND KEYS: MAC KEY2	16 hexadecimal characters

KEYF Screen 2	
Field	Value
OUTBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
OUTBOUND KEYS: KEY COUNTER	6 hexadecimal characters
INBOUND KEYS: MAC KEY1	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
INBOUND KEYS: CURRENT INDEX 1	1 for the value in the MAC KEY1 field, or 2 for the value in the MAC KEY2 field.
INBOUND KEYS: MAC KEY2	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
INBOUND KEYS: KEY COUNTER	6 hexadecimal characters

KEYF Screen 3

KEYF screen 3 contains information on the timer, threshold, and processing parameters used by the interface for Dynamic Key Management (DKM). It also identifies the originator and receiver of DKM messages. The fields on KEYF screen 3 used by the BIC ISO Interface process and their expected values are indicated in the following table.

KEYF Screen 3	
Field	Value
PIN KEY VARIANT	0 for PIN translation directly from the PIN KEK parts, or 1 for translation that includes the variant 1 constant with the PIN KEK parts.
MAC KEY VARIANT	0 for translation directly from the MAC KEK parts, or 3 for translation that includes the variant 3 constant with the MAC KEK parts. This field must be set to 3 when using dynamic key management.
PIN KEY TIMER VALUE	0 or 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days.
MAC KEY TIMER VALUE	0 or 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days.
PIN KEY TIMER INTERVAL	D for days, H for hours, or M for minutes.
MAC KEY TIMER INTERVAL	D for days, H for hours, or M for minutes.
PIN KEY TRAN	0 if a total transaction limit is not used or 50–100000.

KEYF Screen 3	
Field	Value
MAC KEY TRAN	0 if a total transaction limit is not used or 50–100000.
PIN KEY ERROR	0 if an error limit is not used or 5–9999
MAC KEY ERROR	0 if an error limit is not used or 5–9999
CONSECUTIVE PIN KEY ERROR	0 if a consecutive error limit is not used or 5–9999
CONSECUTIVE MAC KEY ERROR	0 if a consecutive error limit is not used or 5–9999
KMAC SYNCHRONIZATION ERROR	0 if a MAC working key synchronization error is not used or 3–9999
CLEAR OLD KEY TIMER VALUE	0 if the old key is not cleared or 1–9999 (seconds)
ORIGINATING ID	ID of sender of dynamic key management messages
RECEIVING ID	ID of receiver of dynamic key management messages
KEY PROCESSING TYPE	C if each interface is only responsible for generating its own outbound keys, M if the interface is responsible for all key generation, S if the interface receives keys but does not generate them, or N if dynamic key management is not used.
NOTARIZATION SUPPORTED	“0”

EMF Screens

This section contains a listing of EMF fields used by the BIC ISO Interface process. In some EMF fields, the BIC ISO Interface process requires that specific values be entered. In these cases, the required values are included in the description. Each description also identifies the EMF fields used by the BIC ISO Interface process for the screen being described.

Note: The information contained in the following descriptions does **not** indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

EMF Screen 1

EMF screen 1 allows the institution to change BASE24 message information. When the key fields are completed appropriately and the **F7** key is pressed, Ms and Cs are displayed in the fields. An M indicates the data is mandatory and is always included in the message. A C indicates the data is not mandatory but can be included in the message. Some fields are left blank. This indicates the field is not included in the message. All fields on EMF screen 1 can be used by the BIC ISO Interface process.

The table below indicates the EMF fields used by the BIC ISO Interface process and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields.

EMF Screen 1	
Field	Value
INTERF TYP	“BIC”
DPC/MOD #	Not used
PROCESS NAM	Symbolic name of the BIC ISO Interface process
PROD #	00 for BASE, 01 for the BASE24-atm product, 02 for the BASE24-pos product
MSG TYP	Allowed external message type or types

EMF Screen 1	
Field	Value
IN-OUT-IND	I for incoming, O for outgoing, or B for both incoming and outgoing
TOKEN GROUP	Not used
FULL MSG MAC	Y to include all elements when computing the MAC, N to use the flag settings for the individual message elements to determine which elements are included when computing the MAC.
MESSAGE ELEMENTS	C for conditional, M for mandatory, or blank if the element is not used.

3: Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications lines between BASE24 and the co-network, and those used to manage PIN and message authentication keys between BASE24 and the co-network.

The BIC ISO Interface process supports the following four types of network management messages:

- Logon messages
- Echo test messages
- Logoff messages
- Dynamic key management (DKM) messages

Network management messages are sent as 0800 messages and require 0810 messages in response.

This section describes the message flows that occur when network management messages are sent.

Monitoring Co-Networks and Stations

The BIC ISO Interface process keeps track of the status of all co-networks and stations with which it communicates using the station table. This station table is built at initialization and is updated continuously during processing.

Stations are marked *up* and *down* in the table depending on whether the BIC ISO Interface process can communicate with them. For instance, if a station fails to respond appropriately to a message, the BIC ISO Interface process checks the maximum retries allowed for the co-network's stations. If the maximum retries allowed is reached, the BIC ISO Interface process marks the station down and takes the necessary network management actions to reestablish the link with the station. Once the link is reestablished, the BIC ISO Interface process marks the station up.

The BIC ISO Interface process also tracks the status of the co-networks. A co-network is marked as *up* if the BIC ISO Interface process can send messages to and receive messages from at least one of the co-network's stations. If all of the stations to a co-network are marked down, the co-network itself is marked down. Additionally, the BIC ISO Interface process tracks such things as whether the co-network is logged off, whether store-and-forward processing is in progress for the co-network, and whether a SAF extract is in progress for the co-network.

The BIC ISO Interface process keeps track of station and co-network information at all times and logs a message when the status of the co-network changes (for example, from up to down or at the start of store-and-forward processing).

Note: The BIC ISO Interchange Interface process only tracks the logical ability of BASE24 to communicate with each station. If a station responds to a message, the BIC ISO Interface process considers it to be up. If a station fails to respond to a message the number of times specified in the MAXIMUM TIMEOUTS field on ICFE screen 3, the BIC ISO Interface process considers the station to be down.

Also, echo test messages do not substitute for logon messages. The BIC ISO Interface process does not consider the co-network to be logged on and available for processing transactions until it receives a logon message.

Logon Messages

Logon messages are network management messages used to manage communications access between BASE24 and a co-network. They are sent as 0800 messages and require 0810 messages in response.

Logon messages are distinguished by a value of 001 in the Network Management Information Code (S-70) field of the BASE24 external message.

Logon Messages From BASE24

The BIC ISO Interface process initiates logon messages at initialization and upon receipt of a LOGON command from a network control facility.

At initialization, the BIC ISO Interface process sends a logon message to each send/receive station.

Upon receiving a LOGON command from the network control facility, the BIC ISO Interface process sends a logon message to each send/receive station associated with the co-network indicated in the command.

Logon Messages From a Co-network

After a period of being logged off, co-networks can send logon messages to log on to BASE24.

If the BIC ISO Interface process receives a logon message after the co-network has been logged off, the BIC ISO Interface process marks the co-network as logged on and begins sending echo test messages to its stations. The BIC ISO Interface process then returns a logon response to the co-network. If the BIC ISO Interface process receives a logon message when the co-network is not logged off, it simply returns a logon response.

Note: In its responses, the BIC ISO Interface process always sets the Response Code (P-39) field in the BASE24 external message to 00 to indicate an approval.

Echo Test Messages

Echo test messages are network management messages used to establish whether a station is available for message processing. They are sent as 0800 messages and require 0810 messages in response.

Echo tests messages are distinguished by a value of 301 in the Network Management Information Code (S-70) field of the BASE24 external message.

Echo Test Messages From BASE24

The BIC ISO Interface process initiates echo tests under the following circumstances:

1. Upon expiration of an Extended Network Management Message timer. This processing is described in section 2.
2. Upon expiration of a Wait-for-Traffic timer. This processing is described in section 2.
3. When a co-network initiates a logon after a period of being logged off. The BIC ISO Interface process sends an echo test to each send/receive station once the co-network has been marked as logged on.

Echo Test Messages From a Co-Network

Co-networks can send echo test messages at any time. The BIC ISO Interface process always returns an echo test response.

Note: In its responses, the BIC ISO Interface process always sets the Response Code (P-39) field in the BASE24 external message to 00 to indicate an approval.

Logoff Messages

Logoff messages are network management messages used to stop communications between BASE24 and a co-network. They are sent as 0800 messages and require 0810 messages in response.

Logoff messages are distinguished by a value of 002 in the Network Management Information Code (S-70) field of the BASE24 external message. They can be sent by either BASE24 or a co-network. In either situation, once a co-network is logged off, BASE24 does not send echo test messages or non-network management messages to any of the co-network's stations. Any non-network management messages that must be forwarded to the co-network are placed in the SAF instead.

BASE24 only attempts to reopen the link to a co-network that is logged off if a logon command is received from a network control facility. The co-network, on the other hand, can reopen the link at any time by sending a logon message.

Note: When a co-network is logged off, any transactions in progress are allowed to complete.

Logoff Messages From BASE24

BASE24 only sends logoff messages to a co-network as a result of a LOGOFF command issued from a network control facility. BASE24 then expects a logoff response from the co-network.

If no logoff response is received, the BIC ISO Interface process still marks the co-network as logged off, but keeps sending logoff messages to the co-network until a response is received.

Logoff Messages From a Co-Network

When a co-network must log off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations.

Upon receiving a logoff message, the BIC ISO Interface process marks the co-network down and logged off and returns a logoff response to the co-network.

Note: In its responses, the BIC ISO Interface process always sets the Response Code (P-39) field in the BASE24 external message to 00 to indicate an approval.

Dynamic Key Management Messages

Key management messages make up a subset of network management (0800) messages. There are four key management messages available:

- Change key
- New key
- Repeat key
- Verify key

The change key and new key messages can be generated automatically as the result of a defined threshold being surpassed. In addition, all of these messages can be generated manually by an operator issuing a text command from a network control facility. This section provides an overview of how dynamic key management messages are used. For more information on dynamic key management, refer to the ***BASE24 Transaction Security Manual***. For more information on using a network control facility to generate key management messages, and for complete syntax of the text commands involved, see the ***BASE24 Text Command Reference Manual***.

Change Key Message

A change key message requests that a key processor generate a new PIN encryption or message authentication key. The receiving processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting process in a new key message.

The BIC ISO Interface process sends this type of message to a co-network when the co-network is responsible for generating a particular key and the BIC ISO Interface process has determined that key needs to be changed. The BIC ISO Interface process could make that determination based on a DKM threshold being passed or in response to a KEY CHANGE command (the KEY CHANGE command has the same effect as passing a DKM threshold).

The BIC ISO Interface process can receive this type of message from a co-network when the BIC ISO Interface process is responsible for generating a particular key and the co-network wants that key to be changed.

New Key Message

A new key message carries a new PIN encryption or message authentication key between processors. The receiving processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

The BIC ISO Interface process sends this type of message to a co-network when the BIC ISO Interface process is responsible for generating a particular key and has either determined that key needs to be changed or been requested by the co-network to change or repeat the key. This could occur as a result of a DKM threshold being passed, notification from the co-network (using a change key or repeat key message), or could occur in response to a KEY CHANGE command (the KEY CHANGE command has the same effect as passing a DKM threshold).

The BIC ISO Interface process can receive this type of message from a co-network when the co-network is responsible for generating a particular key and the co-network has either determined that key needs to be changed or been requested by the BIC ISO Interface process to change or repeat the key (using a change key or repeat key message).

Repeat Key Message

A repeat key message requests that a processor that generated a key resend that key. The processor receiving the repeat key message obtains the requested key from its database and sends it to the requesting process in a new key message. The key itself is not changed, nor are any threshold counters affected.

The BIC ISO Interface process only generates a repeat key message as the result of receiving a KEY REPEAT command. It responds to a repeat key message from the co-network with a new key message carrying the requested key.

Verify Key Message

A verify key message requests that a processor that generated a key verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the processor that sent the verify key message should use a change key message to request a new key.)

The BIC ISO Interface process only generates the verify key message as the result of receiving a KEY VERIFY command. It responds to a verify key message from a co-network, by comparing check digits for the specified key and returning the results.

Dynamic Key Management at Initialization

If BASE24 is set up for dynamic key management, the BIC ISO Interface process must make sure at initialization that its keys are in place prior to marking a co-network up and allowing transactions to be processed. To do this, the BIC ISO Interface process enforces the following sequence of events at initialization as it begins to establish its communication with a co-network. If dynamic key management is not being performed, this sequence is not applicable.

1. Establishes a successful logon with the co-network. At this point, the BIC ISO Interface process can begin exchanging keys, but does not mark the co-network up.
2. Exchanges message authentication keys with the co-network if message authentication is being performed by the BIC ISO Interface process. This allows the message authentication keys to be put in place prior to the exchange of PIN keys.
3. Exchanges PIN keys with the co-network if PIN encryption is being performed by the BIC ISO Interface process. This allows the PIN keys to be put in place prior to transaction processing.
4. Marks the co-network up. At this point, transactions can be accepted from and sent to the co-network.

Note: Since this sequence is only required and only occurs at initialization, it is not included in the logon and key exchange sequences documented in this section.

Interactive Network Management Message Flows

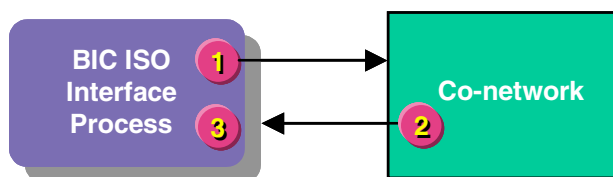
The remainder of this section contains examples of network management and dynamic key management message flows, and describes the processing performed by the BASE24 system for each.

Logon Message Flow

The following pages provide examples of logon sequences and describe the processing performed by the BIC ISO Interface process.

BASE24-Initiated Logon

In this example, the BIC ISO Interface process logs on to the co-network.

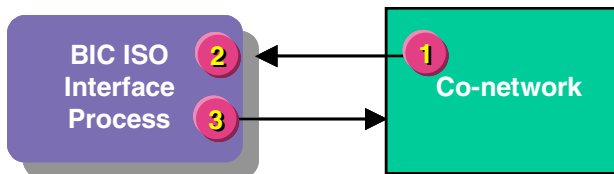


1. If the BIC ISO Interface process cannot send network management messages to the co-network (the NETWORK MANAGEMENT MESSAGE ENABLED field on ICFE screen 3 is set to a value of N), the BIC ISO Interface process marks each station up and the co-network as logged on. If the BIC ISO Interface process can send network management messages to the co-network (the NETWORK MANAGEMENT MESSAGE ENABLED field on ICFE screen 3 is set to a value of Y), the BIC ISO Interface process sends a logon message to each of the co-network's send/receive stations. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message. In addition, the BIC ISO Interface process sets a Network Management Message timer for each station, and marks each receive-only station up.
2. The co-network returns a logon response from each station. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 external message.

3. The BIC ISO Interface process receives each logon response and checks the Response Code (P-39) field in the message.
 - a. If the Response Code field contains zeros, indicating an approval, the BIC ISO Interface process marks the co-network as up and logged on. The BIC ISO Interface process also marks the corresponding station up and sets a Wait-for-Traffic timer for that station.
 - b. If the Response Code field does not contain zeros, the BIC ISO Interface process logs a message, marks the co-network as logged off, and ceases all attempts to communicate with the station. In order to reinstate communications with the station, the co-network must send a message from that station.

Co-Network-Initiated Logon

In this example, the co-network logs on to the BASE24 system.



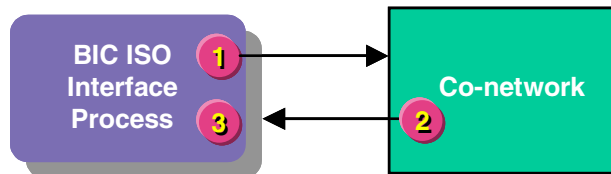
1. The co-network sends a logon message to reestablish communications. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 external message.
2. The BIC ISO Interface process receives the message and performs the following steps:
 - a. Marks the co-network as logged on.
 - b. Sets a Wait-for-Traffic timer for each station defined for the co-network.
 - c. Sends a logon response to the co-network. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.
3. The BIC ISO Interface process sends an echo test message and sets a Network Management Message timer for each of the co-network's send/receive stations that is down. The echo test message is identified by the value 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Logoff Message Flow

The following pages provide examples of logoff sequences and describe the processing performed by the BIC ISO Interface process.

BASE24-Initiated Logoff

In this example, the BIC ISO Interface process logs off from the co-network.

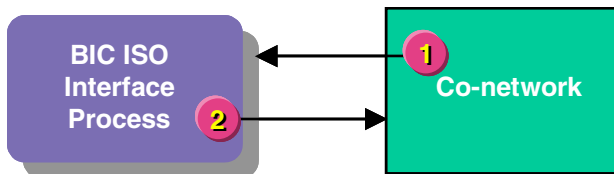


1. The BIC ISO Interface process receives a LOGOFF command from a network control facility. The BIC ISO Interface process sends a logoff message to each of the co-network's send/receive stations and starts a Network Management Message timer for each. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.
2. The co-network returns a logoff response (from one or each station). The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.
3. The BIC ISO Interface process marks the co-network as logged off and clears the following timers running for the co-network and its stations:
 - Network Management Message timers
 - Extended Network Management Message timers
 - Wait-for-Traffic timers

All other timers are left running to ensure that transactions in progress complete.

Co-Network-Initiated Logoff

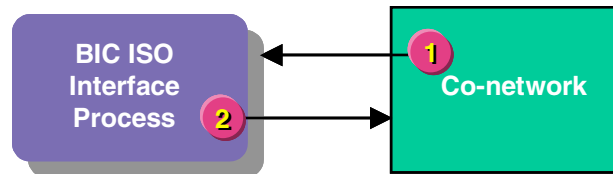
In this example, the co-network logs off from BASE24:



1. The co-network sends a logoff message to BASE24. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.
2. The BIC ISO Interface process receives the logoff message and performs the following steps:
 - a. Marks the co-network as logged off.
 - b. Clears the following timers running for the co-network and its stations:
 - Network Management Message timers
 - Extended Network Management Message timers
 - Wait-for-Traffic timersAll other timers are left running to ensure that current transactions in progress complete.
 - c. Sends a logoff response to the co-network. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Echo Test Message Flow

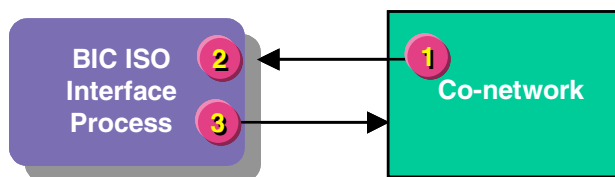
The following example describes how the BIC ISO Interface process handles echo test messages from the co-network. In this case, the co-network is logged on, but BASE24 considers the station to be down.



1. The co-network sends an echo test message from station A. The echo test message is identified by the value 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.
2. The BIC ISO Interface process receives the message and performs the following steps:
 - a. Returns an echo test response. The echo test response is identified by the value 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.
 - b. Sends an echo test message and sets a Network Management Message timer for each of the co-network's send/receive stations that is down.

Change Key Message Flow

The diagram below illustrates the message flow between the co-network and BIC ISO Interface process when the co-network initiates a change key request message. In this case, the BIC ISO Interface process is responsible for generating the key being requested. The steps corresponding to the diagram are also described below.

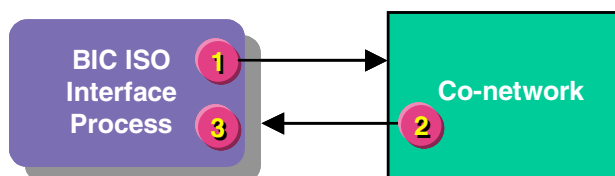


1. The co-network formats and sends an 0800 message with a network management information code of 161 to the BIC ISO Interface process. This message includes the Cryptographic Service Message data element (S-123) containing a Request Service Initiation message (RSI).
2. The BIC ISO Interface process receives the 0800 message and validates it before responding. To validate the message, the BIC ISO Interface process performs as follows:
 - a. Verifies that the RSI is in the correct format. If the RSI is not in the correct format, the BIC ISO Interface process sets an error flag to a value of F to indicate that a format error was found.
 - b. Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the BIC ISO Interface process compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3 or KEY6 screen 4, and the originator ID from the RSI to the value from the RECEIVING ID field on KEYF screen 3 or KEY6 screen 4. If the receiver or originator does not match the corresponding KEYF or KEY6 values for this co-network, the BIC ISO Interface process sets an error flag to a value of H to indicate that the receiver of originator ID is incorrect.
 - c. Verifies that there is not already a key exchange action in progress for the type of key identified in the RSI. The BIC ISO Interface process checks an internal flag to verify that a key exchange is not currently in progress. If a key exchange is in progress, the BIC ISO Interface process sets an error flag to a value of C to indicate that the current Change Key request cannot be processed.

- d. Formats an 0810 response message and returns it to the co-network. To format the response message, the BIC ISO Interface process performs as follows:
 - 1) If one or more of the above checks failed with an error, the BIC ISO Interface process formats an 0810 response message with the response code set to a value of 05 (denied). This message includes the Cryptographic Service Message data element (S-123) containing an Error Service Message (ESM). The ERF field in the ESM includes any error flag values set during the checks. No further processing is performed.
 - 2) If no errors were encountered in the above checks, the BIC ISO Interface process formats an 0810 response message with the response code set to 00 (approved) and a network management information code of 161. This message includes the Cryptographic Service Message data element (S-123) containing a copy of the original RSI. Processing continues with the next step.
3. The BIC ISO Interface process creates a new key and sends it to the co-network in a New Key message. For a description of this processing, refer to the “New Key Message Flow” procedure immediately following.

New Key Message Flow

The diagram below illustrates the message flow between the BIC ISO Interface process and the co-network when the BIC ISO Interface process sends a new key message. The new key message can be sent manually using text commands or programmatically when the threshold for a defined working key has been surpassed. In this scenario, the BIC ISO Interface process is responsible for generating the key being sent. The steps corresponding to the diagram are also described below.

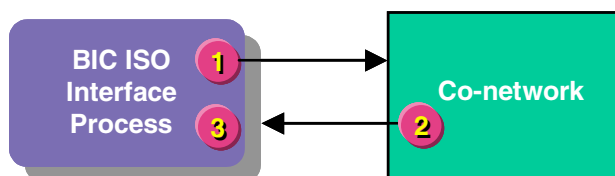


1. The BIC ISO Interface process creates a New Key message and sends it to the co-network. To do this, the BIC ISO Interface process performs as follows:
 - a. Generates the new key and check digits.
 - b. Formats a Key Service Message (KSM) containing the new key and the check digits.
 - c. Sends an 0800 message to the co-network with the network management information code set to 162. This message includes the Cryptographic Service Message data element (S-123) containing the KSM.
 - d. Sets an internal flag indicating that a key exchange is in progress for the type of key being sent.
 - e. Starts a Network Management Message timer for the message. If this timer expires before an 0810 response is received from the co-network, the BIC ISO Interface process starts an Extended Network Management Message timer for the message. If the Extended Network Management Message timer expires, the BIC ISO Interface process resends the New Key message.
2. The co-network receives the New Key message, processes it, and returns an 0810 response message with a network management information code of 162 to the BIC ISO Interface process. This message includes the Cryptographic Service Message data element (S-123) containing either a Response Service Message (RSM) or an Error Service Message (ESM).

3. The BIC ISO Interface process receives the 0810 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out.
 - 1) If the 0800 message timed out before the 0810 response was received, the BIC ISO Interface process drops the message, and no further steps are performed.
 - 2) If the 0800 message has not timed out, processing continues with the next step.
 - b. Clears the internal flag indicating that a key exchange is in progress.
 - c. Continues processing depending on the response code sent in the 0810 response message.
 - 1) If the transaction was denied, the BIC ISO Interface process parses the Cryptographic Service Message data element (S-123) to determine why the new key was denied. In this case, the Cryptographic Service Message data element contains an Error Service Message (ESM).
 - a) If the ESM indicates the message was denied because the BASE24 key counter does not match the key counter maintained by the co-network, the BIC ISO Interface process resets its key counter in the KEYF or KEY6 to the value sent by the co-network. The BIC ISO Interface process then resends the new key. Processing returns to step 1.
 - b) If the ESM indicates that any other error occurred, the BIC ISO Interface process sends a message indicating the error to the network logging facility and drops the 0810 response.
 - 2) If the transaction was approved, the BIC ISO Interface process updates the KEYF or KEY6 with the new key, check digit, and counter information. The BIC ISO Interchange Interface processBIC ISO Interchange Interface process also reinitializes its internal key change threshold counters and timers.

Repeat Key Message Flow

The diagram below illustrates the message flow between the BIC ISO Interface process and the co-network when the BIC ISO Interface process initiates a repeat key message. The BIC ISO Interface process only initiates a repeat key message if it receives a KEY REPEAT command from a network control facility. The processing described below starts from the point where the BIC ISO Interface process receives the text command. The steps corresponding to the diagram are also described below.

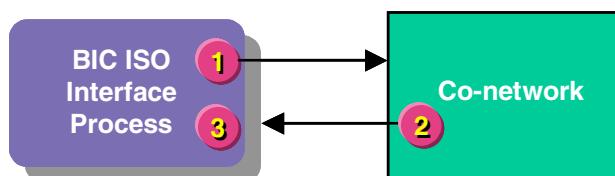


1. When the BIC ISO Interface process receives the KEY REPEAT command, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the BIC ISO Interface process logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the BIC ISO Interface process continues with the next step (step 1b).
 - b. Determines whether a key exchange is in progress for the key in the message. If a key exchange is in progress, the BIC ISO Interface process logs a message and performs no further steps.
 - c. Sends an 0800 message to the co-network with the network management information code set to 163. This message includes a blank Cryptographic Service Message data element (S-123).
 - d. Starts a Network Management Message timer for the message.
2. The co-network receives the Repeat Key request and processes the message. The co-network returns an 0810 with a network management information code of 163 response to the BIC ISO Interface process. This message includes the Key Service Message (KSM) in the Cryptographic Service Message data element (S-123).

3. The BIC ISO Interface process receives the 0810 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out.
 - 1) If the 0800 message timed out before the 0810 response was received, the BIC ISO Interface process drops the message, and no further steps are performed.
 - 2) If the 0800 message has not timed out, processing continues with the next step.
 - b. Continues processing depending on the response code sent in the 0810 response message.
 - 1) If the transaction was denied, the BIC ISO Interface process logs a message and drops the 0810 message.
 - 2) If the transaction was approved, the BIC ISO Interface process performs as follows:
 - a) Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the BIC ISO Interface process compares the receiver ID from the KSM to the value from the ORIGINATING ID field on KEYF screen 3 or KEY6 screen 4, and the originator ID from the KSM to the value from the RECEIVING ID field on KEYF screen 3 or KEY6 screen 4. If the receiver or originator does not match the corresponding KEYF or KEY6 values for this co-network, the BIC ISO Interface process logs a message and drops the 0810 response.
 - b) Verifies the keys. To do this, the BIC ISO Interface process translates the working key sent by the co-network and verifies the check digits. The check digits generated as a result of translating the key must match the check digits sent by the co-network. If the BIC ISO Interface process cannot translate the key or the check digits do not match, the BIC ISO Interface process logs a message and drops the 0810 response.
 - c) Updates the KEYF or KEY6 with the new key and check digit information.

Verify Key Message Flow

The diagram below illustrates the message flow between the BIC ISO Interface process and the co-network when the BIC ISO Interface process initiates a verify key request message. The BIC ISO Interface process only initiates a verify key message if it receives a KEY VERIFY command from a network control facility. The processing described below starts from the point where the BIC ISO Interface process receives the text command. The steps corresponding to the diagram are also described below.

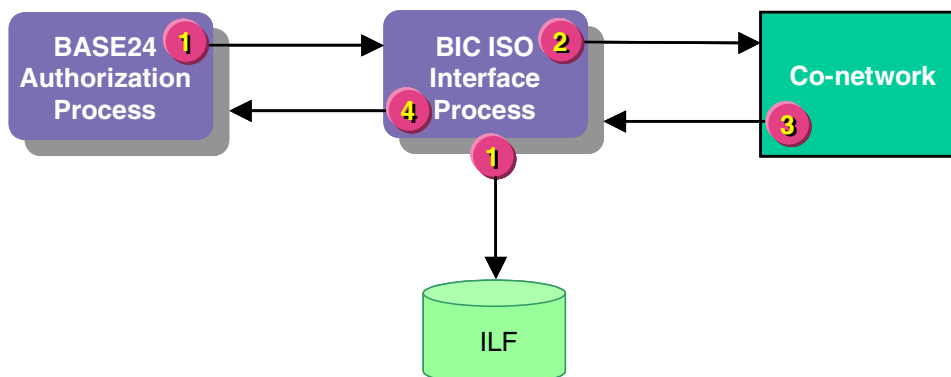


1. When the BIC ISO Interface process receives the KEY VERIFY command, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the BIC ISO Interface process logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the BIC ISO Interface process continues with the next step (step 1b).
 - b. Determines whether a key exchange is in progress for the key in the message. If a key exchange is in progress, the BIC ISO Interface process logs a message and performs no further processing.
 - c. Sends an 0800 message to the co-network with the network management information code set to 164. This message does not include the Cryptographic Service Message data element (S-123).
 - d. Starts a Network Management Message timer for the message.
2. The co-network receives the Verify Key request and processes the message. The co-network returns an 0810 with a network management information code of 164 response to the BIC ISO Interface process.
3. The BIC ISO Interface process receives the 0810 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out.
 - 1) If the 0800 message timed out before the 0810 response was received, the BIC ISO Interface process drops the message, and no further steps are performed.

- 2) If the 0800 message has not timed out, processing continues with the next step.
- b. Continues processing depending on the response code sent in the 0810 response message.
 - 1) If the transaction was denied (the Response Code data element (P-39) contained the value 05), the BIC ISO Interface process logs a message indicating the co-network denied the request and drops the 0810 message.
 - 2) If the transaction was approved (the Response Code data element (P-39) contained the value 00), the BIC ISO Interface process logs a message indicating that the command was successful and drops the 0810 message.
 - 3) If the Response Code data element (P-39) contained any other value, the BIC ISO Interface process logs a message indicating that the request failed and a Change Key request is required. The BIC ISO Interface process drops the 0810 message.

BASE24-Initiated Request—MAC Failure

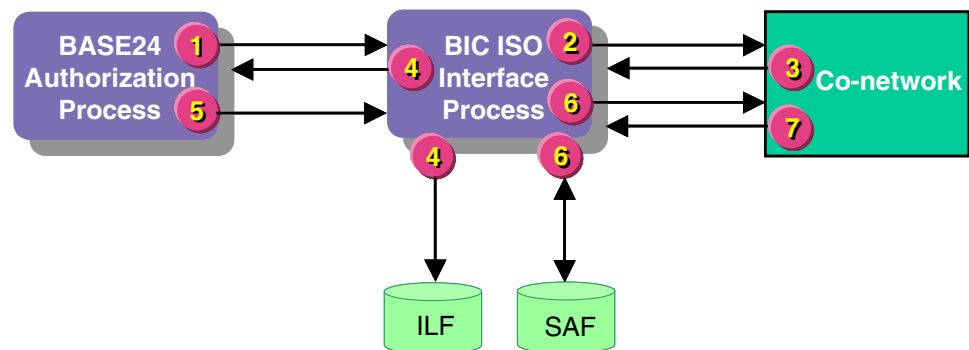
The diagram below illustrates the message flow when a MAC generation failure occurs on a transaction request initiated by the BASE24 network destined for a co-network. In this case, a denial response is sent to the BASE24 Authorization process indicating a MAC security failure condition.



1. A BASE24 Authorization process sends a request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the BIC ISO Interface process. The message fails due to a MAC generation error, and the co-network formats the response into the 0110 or 0210 external message format and sends it to the BIC ISO Interface process.
4. The BIC ISO Interface process deletes the outbound request timer, formats a 0210 internal response message, sends the response to the BASE24 Authorization process to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).

BASE24-Initiated Transaction Response—MAC Failure

The diagram below illustrates the message flow when a MAC generation failure occurs on a transaction response initiated by the BASE24 network destined for a co-network and the financial transaction was approved. In this case, a reversal message is sent to the BASE24 Authorization process indicating a MAC security failure condition.

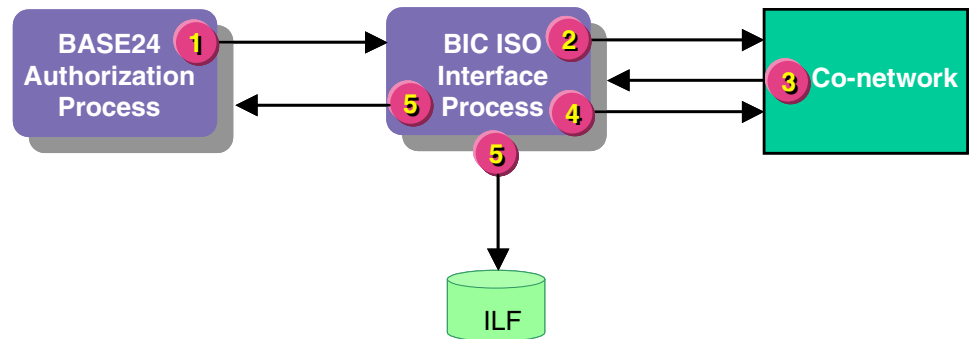


1. A BASE24 Authorization process sends a request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the BIC ISO Interface process. After the transaction is approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the BIC ISO Interface process.
4. The BIC ISO Interface process deletes the outbound request timer, formats an 0210 internal response message, sends the response to the BASE24 Authorization process to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).
5. The BASE24 Authorization process returns a 0420 message to the BIC ISO Interface process, indicating the transaction failed to complete as authorized due to a MAC generation error. The BIC ISO Interface process locates the ILF record for the original transaction and updates the ILF with the reversal information.

6. The BIC ISO Interface process formats an 0420 external message and logs this message to the Store-and-Forward File (SAF) to be sent to the co-network during store-and-forward processing.
7. If the text-level acknowledgment option is being used, the co-network sends an 0430 message to the BIC ISO Interface process. Upon receipt of the 0430 message, the BIC ISO Interface process deletes the record from the SAF.

BASE24-Initiated Transaction Request—MAC Verification Failure

The diagram below illustrates the message flow when a MAC verification failure occurs on a transaction request initiated by the BASE24 network destined for a co-network. In this case, the transaction is rejected and a denial response is sent to the BASE24 Authorization process indicating the message was rejected.



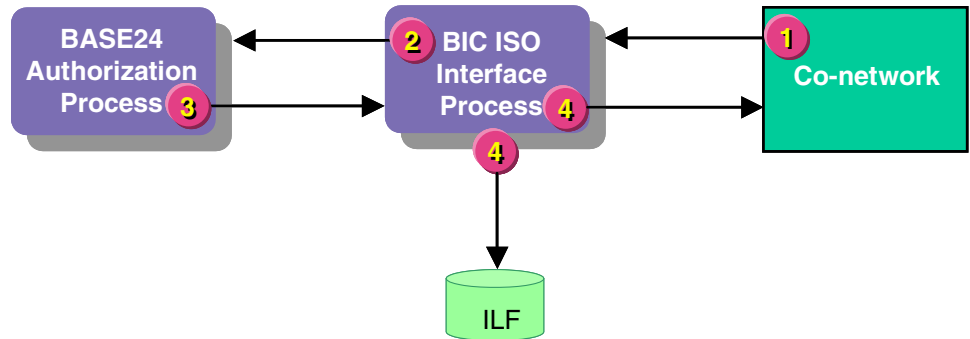
1. A BASE24 Authorization process sends a request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the BIC ISO Interface process. After the transaction is approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the BIC ISO Interface process.
4. The BIC ISO Interface process attempts to verify the MAC on the 0110 or 0210 message, but it fails to verify.

The BIC ISO Interface process rejects the message by placing a 9 in the first position of the message type making it 9210. The BIC ISO Interface process also places the bit number (i.e., 64 or 128, as appropriate) of the data element causing rejection in the STATUS field of the BASE24 External Message header. The STATUS field indicates the message failed due to a MAC verification error. The BIC ISO Interface process sends the 9210 external message to the co-network.

5. The BIC ISO Interface process deletes the outbound request timer, formats an 0210 internal message with a response code of 70 or 74, and sends it to the BASE24 Authorization process. A response code of 70 (system failure) is sent for BASE24-atm messages and a response code of 74 (unable to authorize) is sent for BASE24-pos messages. The BIC ISO Interface process also logs a copy of the transaction to the Interchange Log File (ILF).

Co-Network-Initiated Transaction Request—MAC Failure

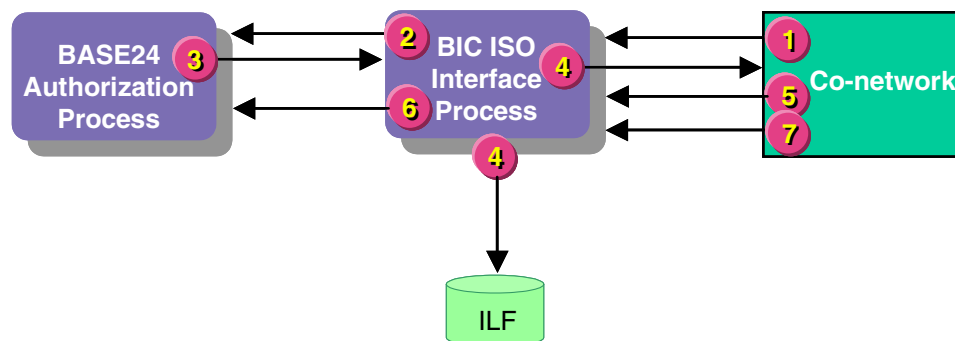
The diagram below illustrates the message flow when a MAC generation failure occurs on a transaction request initiated by the co-network to the BASE24 system for authorization. In this case, a denial response is sent to the BIC ISO Interface process indicating a MAC security failure condition.



1. A co-network sends a request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to the BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response.
3. The BASE24 Authorization process sends the response to the BIC ISO Interface process in the 0210 internal message format, indicating the message failed due to a MAC generation error.
4. The BIC ISO Interface process deletes the inbound request timer, formats a 0210 external response message, sends the response to the co-network to be completed, and logs a copy of the transaction to the Interchange Log File (ILF).

Co-Network-Initiated Transaction Response—MAC Failure

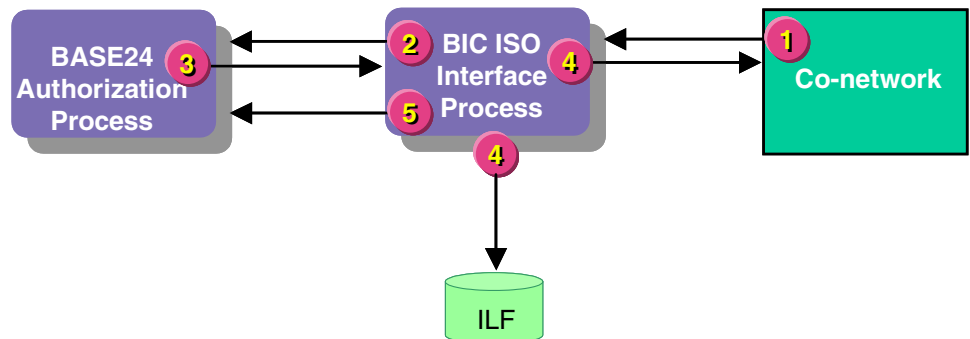
The diagram below illustrates the message flow when a MAC generation failure occurs on transaction response initiated by the co-network to the BASE24 system for authorization. In this case, a reversal message is sent to the BASE24 Authorization process indicating a MAC security failure condition.



1. A co-network sends a request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to the BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response.
3. The BASE24 Authorization process sends the transaction response to the BIC ISO Interface process in the 0210 internal message format.
4. The BIC ISO Interface process deletes the inbound request timer, formats a 0110 or 0210 external response message, returns the response to the co-network to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).
5. The co-network returns a 0420 message to the BIC ISO Interface process, indicating the transaction failed to complete as authorized due to a MAC verification error.
6. The BIC ISO Interface process locates the ILF record for the original transaction, updates the ILF with the reversal information, formats a 0420 internal message, and sends the internal message to the BASE24 Authorization process.
7. If the text-level acknowledgment option is being used, the co-network sends an 0430 message to the BIC ISO Interface process.

Co-Network-Initiated Request—MAC Verification Failure

The diagram below illustrates the message flow when a MAC verification failure occurs on a transaction request initiated by the co-network and sent to the BASE24 system for authorization. In this case, the transaction is rejected and a denial response is sent to the BASE24 Authorization process indicating the message was rejected.



1. A co-network sends a request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to the BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response.
3. The BASE24 Authorization process sends the response to the BIC ISO Interface process in the 0210 internal message format, indicating the message failed due to a MAC verification error.
4. The BIC ISO Interface process rejects the message by placing a 9 in the first position of the message type making it 9210. The BIC ISO Interface process also places the bit number of the data element causing rejection in the STATUS field of the BASE24 External Message header. The STATUS field indicates the message failed due to a MAC verification error. The BIC ISO Interface process sends the 9210 external message to the co-network. The BIC ISO Interface process also logs a copy of the transaction to the Interchange Log File (ILF).

Denied Request Initiated by BASE24—MAC Failure

The diagram below illustrates the message flow when a MAC generation failure occurs on a transaction request that was denied.

Note: Similar MAC verification failures can occur for all message types supported by the BIC ISO Interface process, not just 0200 messages (e.g., 0100, 0110, 0120, 0130, 0210, etc.).



1. A BASE24 Authorization process sends a 0200 message to the BIC ISO Interface process.
2. The BIC ISO Interface process reformats the message and sends the 0200 to the co-network. MAC generation occurs at this point.
3. The co-network cannot verify the MAC so it rejects the transaction and sends a 9200 message to the BIC ISO Interface process.
4. The BIC ISO Interface process reformats the 9200 message and sends a 0210 message to the Authorization process, indicating the transaction has been denied.

4: Store-and-Forward Processing

Store-and-forward processing allows for storing transactions that the BASE24 system has processed when a co-network is down or unavailable.

The BIC ISO Interface process keeps track of each station and co-network. If a message bound for a co-network cannot be transmitted, the BIC ISO Interface process places the message in the Store-and-Forward File (SAF) for later transmission. When communications with the co-network resume, the messages in the SAF are sent to the co-network.

This section describes store-and-forward processing. It includes information on what messages can be placed in the SAF, when messages are placed in the SAF, and when messages from the SAF are sent to the co-network. It also contains messages flows that illustrate store-and-forward processing.

The Store-and-Forward File (SAF)

The Store-and-Forward File (SAF) temporarily stores messages bound for a co-network until those messages can be sent to the intended co-network. The SAF is a key-sequenced file, and, in general, the messages are processed as first-in, first-out, as described below.

The store-and-forward cycle used by the BIC ISO Interface process is single-threaded, so the oldest SAF records are sent to the co-network first. Records arrive at the co-network in the same order in which they were added to the SAF.

Messages in the SAF are stored in the format that they are to be sent to the co-network.

Co-networks can have store-and-forward messages sent to them using communications lines, or following long periods of down-time, the SAF can be extracted to tape in order to keep from clogging communications once communications have been restored.

Messages Placed in the SAF

The BIC ISO Interface process places the following message types in the SAF:

- 0120 — Authorization Advice
- 0121 — Authorization Advice Repeat
- 0220 — Financial Transaction Advice
- 0221 — Financial Transaction Advice Repeat
- 0320 — File Inquiry/Update Advice
- 0321 — File Inquiry/Update Advice Repeat
- 0402 — Card Issuer Reversal Request
- 0420 — Reversal Advice
- 0421 — Reversal Advice Repeat
- 0520 — Acquirer Reconciliation Advice
- 0620 — Administrative Advice
- 0621 — Administrative Advice Repeat

0120, 0220, 0320, 0402, 0402, 0420, 0520 and 0620 messages are placed in the SAF if the intended co-network is marked down at the time the message is to be sent. If the BIC ISO Interface process receives one of these message to be sent to a co-network, but discovers the co-network is marked down, the BIC ISO Interface process places the message in the SAF without attempting to send it to the co-network.

In addition, these messages can be added to the SAF as the result of a recoverable message authentication error. In this case, the link to the co-network may not be down.

If the BIC ISO Interface process sends a 0120, 0220, 0320, 0402, 0420, or 0620 message to a co-network, but the message times out, the BIC ISO Interface process changes the 0120 message to a 0121, 0220 to a 0221, 0320 to a 0321, 0420 to a 0421, or 0620 to a 0621 message and places it in the SAF. The 0402 is not changed before it is placed in the SAF.

Messages Not Placed in the SAF

The BIC ISO Interface process never places the following message types in the SAF:

- 0100 — Authorization Request
- 0110 — Authorization Request Response
- 0130 — Authorization Advice Response
- 0200 — Financial Transaction Request
- 0210 — Financial Transaction Request Response
- 0230 — Financial Transaction Advice Response
- 0300 — File Inquiry/Update Request
- 0310 — File Inquiry/Update Response
- 0330 — File Inquiry/Update Advice Response
- 0412 — Card Issuer Reversal Request Response
- 0430 — Reversal Advice Response
- 0500 — Acquirer Reconciliation Request
- 0510 — Acquirer Reconciliation Response
- 0530 — Acquirer Reconciliation Advice Response
- 0600 — Administrative Request
- 0610 — Administrative Response
- 0630 — Administrative Advice Response

0100, 0110, 0200, 0210, 0300, 0310, 0600, and 0610 messages are interactive messages and their timing is crucial. If they are not processed within a given period of time, the transactions they represent are declined. Therefore, these messages are not placed in the SAF.

0130, 0230, 0330, 0412, 0430, 0530, and 0630 messages are acknowledgments or responses to other messages. Here again, timing is crucial. If a co-network does not receive a required acknowledgment or response within a given time, the co-network assumes that BASE24 never received the message and might then send it again as a duplicate.

Deleting Messages from the SAF

Records are deleted from the SAF under the following circumstances:

- When the BIC ISO Interface process determines that the SAF record was processed successfully by the co-network.

If a co-network must send text-level acknowledgments, the BIC ISO Interface process uses the text-level acknowledgments to determine when the co-network has processed the SAF records. If the co-network is not required to send text-level acknowledgments, the XPNET process notifies the BIC ISO Interface process that the message was sent successfully. In both of these cases, the BIC ISO Interface process assumes the record was processed successfully by the co-network and deletes the record.

- When the BIC ISO Interface process determines that the SAF record cannot be processed by the co-network.

If the message is returned as failed from the co-network or the same SAF message has timed out the number of times specified in the MAX SAF RETRY field on ICFE screen 3, the BIC ISO Interface process assumes the co-network cannot process the message. In this case, the BIC ISO Interface process writes the message to its log and deletes it from the SAF.

- When the record is extracted from the SAF using the SAF extract.

Once a record is extracted to tape, the Super Extract process deletes it from the SAF.

Forwarding SAF Messages to a Co-Network

The BIC ISO Interface process checks for the presence of store-and-forward messages at the following times:

- When a text-level or logical acknowledgment is received for a store-and-forward message.
- Each time a message is received from the co-network.

If there are store-and-forward messages to send, the BIC ISO Interface process determines whether a SAF extract is in progress for the co-network. If a SAF extract is not in progress, the BIC ISO Interface process begins sending the store-and-forward messages.

Interspersed SAF Messages

Co-networks have their store-and-forward messages sent to them interspersed with live transaction messages. This means that the BIC ISO Interface process continues sending 0100, 0120, 0200, 0220, 0300, 0320, 0600, and 0620 messages along with its store-and-forward messages.

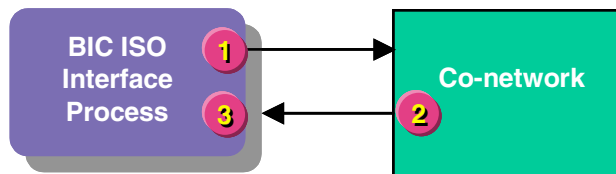
The only messages affected in this case are 0402 and 0420 messages going to the co-network. Instead of being sent to the co-network, these messages are placed in the SAF to be sent at the end of the store-and-forward processing. This ensures that a reversal will not reach the co-network before the original transaction.

Store-and-Forward Message Processing

The general processing flow for SAF messages is the same as for normal messages. To the co-network, the transactions appear to be happening as the messages are sent—they are indistinguishable from normal messages. Internally, the BIC ISO Interface process uses different timers that identify the messages as SAF records. If the specified timer expires, these messages are left on the SAF to be resent during store-and-forward processing.

Normal Store-and-Forward Messages

The following diagram illustrates the message exchanges expected by BASE24 when store-and-forward messages are sent to a co-network.



1. The BIC ISO Interface process sends a store-and-forward message to the co-network. To do this, it performs the following steps:
 - a. Reads a record from the SAF.

The BIC ISO Interface process selects the oldest record in the file for the co-network. The record is a 0120, 0121, 0220, 0221, 0320, 0321, 0402, 0420, 0421, 0520, 0521, 0620, or 0621 message that is already in external message format.
 - b. Sets a flag for the co-network to indicate that store-and-forward processing is in progress. The BIC ISO Interface process performs this step only on the first store-and-forward message processed.
 - c. Locates an available send/receive station and sends the message.
 - d. Sets a Store-and-Forward Message timer for the message.
 - e. Adds one to the count of store-and-forward messages outstanding.

2. The co-network processes the 0120, 0121, 0220, 0221, 0320, 0321, 0402, 0420, 0421, 0520, 0521, 0620, or 0621 message and returns a 0130, 0230, 0330, 0412, 0430, 0530, or 0630 message if text-level acknowledgments are required. If text-level acknowledgments are not required, the XPNET process notifies the BIC ISO Interface process when it has successfully sent the message to the co-network.
3. Upon receipt of a 0130, 0230, 0330, 0412, 0430, 0530, or 0630 message or notification from the XPNET process that the store-and-forward message was sent, the BIC ISO Interface process performs the following steps:
 - a. Locates and deletes the corresponding record in the SAF.
 - b. Subtracts one from the number of store-and-forward messages outstanding.
 - c. Determines whether more SAF records need to be sent. If more SAF records need to be sent, the BIC ISO Interface process begins again with step 1. Otherwise, the BIC ISO Interface process continues processing acknowledgments as they are received until no more store-and-forward messages are outstanding. The BIC ISO Interface process then terminates the store-and-forward processing and resets the flag for the co-network, indicating that store-and-forward processing is no longer in progress.

Failed Messages

If a message cannot be processed by a co-network, the message must be returned with the first position of the message type set to 9.

The co-network can reject the message for several reasons, including retryable message authentication errors (that is, key synchronization errors and security device failures). If the message was rejected because of a retryable message authentication error, the Status field in the external message header contains the value 196 or 199. In this case, the BIC ISO Interface process resends the message to the co-network.

When the BIC ISO Interface process receives a message from the co-network that failed for any other reason, it performs the following steps:

1. Writes the message to its log.
2. Deletes the record from the Store-and-Forward File (SAF).
3. Clears the SAF retries count.

4. Continues store-and-forward processing with the next record in the Store-and-Forward File (SAF).

Messages marked as failed by the XPNET process, rather than by the co-network, are handled as if the messages timed out. For a description of this processing, refer to the discussion on the Store-and-Forward Message timer in section 2.

5: Transaction Handling

This section describes how the BIC ISO Interface process handles various types of transactions. It describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from the co-network are handled.

Note: Transaction flow diagrams for cutover messages are provided in section 6.

Introduction

The following transaction flows are described in this section. In most cases, these flows are described twice—once when the message flow is initiated by BASE24 and once when the message flow is initiated by the co-network.

- Requests
- Reversals
- Timeouts
- Force Posts
- Logon
- Logoff
- Echo Test

Logging to the ILF

The BIC ISO Interface process logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- Force posts
- Reconciliation totals

Whenever a reversal occurs, the BIC ISO Interface process attempts to update the ILF record already logged for the transaction. If MACs are used, the BIC ISO Interface process attempts to verify the MAC.

Rejected Messages

If the BIC ISO Interface process receives an external message that it cannot process or reformat for internal use, it rejects the message as follows:

1. Changes the first position of the message type to 9. For example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message.
2. Places the bit number of the data element causing rejection or the MAC verification error encountered in the Status field of the BASE24 External Message header. For example, if the Track 2 data in the P-35 data element cannot be parsed, the Status field in the header would be set to 035.
3. Returns the message to the co-network.

These actions are taken on any message that cannot be processed, not just those requiring a response.

Tracing Transactions

In order to trace transactions, the BIC ISO Interface process uses the Retrieval Reference Number (P-37) from the external message. The Retrieval Reference Number is assigned by a message initiator to uniquely identify a transaction. It is a number that remains unchanged for all messages throughout the life of a transaction.

Other data elements from the external message that can be used to trace transactions are listed below. For additional information on these data elements, refer to the ***BASE24 BIC ISO Standards Manual***.

- Transaction Amount (P-4)
- Local Transaction Time (P-12)
- Local Transaction Date (P-13)
- Acquiring Institution Identification Code (P-32)
- Track 2 Data (Primary Account Number) (P-35)

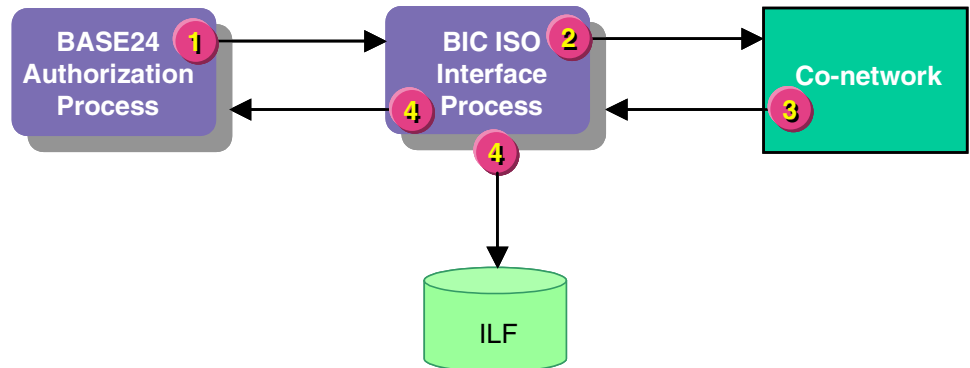
Message Flows to the Co-Network

The following pages describe the message flows for the following transactions outbound from BASE24 to the co-network.

- Requests
- Reversals
- Force Posts
- Timeouts

Request Message to the Co-Network

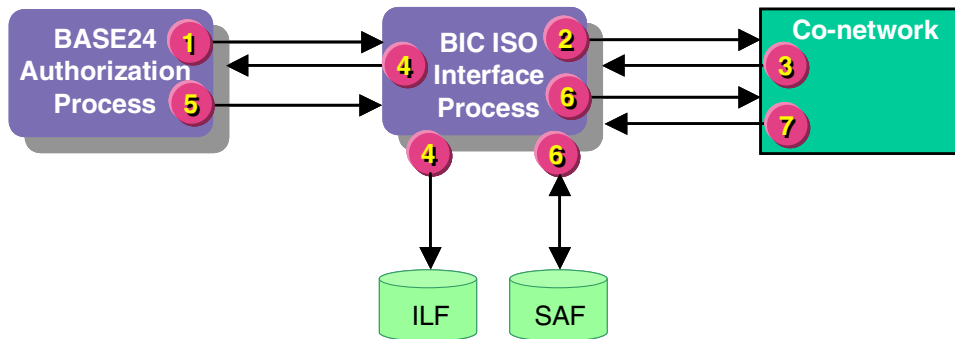
The diagram below illustrates the message flow when a transaction is initiated by the BASE24 network and must be sent to the co-network for authorization. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authorizer. After the transaction has been approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the BIC ISO Interface process.
4. The BIC ISO Interface process deletes the outbound request timer, formats a 0210 internal response message, sends the response to the BASE24 Authorization process to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).

Reversal to the Co-Network

When an ATM or POS device connected to the local BASE24 network fails to complete a transaction as authorized, a reversal message must be sent to the co-network. The diagram below illustrates message routing when a reversal is sent to the co-network. The steps in the diagram are described below.

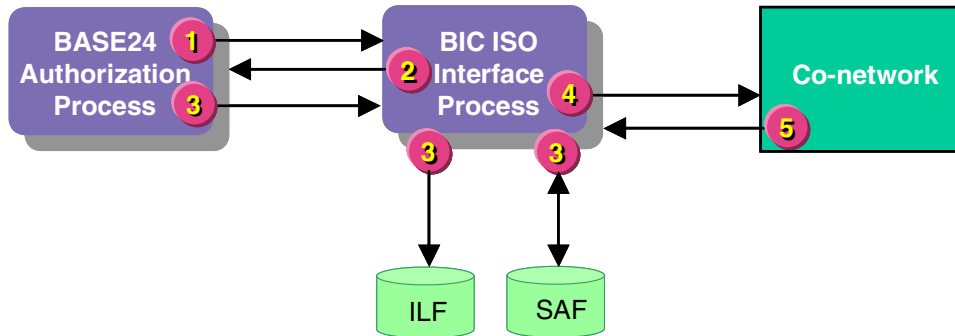


1. A BASE24 Authorization process sends the request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authorizer. After the transaction is approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the BIC ISO Interface process.
4. The BIC ISO Interface process deletes the outbound request timer, formats a 0210 internal response message, returns the response to the BASE24 Authorization process to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).
5. The BASE24 Authorization process returns a 0420 message to the BIC ISO Interface process, indicating that the transaction failed to complete as authorized. The BIC ISO Interface process locates the ILF record for the original transaction and updates the ILF record with the reversal information.
6. The BIC ISO Interface process formats a 0420 external message and logs this message to the Store-and-Forward File (SAF) to be sent to the co-network during store-and-forward processing.
7. If text-level acknowledgment are required, the co-network sends a 0430 message to the BIC ISO Interface process. Upon receipt of the 0430 message, the BIC ISO Interface process deletes the record from the SAF. If

text-level acknowledgments are not required, a logical acknowledgment must be received. If a logical acknowledgment is not received, the record is deleted from the SAF after the 0420 message is sent.

Force Post Messages to the Co-Network

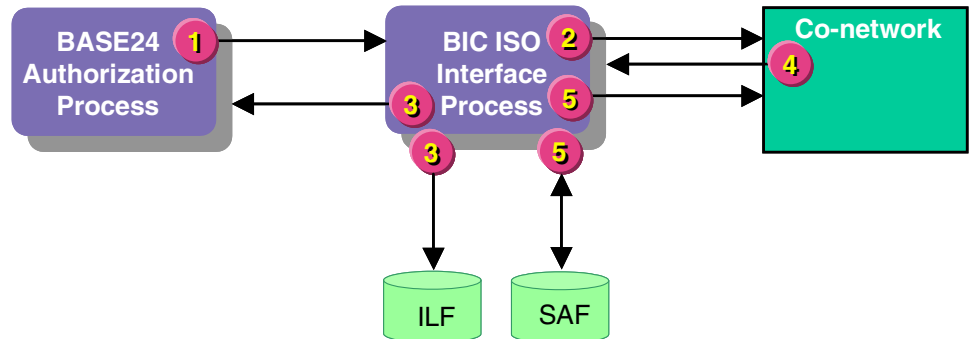
When the local BASE24 network performs stand-in authorization, it sends the co-network a force post message for each transaction it authorizes. The diagram below illustrates message routing when a force post message is sent to the co-network. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process determines that the co-network is not available and sends the message back to the BASE24 Authorization process to be authorized.
3. The BASE24 Authorization process sends the BIC ISO Interface process a force post message in the 0220 internal message format. If the transaction was approved, the BIC ISO Interface process reformats the message into the 0220 external message format and adds the external message to the Store-and-Forward File (SAF). A copy of the transaction is added to the Interchange Log File (ILF). If the transaction was declined, the message is dropped.
4. The BIC ISO Interface process sends the force post transaction to the co-network using store-and-forward processing when the co-network becomes available.
5. If the text-level acknowledgment option is being used, the co-network sends a 0230 message to the BIC ISO Interface process. Upon receipt of the 0230 message, the BIC ISO Interface process deletes the record from the SAF. If text-level acknowledgments are not required, a logical acknowledgment must be received. If a logical acknowledgment is not received, the record is deleted from the SAF after the 0220 message is received.

Timeouts of Requests to the Co-Network

Each time the BIC ISO Interface process sends a 0100 or a 0200 financial transaction request to the co-network for authorization, the BIC ISO Interface process starts an outbound request timer. If the outbound request timer expires before a response is received from the co-network, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. The BASE24 Authorization process sends the request to the BIC ISO Interface process in the 0200 internal message format.
2. The BIC ISO Interface process reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorization, and starts an outbound request timer to wait for the response from the co-network.
3. When the timer expires before a response is received from the co-network, the BIC ISO Interface process checks to determine whether stand-in authorization is allowed.
 - a. If stand-in authorization is allowed, the BIC ISO Interface process returns the transaction to a BASE24 Authorization process as a 0200 message with the RTE-STAT field set to indicate this condition.
 - b. If stand-in authorization is not allowed and this is a BASE24-pos transaction, the BIC ISO Interface process checks the TIMEOUT ACTION field in the Enhanced Interchange Configuration File (ICFE). If this field is set to 2, the BIC ISO Interface process attempts to route the transaction to an alternate destination.
 - c. If stand-in authorization is not allowed and this is a BASE24-atm transaction, the BIC ISO Interface process attempts to route the transaction to an alternate destination.

- d. If stand-in authorization is not allowed and the alternate destination is not available, the BIC ISO Interface process formats a 0210 internal response message denying the transaction, returns the response to the BASE24 Authorization process, and logs a copy of the transaction to the Interchange Log File (ILF).
4. The co-network sends a 0110 or 0210 late response to the BIC ISO Interface process. If the transaction was denied by the co-network, the BIC ISO Interface process drops the late response.
5. If the transaction was approved by the co-network, the BIC ISO Interface process formats a 0420 external reversal message and adds it to the Store-and-Forward File (SAF) to be sent to the co-network during store-and-forward processing.

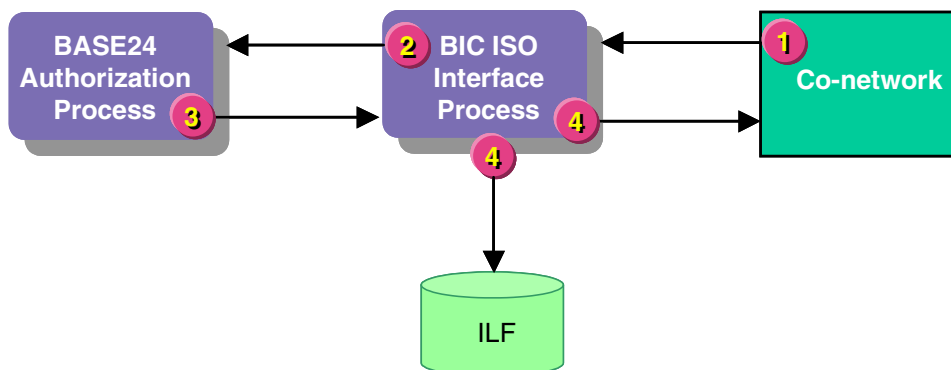
Message Flows from the Co-Network

The following pages describe the message flows for the following transactions inbound to the BASE24 network from the co-network.

- Requests
- Reversals
- Force Posts
- Timeouts

Request Message from the Co-Network

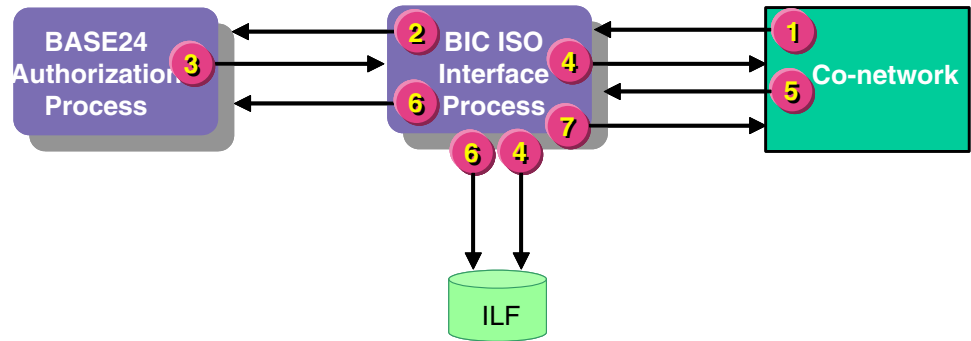
The diagram below illustrates message routing when a transaction is sent from the co-network to the BASE24 network for authorization. The steps in the diagram are described below.



1. The co-network sends the request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorization process for authorization, and starts an inbound request timer to wait for the response.
3. The BASE24 Authorization process sends the response to the BIC ISO Interface process in the 0210 internal message format.
4. The BIC ISO Interface process deletes the inbound request timer, formats a 0110 or 0210 external response message, returns the response to the co-network to be completed, and logs a copy of the transaction to the Interchange Log File (ILF).

Reversal from the Co-Network

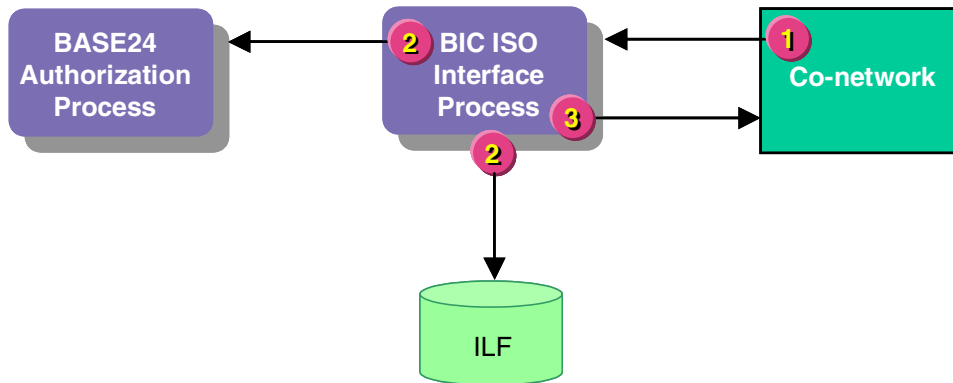
When an ATM or POS device connected to the co-network fails to complete a transaction as authorized, a reversal message must be sent to the BASE24 network. The diagram below illustrates message routing when a reversal is sent to the BASE24 network. The steps in the diagram are described below.



1. The co-network sends the request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response from the BASE24 Authorization process that received the internal message.
3. The BASE24 Authorization process sends the transaction response to the BIC ISO Interface process in the 0210 internal message format.
4. The BIC ISO Interface process deletes the inbound request timer, formats a 0110 or 0210 external response message, returns the response to the co-network to be delivered to the ATM or POS device, and logs a copy of the transaction to the Interchange Log File (ILF).
5. The co-network returns a 0420 message to the BIC ISO Interchange Interface process, indicating that the transaction failed to complete as authorized.
6. The BIC ISO Interface process updates the ILF record for the original transaction, formats a 0420 internal message, and sends the internal message to the BASE24 Authorization process.
7. If the text-level acknowledgment option is being used, the BIC ISO Interface process sends a 0430 message to the co-network.

Force Post Messages from the Co-Network

The diagram below illustrates message routing when a force post message is sent to the BASE24 network. The steps in the diagram are described below.



1. The co-network sends the BIC ISO Interface process a force post message in the 0220 external message format.
2. The BIC ISO Interface process reformats the message into the 0220 internal message format, sends the internal message to a BASE24 Authorization process, and logs a copy of the transaction to the Interchange Log File (ILF).
3. If completion acknowledgments are required, the BIC ISO Interface process sends the co-network a 0230 completion acknowledgment.

Timeouts of Request Messages from the Co-Network

Each time the BIC ISO Interface process sends a financial transaction to a BASE24 Authorization process for authorization, the BIC ISO Interface process starts an inbound request timer. If the inbound request timer expires before a response is received from the BASE24 network, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. The co-network sends the request to the BIC ISO Interface process in the 0100 or 0200 external message format.
2. The BIC ISO Interface process reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response from the Authorization process.
3. When the inbound request timer expires, the BIC ISO Interface process formats a 0110 or 0210 external response message denying the transaction and sends the 0110 or 0210 message to the co-network.
4. The BASE24 Authorization process sends a late 0210 response to the BIC ISO Interface process. If the transaction was denied by the BASE24 Authorization process, the BIC ISO Interface process drops the late response.
5. If the transaction was approved, the BIC ISO Interface process formats a 0420 internal reversal message and sends the reversal to a BASE24 Authorization process.

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6: Cutover and Reporting

This section describes BIC ISO Interface process cutover processing and report generation. This includes detailed descriptions of the cutover message flows. This section also provides samples of the reports generated for the BIC ISO Interface process.

Introduction

Within the BASE24 system, interchange cutover processing is a management and reporting function that produces reports for each business day. These reports are provided to aid in reconciling the BASE24 network with the co-network. Cutover processing occurs seven days a week.

Interchange cutover is initiated by the BASE24-atm Settlement Initiator process or the BASE24-pos Main Settlement process. As part of product cutover, the Settlement process sends the BIC ISO Interface process a cutover message. If only one BASE24 product is present, interchange cutover occurs at the same time as BASE24 cutover.

If both BASE24-atm and BASE24-pos are present, the BIC ISO Interface process determines when cutover should occur by comparing the cutover times set in the LCONF for BASE24-atm and BASE24-pos. For example, if BASE24-atm cuts over at 3:00 p.m. and BASE24-pos cuts over at 9:00 p.m., the BASE24-atm Settlement Initiator process sends the BIC ISO Interface process a cutover message at 3:00 p.m. When the BIC ISO Interface process receives the cutover message, it updates the BASE24-atm processing date and compares the two cutover times. Since BASE24-atm does not have the last cutover time for the day, the interchange is not cut over. At 9:00 p.m., the BASE24-pos Main Settlement process sends the BIC ISO Interface process a cutover message. When the BIC ISO Interface process receives the cutover message, it updates the BASE24-pos processing date and compares the two cutover times. Since BASE24-pos has the last cutover time, the interchange is also cut over.

Note: Both the BASE24-atm Settlement Initiator process and the BASE24-pos Main Settlement process are referred to as the “Settlement Initiator process” throughout the message flows presented in this section.

BIC ISO Totals

The BIC ISO Interface process maintains a number of running totals that reflect the approved transactions processed between the BASE24 network and the co-network since the last business day cutover. The totals maintained by the BIC ISO Interface process include the following:

Debit amount — The total amount of all debit transactions

Debit count — The total number of debit transactions

Credit amount — The total amount of all credit transactions

Credit count — The total number of credit transactions

Reversal debit amount — The total amount of all reversal debit transactions

Reversal debit count — The total number of all reversal debit transactions

Reversal credit amount — The total amount of all reversal credit transactions

Reversal credit count — The total number of all reversal credit transactions

Authorization count — The total number of authorization transactions

Inquiries count — The total number of inquiry transactions

Transfers count — The total number of transfer transactions

Transfers reversal count — The total number of transfer reversal transactions

Net settlement amount — The net settlement amount for the cutover date just completed.

These totals are kept for the current processing date, the previous processing date, and the next processing date. The totals kept by the BIC ISO Interface process are sent to the co-network in the 0500, 0502, 0520, and 0522 reconciliation messages and are used by the interchange report programs.

As each transaction is processed by the BIC ISO Interface process, the appropriate totals are updated based on the type of transaction and the posting date for the transaction. The following tables summarize how each transaction impacts the totals listed above. For the purposes of impacting totals, force post transactions are treated the same as normal transactions, and adjustments are treated the same as reversals. A key to notations used in each table is provided immediately following each table.

BASE24-atm Transactions

Transaction	Reconciliation Totals											
	Amounts				Counts							
	Debits	Credits	Debit Reversals	Credit Reversals	Debits	Credits	Debit Reversals	Credit Reversals	Inquiries	Transfers	Transfer Reversals	Authorizations
Withdrawal	O				+1							
Withdrawal Reversal	CA			O				+1				
Deposit		O				+1						
Deposit Reversal		CA	O				+1					
Deposit W/Cash Back	CB	O			+1	+1						
Deposit W/Cash Back Reversal (Entire Transaction)			O	CB			+1	+1				
Deposit W/Cash Back Reversal (Cash Back Amount Only)	RA			CB				+1				
Split Deposit		O				+1						
Transfer										+1		
Transfer Reversal											+1	
Payment From/To										+1		
Payment From/To Reversal											+1	
Payment Enclosed		O				+1						
Balance Inquiry									+1			

Key

O = Original transaction amount

CB = Cash back amount

CA = Actual completion amount of the transaction

RA = Replacement amount

+1 = Adds one to specified transaction count

BASE24-pos Transactions

Transaction	Reconciliation Totals											
	Amounts				Counts							
	Debits	Credits	Debit Reversals	Credit Reversals	Debits	Credits	Debit Reversals	Credit Reversals	Inquiries	Transfers	Transfer Reversals	Authorizations
Normal Purchase	O				+1							
Normal Purchase Reversal				O				+1				
Preauthorization												+1
Preauthorization Completion	CA				+1							
Preauthorization Completion Reversal				CA				+1				
Mail/Phone Order	O				+1							
Mail/Phone Order Reversal				O				+1				
Merchandise Return		O				+1						
Merchandise Return Reversal			O				+1					
Cash Advance	O				+1							

Transaction	Reconciliation Totals											
	Amounts				Counts							
	Debits	Credits	Debit Reversals	Credit Reversals	Debits	Credits	Debit Reversals	Credit Reversals	Inquiries	Transfers	Transfer Reversals	Authorizations
Cash Advance Reversal				O				+1				
Purchase W/Cash Back	T				+1							
Purchase W/Cash Back Reversal	CB			T				+1				
Purchase Adjustment	N	OC				+1						
Purchase Adjustment Reversal		N	OC				+1					
Merchandise Return Adjustment	OC	N			+1							
Merchandise Return Adjustment Reversal	N			OC				+1				
Cash Advance Adjustment	N	OC				+1						
Cash Advance Adjustment Reversal		N	OC				+1					
Check Guarantee	O				+1							
Check Verification	O				+1							
Balance Inquiry									+1			

Key

O = Original transaction amount

T = Total transaction amount—Purchase with cash back transaction (Purchase amount + cash back)

OC = Original completed amount—Adjustment transactions

Key

CA = Completed amount—Preauthorization completion transaction

CB = Cash back amount—Purchase with cash back transaction

N = New amount—Adjustment transactions

+1 = Adds one to the specified transaction count

Interchange Totals File (ITF)

The BIC ISO Interface process accumulates its reconciliation totals in memory during the business day. These accumulated totals are stored in the Interchange Totals File (ITF). The BIC ISO Interface process updates the ITF when it has processed the number of transactions specified in the ITF UPDATE MAX TRANSACTIONS field on ICFE screen 12 or when the number of minutes specified in the ITF UPDATE INTERVAL field on ICFE screen 12 have elapsed.

Each time the totals in the ITF are updated, the keys (i.e., addresses) of the last ILF records included in the accumulated totals are recorded in the ITF with the totals figures. (Since one ITF can represent totals from up to four ILFs, the totals are accompanied by up to four ILF dates as well as the record address of the last accumulated transaction in each of the ILFs.) During warmboot or process initialization, the BIC ISO Interface process locates the last accumulated transaction for each of the ILFs indicated in the ITF and reads from that record to the end of the file, updating the totals in memory for the previous, current, and next local BASE24 business days. In this way, the BIC ISO Interface process being restarted can rebuild its memory-resident totals without having to read any ILF in its entirety.

Logging Reversals to the ITF

Because the transaction that is being updated in place may have already been included in the last ITF update, a reversal could be missed in restart processing after a process failure. For this reason, a reversal causes all fields in the ITF to be updated.

ITF Status Flags

During cutover processing, the BIC ISO Interface process uses three status flags in the ITF to keep track of the steps that have been performed by the process during the cutover period. The use of these status flags enables the BIC ISO Interface process to recover in the event that it is stopped during the cutover period. The ITF status flags used by the BIC ISO Interface process are described as follows.

The ITF status flag tracks the overall progression of events during cutover processing. Valid values are as follows:

- 0 = Normal
- 1 = Waiting for cutover acknowledgment
- 2 = Cutover message placed in the Store-and-Forward File (SAF)
- 3 = Ready to send totals
- 4 = Totals complete

The acquirer status flag tracks the status of acquirer processing. Valid values are as follows:

- 0 = Normal
- 1 = Acquirer totals sent
- 2 = Acquirer settlement message placed in SAF
- 3 = Received acquirer acknowledgment

The issuer status flag tracks the status of issuer processing. Valid values are as follows:

- 0 = Normal
- 1 = Issuer totals sent
- 2 = Issuer settlement message placed in SAF
- 3 = Received issuer acknowledgment

Cutover Message Processing

The following paragraphs describe the processing performed by the BIC ISO Interface process when it receives a cutover message from the BASE24 network or the co-network.

BASE24 Settlement Initiator Process Sends Cutover Message

The following steps are taken by the BIC ISO Interface process when a cutover message is received from a BASE24 Settlement Initiator process. To cut over, the BIC ISO Interface process performs as follows:

1. Updates the current business date for the product whose Settlement Initiator process sent the cutover message.
2. Compares the cutover times found in the LCONF for the BASE24-pos and BASE24-atm products to determine if the BIC ISO Interface process should be cut over at this time. BIC ISO Interface process cutover occurs at the later of the two cutover times.

If the BIC ISO Interface process should not be cut over at this time, no further cutover steps are taken.

3. Checks to see if the cutover message has already been sent for this date by comparing the date sent in the cutover message with the previous cutover date the BIC ISO Interface process keeps in memory.

If the cutover message has already been sent, the BIC ISO Interface process has already been cut over and no further cutover steps are taken.

4. Updates the reconciliation totals that are kept in memory. To do this, the BIC ISO Interface process performs the following steps:
 - a. Moves the current day totals to the previous day totals.
 - b. Moves the next day totals to the current day totals.
 - c. Initializes the next day totals to zero.
5. Updates the reconciliation totals dates that are kept in memory. To do this, the BIC ISO Interface process performs the following steps:
 - a. Sets the previous reconciliation totals date to the current processing date.
 - b. Initializes the next reconciliation totals date to blanks.

6. Checks the ICFE information stored in the Process Control Table (PCT) to determine whether this BIC ISO Interface process is secondary to the co-network.

The secondary network is not responsible for sending a cutover message to the co-network. If this process is designated as a secondary network, no further cutover steps are taken.

7. Formats a cutover message and sends it to the co-network.
8. Determines whether this BIC ISO Interface process is in the main network of the co-network.

If this BIC ISO Interface process is not in the main network of the co-network, equal partner settlement has been performed and no further steps are needed.

If the BIC ISO Interface process is the main network, the BIC ISO Interface process opens a new ILF and begins logging transaction records to the new ILF.

Co-Network Sends Cutover Message

The following steps are taken by the BIC ISO Interface process when a cutover message is received from the co-network. When the cutover message is received, the BIC ISO Interface process performs as follows:

1. If this BASE24 network is the main network, no further steps are taken.
2. Determines if a cutover message has already been processed for the co-network's processing date by comparing the date from the external message with the co-network's posting date kept in the PCT. If so, a message is logged and no further cutover steps are taken.
3. Updates the current business date for the co-network.
4. Opens and begins logging transaction records to a new ILF.
5. Sends a cutover acknowledgment message to the co-network.

Cutover Message Flows

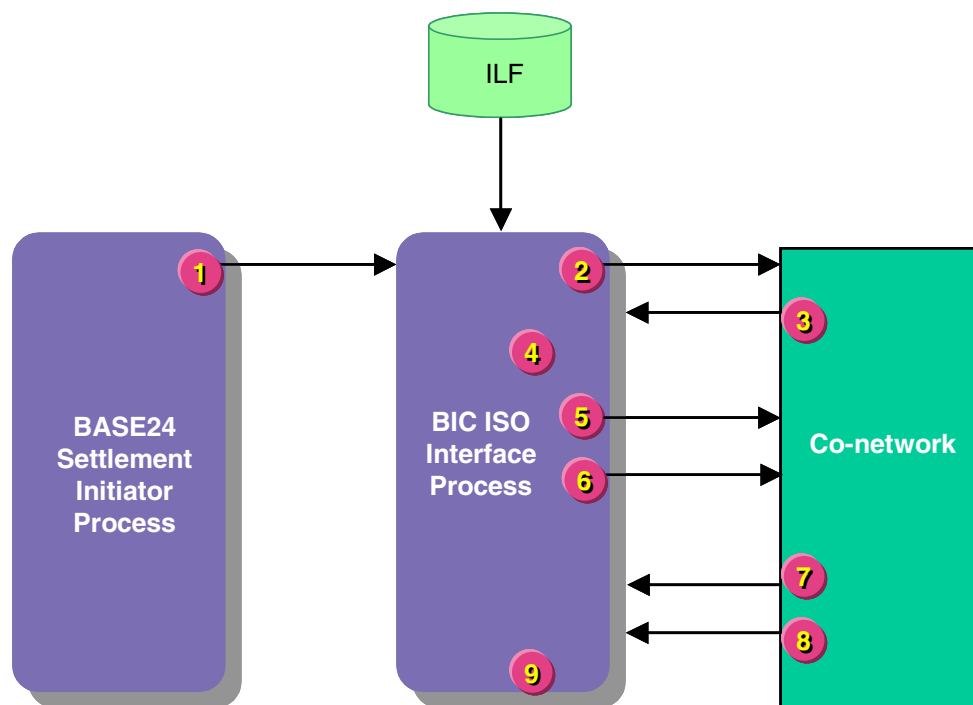
Cutover processing occurs five or seven days a week. The BIC ISO Interface process maintains an ILF, which reflects the transaction activity for the co-network's business day. For example, when the co-network cuts over to a new business day, the BIC ISO Interface process closes its ILF and opens a new ILF reflecting the current business date of the co-network.

When both the BASE24 network and the co-network are equal partners, each network informs the other network when cutover occurs. In addition, each network must send the other network reconciliation totals.

When both networks are not equal partners, BASE24 must be identified as either the main network or the secondary network in the ICFE. The co-network is either main or secondary, whichever the BASE24 network is not. Secondary network cutover must occur before the main network cutover. If it does not, incoming transactions from the secondary network are denied by the main network.

Equal Partner Reconciliation and BASE24 Cutover

When the BASE24 network and the co-network are equal partners regarding reconciliation and cutover, the same processing occurs for both. The diagram below illustrates the message flow when the BASE24 network cuts over. The steps in the diagram are described below.



1. The BIC ISO Interface process receives notification of cutover from the main Settlement Initiator process on its network. This is the last cutover message the BIC ISO Interface process receives for this business day; therefore, it is BASE24 Interchange node cutover.
2. The BIC ISO Interface process sends an 0800 message with an information code of 201 to the co-network. This 0800 message contains the business date for the processing day just ended. The BIC ISO Interface process sets the ITF status flag to a value of 1 (waiting for a cutover acknowledgment), sets a cutover acknowledgment timer, and waits for a cutover acknowledgment. All new requests are denied until the BIC ISO Interface process receives a cutover acknowledgment message.

If no stations are available or if the cutover message times out, the cutover message is placed in the Store-and-Forward File (SAF) as an 0820 network management advice message. If this occurs, the BIC ISO Interface process sets the ITF status flag to a value of 2 (cutover message placed in SAF).

3. The co-network sends an 0810 cutover acknowledgment message to the BIC ISO Interface process with the information code set to 201. The BIC ISO Interface process logs a message stating that the cutover acknowledgment message has been received, deletes the cutover acknowledgment timer, and sets the ITF status flag to a value of 3 (ready to send totals).

In the event the co-network responds late with the 0810 network management response to a cutover request, it is not processed. The BIC ISO Interface process sends out the 0820 advice message and the co-network sends back an 0830 advice response message. When the BIC ISO Interface process receives the 0830 response, it changes the ITF status flag to a value of 3 (ready to send totals).

4. The BIC ISO Interface process sets a settlement interval timer to expire. This allows all of the transactions for the old business day to finish. This time interval is from the SETTLEMENT TIMER field on ICFE screen 12.
5. When the settlement interval timer expires, reconciliation totals are sent. The BIC ISO Interface process sends a 0500 acquirer reconciliation totals message to the co-network and logs a message stating that the acquirer reconciliation totals were sent. The BIC ISO Interface process sets the acquirer status flag in the ITF to a value of 1 (acquirer totals sent), sets a network management timer, and waits for an acquirer settlement acknowledgment message.

If no stations are available or if the acquirer reconciliation totals message times out, the acquirer reconciliation message is placed in the Store-and-Forward File (SAF). If this occurs, the BIC ISO Interface process sets the acquirer status flag in the ITF to a value of 2 (acquirer settlement message placed in the SAF). The message type is changed to indicate that this reconciliation message is a store-and-forward advice message. That is, a 0500 acquirer reconciliation message is changed to a 0520 acquirer reconciliation advice message.

6. The BIC ISO Interface process sends a type 0502 issuer reconciliation totals message to the co-network and logs another message stating that the issuer reconciliation totals were sent. The BIC ISO Interface process sets the issuer status flag in the ITF to a value of 1 (issuer totals sent), sets a network management timer, and waits for an issuer settlement acknowledgment message.

Both the 0500 and 0502 totals messages contain totals for all shared activity for the BASE24 date that is being cut over.

If no stations are available or if the issuer reconciliation message times out, the issuer reconciliation message is placed in the Store-and-Forward File (SAF). If this occurs, the BIC ISO Interface process sets the issuer status flag in the ITF to a value of 2 (issuer settlement message placed in SAF). The

message type is changed to indicate that this reconciliation message is a store-and-forward advice message. That is, the 0502 issuer reconciliation message is changed to a 0522 issuer reconciliation advice message.

7. The co-network replies with a type 0510 acquirer reconciliation acknowledgment message. Upon the receipt of the 0510 message, the BIC ISO Interface process finds and deletes the acquirer network management timer and sets the acquirer status flag in the ITF to 3 (acquirer acknowledgment received).

In the event the co-network responds late with the 0510 acquirer reconciliation response message, it is not processed. The BIC ISO Interface process sends out a 0520 acquirer reconciliation advice message and the co-network sends back a 0530 advice response message. When the BIC ISO Interface process receives the 0530 response, it changes the acquirer status flag in the ITF to a value of 3 (received acquirer acknowledgment).

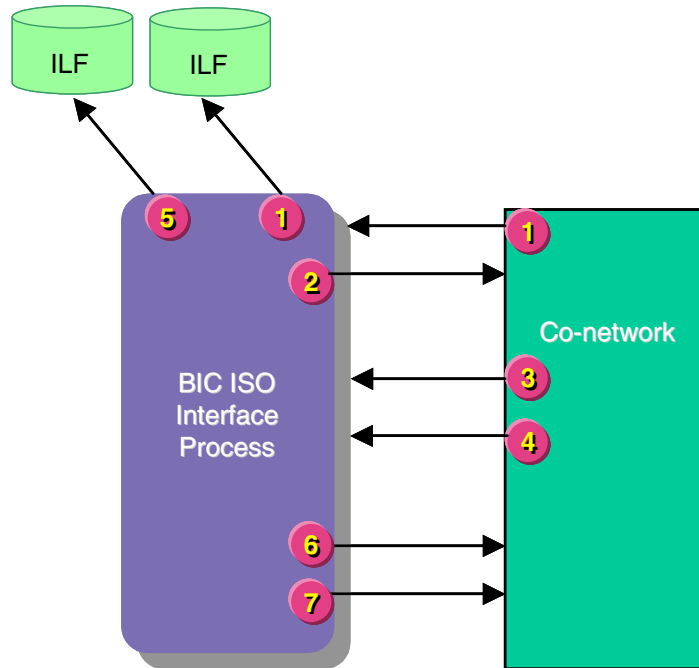
8. The co-network replies with a type 0512 issuer reconciliation acknowledgment message. Upon the receipt of the 0512 message, the BIC ISO Interface process finds and deletes the issuer network management timer and sets the issuer status flag in the ITF to a value of 3 (issuer acknowledgment received).

In the event the co-network responds late with the 0512 issuer reconciliation response message, it is not processed. The BIC ISO Interface process sends out a 0522 issuer reconciliation advice message, and the co-network sends back a 0532 advice response message. When the BIC ISO Interface process receives the 0532 response, it changes the issuer status flag in the ITF to a value of 3 (received issuer acknowledgment).

9. The BIC ISO Interface process checks the issuer and acquirer status flags in the ITF to see if they are both set to a value of 3 (acknowledgments received). If they are both set to a value of 3, the BIC ISO Interface process sets the ITF status flag to a value of 4 (totals complete).

Equal Partner Reconciliation and Co-Network Cutover

When the BASE24 network and the co-network are equal partners regarding reconciliation and cutover, the same processing occurs for both. The diagram below illustrates the message flow when the co-network cuts over. The steps in the diagram are described below.



1. The co-network sends an 0800 message with an information code of 201 to the BIC ISO Interface process. This 0800 message contains the business date for the processing day just ended.

The BIC ISO Interface process creates and opens a new Interchange Log File (ILF) the first time it receives the business date for the processing day just ended from the co-network. When the BIC ISO Interface process receives the 0800 cutover message, it creates a new ILF with the new business date, if this has not yet been done. The current day's ILF becomes the previous day's ILF, and the next day's ILF becomes the current day's ILF. In addition to the ILF parameters in memory and in the ICFE, the BIC ISO Interface process also updates the ILF information in the Interchange Totals File (ITF): ILF2 information moves to ILF1, ILF3 information moves to ILF2, and ILF4 information moves to ILF3. A new date is calculated for ILF4, but the file is not created until the first transaction for that date arrives. During this update of the ITF, the totals update is also performed.

2. The BIC ISO Interface process sends an 0810 cutover acknowledgment message to the co-network with the information code set to 201.

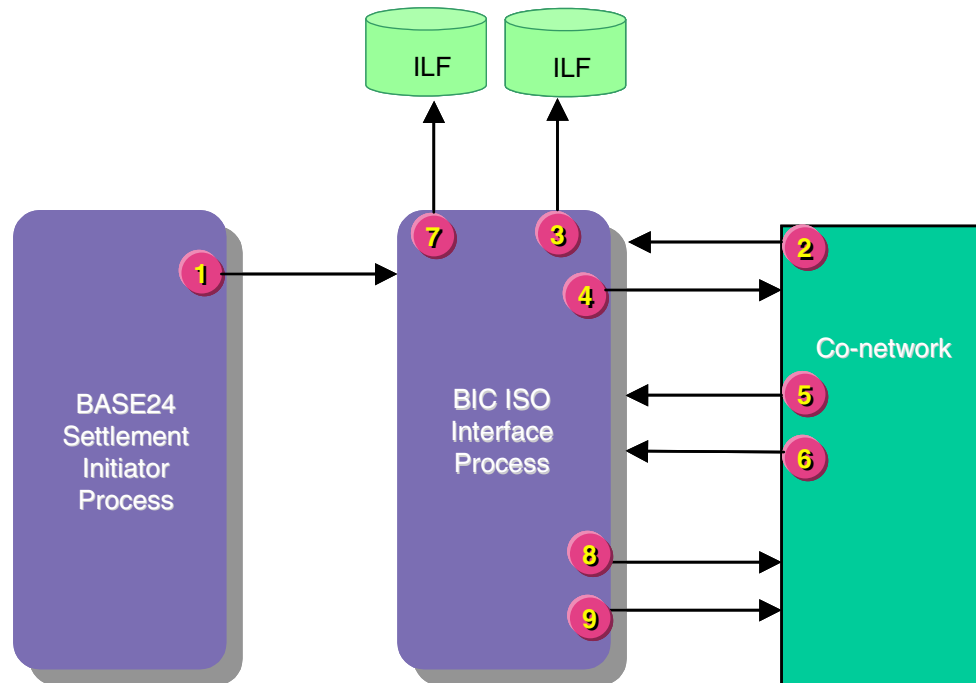
3. At some point, the co-network sends reconciliation totals to the BIC ISO Interface process. The co-network sends a type 0500 acquirer reconciliation totals message to the BIC ISO Interface process.
4. The co-network sends a type 0502 issuer reconciliation totals message to the BIC ISO Interface process.

Both the 0500 and 0502 totals messages contain totals for all shared activity for the co-network date that is being cut over.

5. The BIC ISO Interface process logs the acquirer reconciliation totals to the ILF with a record type of 90 for the business day just closed. The issuer reconciliation totals are also logged to the same ILF with a record type of 91. After the BIC ISO Interface process receives both acquirer and issuer totals, it closes the ILF. The BIC ISO Interface process checks to make sure that no duplicate reconciliation records are logged to the ILF.
6. The BIC ISO Interface process replies with a type 0510 acquirer reconciliation acknowledgment message.
7. The BIC ISO Interface process replies with a type 0512 issuer reconciliation acknowledgment.

Main and Secondary Reconciliation when the BASE24 Network is Secondary

When the two networks are not equal partners regarding reconciliation and cutover, the secondary network must cut over before the main network. The diagram below shows the message flow when the BASE24 network is the secondary network. The steps in the diagram are described below.



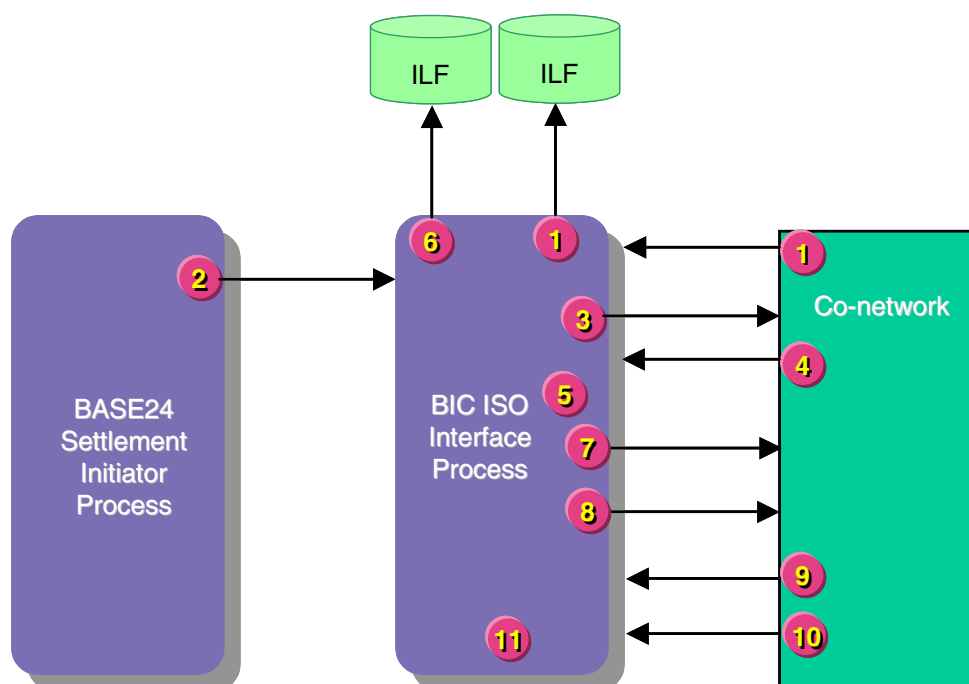
1. The BIC ISO Interface process receives notification of cutover from the main Settlement Initiator process on its network. This is the last cutover message the BIC ISO Interface process receives for this business day; therefore, it is BASE24 Interchange node cutover.
2. At some later time, the co-network cuts over and sends an 0800 message with the information code set to 201 to the BIC ISO Interface process. This 0800 message contains the business date for the processing day just ended.
3. The BIC ISO Interface process creates and opens a new Interchange Log File (ILF) the first time it receives the business date for the processing day just ended from the co-network. When the BIC ISO Interface process receives the 0800 cutover message, it creates a new ILF with the new business date, if this has not yet been done. The current day's ILF becomes the previous day's ILF, and the next day's ILF becomes the current day's ILF. In addition to the ILF parameters in memory and in the ICFE, the BIC ISO Interface process also updates the ILF information in the ITF: ILF2 information moves to ILF1, ILF3 information moves to ILF2, and ILF4 information moves to ILF3. A

new date is calculated for ILF4, but the file is not created until the first transaction for that date arrives. During this update of the ITF, the totals update is also performed.

4. The BIC ISO Interface process sends an 0810 cutover acknowledgment with the information code set to 201 to the co-network.
5. At some point, the co-network sends reconciliation totals to the BIC ISO Interface process. The co-network sends a type 0500 acquirer reconciliation totals message to the BIC ISO Interface process.
6. The co-network also sends a type 0502 issuer reconciliation totals message to the BIC ISO Interface process.
7. The BIC ISO Interface process logs the acquirer reconciliation totals to the ILF with a record type of 90 for the business day just closed. The issuer reconciliation totals are also logged to the same ILF with a record type of 91. After the BIC ISO Interface process receives both acquirer and issuer totals, it closes the ILF. The BIC ISO Interface process checks to make sure that no duplicate reconciliation records are logged to the ILF.
8. The BIC ISO Interface process replies with a type 0510 acquirer reconciliation acknowledgment message.
9. The BIC ISO Interface process also replies with a type 0512 issuer reconciliation acknowledgment message.

Main and Secondary Reconciliation when the BASE24 Network is Main

When the two networks are not equal partners regarding reconciliation and cutover, the secondary network must cut over before the main network. The diagram below shows the message flow when the BASE24 network is the main network. The steps in the diagram are described below.



1. The co-network cuts over. At some point after cutover, a 0200 financial request message is sent to the BIC ISO Interface process containing the business date for the processing day just started. The BIC ISO Interface process opens a new Interchange Log File (ILF) when it receives the business date for the processing day just started.

The current day's ILF becomes the previous day's ILF, and the next day's ILF becomes the current day's ILF. In addition to the ILF parameters in memory and in the ICFE, the BIC ISO Interface process also updates the ILF information in the Interchange Totals File (ITF): ILF2 information moves to ILF1, ILF3 information moves to ILF2, and ILF4 information moves to ILF3. A new date is calculated for ILF4, but the file is not created until the first transaction for that date arrives. During this update of the ITF, the totals update is also performed.

2. The BIC ISO Interface process receives notification of cutover from the main Settlement Initiator process on its network. This is the last cutover message the BIC ISO Interface process receives for this business day; therefore, it is BASE24 Interchange node cutover.
3. The BIC ISO Interface process sends an 0800 message with the information code set to 201 to the co-network. This 0800 message contains the business date for the processing day just ended. The BIC ISO Interface process sets the ITF status flag to a value of 1 (waiting for a cutover acknowledgment), sets a network management timer, and waits for a cutover acknowledgment. All new requests are denied until the BIC ISO Interface process receives a cutover acknowledgment message.

If no stations are available or if the cutover message times out, the cutover message is placed in the Store-and-Forward File (SAF) as an 0820 network management advice message. If this occurs, the BIC ISO Interface process sets the ITF status flag to a value of 2 (cutover message placed in SAF).

4. The co-network sends an 0810 cutover acknowledgment message to the BIC ISO Interface process with the information code set to 201. The BIC ISO Interface process logs a message stating that the cutover acknowledgment message has been received, deletes the network management timer, and sets the ITF status flag to a value of 3 (ready to send totals).

In the event the co-network responds late with the 0810 network management response to a cutover request, it is dropped. The BIC ISO Interface process sends out an 0820 advice message, and the co-network sends back an 0830 advice response message. When the BIC ISO Interface process receives the 0830 response, it changes the ITF status flag to a value of 3 (ready to send totals).

5. The BIC ISO Interface process sets a settlement interval timer to expire. This allows all of the transactions for the old business day to finish. This time interval is from the SETTLEMENT TIMER field on ICFE screen 12.
6. The BIC ISO Interface process closes the ILF.
7. When the settlement interval timer expires, the BIC ISO Interface process sends reconciliation totals to the co-network. The BIC ISO Interface process sends a 0500 acquirer reconciliation totals message to the co-network and logs a message stating that the acquirer reconciliation totals were sent. The BIC ISO Interface process sets the acquirer status flag in the ITF to a value of 1 (acquirer totals sent), sets a network management timer, and waits for an acquirer settlement acknowledgment message.

If no stations are available or if the acquirer reconciliation totals message times out, the acquirer reconciliation message is placed in the Store-and-Forward File (SAF). If this occurs, the BIC ISO Interface process sets the

acquirer status flag in the ITF to a value of 2 (acquirer settlement message placed in SAF). The message type is changed to indicate that this reconciliation message is a store-and-forward advice message. That is, the 0500 acquirer reconciliation message is changed to a 0520 acquirer reconciliation advice message.

8. The BIC ISO Interface process also sends a type 0502 issuer reconciliation totals message to the co-network and logs another message stating that the issuer reconciliation totals were sent. The BIC ISO Interface process sets the issuer status flag in the ITF to a value of 1 (issuer totals sent), sets a network management timer, and waits for an issuer settlement acknowledgment message.

Both the 0500 and 0502 totals messages contain totals for all shared activity for the BASE24 date that is being cut over.

If no stations are available or if the issuer reconciliation message times out, the issuer reconciliation message is placed in the Store-and-Forward File (SAF). If this occurs, the BIC ISO Interface process sets the issuer status flag in the ITF to a value of 2 (issuer settlement message placed in SAF). The message type is changed to indicate that this reconciliation message is a store-and-forward advice message. That is, the 0502 issuer reconciliation message is changed to a 0522 issuer reconciliation advice message.

9. The co-network replies with a type 0510 acquirer reconciliation acknowledgment message. Upon the receipt of the 0510 message, the BIC ISO Interface process finds and deletes the acquirer network management timer and sets the acquirer status flag in the ITF to a value of 3 (acquirer acknowledgment received).

In the event the co-network responds late with the 0510 acquirer reconciliation response message, it is dropped. The BIC ISO Interface process sends out a 0520 acquirer reconciliation advice message and the co-network sends back a 0530 advice response message. When the BIC ISO Interface process receives the 0530 response, the acquirer status flag in the ITF is changed to a value of 3 (received acquirer acknowledgment).

10. The co-network also replies with a type 0512 issuer reconciliation acknowledgment. Upon the receipt of the 0512 message, the BIC ISO Interface process finds and deletes the issuer network management timer and sets the issuer status flag in the ITF to a value of 3 (issuer acknowledgment received).

In the event the co-network responds late with the 0512 issuer reconciliation response message, it is dropped. The BIC ISO Interface process sends out a 0522 issuer reconciliation advice message, and the co-network sends back a

0532 advice response message. When the BIC ISO Interface process receives the 0532 response, it changes the issuer status flag in the ITF to a value of 3 (received issuer acknowledgment).

11. The BIC ISO Interface process checks the issuer and acquirer status flags to see if they are both set to 3 (acknowledgments received). If they are both equal to a value of 3, the BIC ISO Interface process sets the ITF status flag to a value of 4 (totals complete).

Starting Reports

The BIC ISO Interface process is responsible for starting the interface reports. It does so by creating an obey file to start the report programs and then running the obey file automatically at the appropriate time.

Using the time specified in the LCONF param SW-RPT-STARTUP, the BIC ISO Interface process sets a timer to expire when the reports are to be run. When the timer expires, the BIC ISO Interface process creates and runs an obey file containing the Tandem Advanced Command Language (TACL) commands necessary to run the report programs.

Under normal circumstances, the report programs start and stop without operator intervention. However, if the BIC ISO Interface process does not start the programs, they can be started manually by running the report obey file created by the BIC ISO Interface process from a TACL prompt.

If the report programsabend or fail, they can be restarted in the same way.

Report Overview

The BIC ISO Interface process produces two reports of interchange transaction activity. These reports are the SETL01-BIS Clearings Statement Report and the STAT09-BIC Transaction Detail Report. These reports can be used in conjunction with BASE24-atm and BASE24-pos reports to reconcile the differences between the BASE24 processing day and the co-network processing day.

This section provides an overview of the information provided in the interchange reports. Sample reports are also provided.

SETL01-BIS Clearings Statement Report for BIC ISO

The Clearings Statement Report for BIC ISO reflects the debit, credit, and net position of the co-network to the BASE24 network.

This report contains the totals of all transactions logged to the ILF. Reconciliation records are logged to the ILF with a record type of 90 for acquirer reconciliation records and a record type of 91 for issuer reconciliation records.

The BIC ISO Interface process calculates totals as an acquirer and as an issuer, and the transaction type determines whether it is a debit or a credit. Withdrawals and normal purchases are always counted as debits, and deposits are always treated as credits.

The report includes a section for transactions that occurred during the BASE24 processing day that did not occur during the co-network processing day. These transactions are subtracted from the transaction totals for the BASE24 processing day. This section provides two pieces of information: the net amount of transactions and the dollar difference between transactions occurring during the BASE24 processing day that did not occur during the co-network processing day.

Next, a section shows transactions that occurred during the co-network processing day that did not occur during the BASE24 processing day. These transactions are added to the transaction totals for the BASE24 processing day. This section provides two pieces of information: the net amount of transactions and the dollar difference between transactions occurring during the co-network processing day that did not occur during the BASE24 processing day.

Once the report program has calculated the debit and credit positions, it compares the report credits and debits to the reconciliation records. If there are any discrepancies, a separate page is printed showing the calculated figures and all the figures received from the co-network.

A sample of the SETL01-BIS Clearings Statement Report for BIC ISO is shown on the following pages, followed by descriptions of the fields that appear on the report.

SETL01-BIS				CLEARINGS STATEMENT REPORT FOR BIC ISO 1				DATE YY/MM/DD	
								PAGE 1	
				TRAN DATE	DEBIT	CREDIT	NET		
***** PRO1 TODAY *****				YY/MM/DD	300.00	.00	300.00-		
PRO1									
** MINUS INTERCHANGE NOT-TODAY **									
SETTLEMENT SUBTOTAL				YY/MM/DD	300.00	.00	300.00-		
**** ADD INTERCHANGE TODAY ****									
SETTLEMENT SUBTOTAL				YY/MM/DD	300.00	.00	300.00-		

SETL01-BIS		CLEARINGS STATEMENT REPORT FOR BIC ISO 1		DATE YY/MM/DD	
				PAGE	
				2	
		TRAN DATE	DEBIT	CREDIT	NET
***** PRO2 TODAY *****		YY/MM/DD	450.00	.00	450.00-
PRO2					
** MINUS INTERCHANGE NOT-TODAY **					
SETTLEMENT SUBTOTAL		YY/MM/DD	450.00	.00	450.00-
**** ADD INTERCHANGE TODAY ****					
SETTLEMENT SUBTOTAL		YY/MM/DD	450.00	.00	450.00-

CLEARINGS STATEMENT REPORT FOR BIC ISO 1				DATE YY/MM/DD
				PAGE 3
SETL01-BIS	TRAN DATE	DEBIT	CREDIT	NET
***** PRO3 TODAY *****				
PRO3	YY/MM/DD	50.00	.00	50.00-
** MINUS INTERCHANGE NOT-TODAY **				
SETTLEMENT SUBTOTAL	YY/MM/DD	50.00	.00	50.00-
**** ADD INTERCHANGE TODAY ****				
SETTLEMENT SUBTOTAL	YY/MM/DD	50.00	.00	50.00-

CLEARINGS STATEMENT REPORT FOR BIC ISO 1				DATE YY/MM/DD
				PAGE 4
TRAN DATE				NET
***** SPAN TODAY *****				
SPAN	YY/MM/DD	800.00	.00	800.00-
** MINUS INTERCHANGE NOT-TODAY **				
SETTLEMENT SUBTOTAL	YY/MM/DD	800.00	.00	800.00-
**** ADD INTERCHANGE TODAY ****				
SETTLEMENT SUBTOTAL	YY/MM/DD	800.00	.00	800.00-

SETL01-BIS				CLEARINGS STATEMENT REPORT FOR BIC ISO 1				DATE YY/MM/DD	
								PAGE	
								5	

CLEARINGS EXCEPTION REPORT									
***** COMPUTED TOTALS *****									
DEBITS		CREDITS		RVSL/ADJ DEBITS		RVSL/ADJ CREDITS			
COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT
23	1,150.00	7	350.00	0	.00	7	.00	7	350.00

***** ACQUIRER SETTLEMENT TOTALS *****									
DEBITS		CREDITS		RVSL/ADJ DEBITS		RVSL/ADJ CREDITS			
COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT
15	600.00	0	.00	0	.00	0	.00	0	.00

***** ISSUER SETTLEMENT TOTALS *****									
DEBITS		CREDITS		RVSL/ADJ DEBITS		RVSL/ADJ CREDITS			
COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT	COUNT	AMOUNT
0	.00	0	.00	0	.00	0	.00	0	.00

***** END OF REPORT *****									

DATE — The reporting date (YY/MM/DD) to which these reconciliation figures apply.

<LNET> TODAY

The settlement position of the logical network to the co-network for the day this report is produced.

TRAN DATE — The Interchange Log File (ILF) date (YY/MM/DD) to which these transactions belong.

DEBIT — The dollar amount the logical network owes the co-network according to the co-network. Co-network debits include the following:

- Deposits by co-network cardholders at terminals owned by a BASE24 institution.
- Withdrawals and cash advances by cardholders belonging to a BASE24 institution at co-network terminals.

CREDIT — The dollar amount the co-network owes the logical network according to the co-network. Co-network credits include the following:

- Withdrawals and cash advances by co-network cardholders at terminals owned by a BASE24 institution.
- Deposits by cardholders belonging to a BASE24 institution at co-network terminals.

NET — The net position of the logical network to the co-network according to the co-network, (credits minus debits). A minus sign (–) indicates a debit position, or the amount owed to the co-network.

MINUS INTERCHANGE NOT-TODAY

The debits, credits, and net amounts not considered part of the co-network reporting day. These amounts are considered part of the BASE24 network reporting day.

SETTLEMENT SUBTOTAL — The debit, credit, and net figures not considered part of the co-network reporting day are subtracted from today's business. The result is a reconciliation subtotal for all activity considered as today's business by the co-network.

ADD INTERCHANGE TODAY

The debit, credit, and net amounts considered as part of the co-network reporting day. These amounts are not considered part of the BASE24 network reporting day.

SETTLEMENT SUBTOTAL — The debit, credit, and net figures considered part of the co-network reporting day are added to today's business. The result is a reconciliation subtotal for all activity considered as today's business by the co-network.

SPAN TODAY

This section summarizes the net position of the logical network to the co-network. However, if there are multiple logical networks present in the BASE24 system, this section summarizes the net position of each logical network to the co-network.

CLEARINGS EXCEPTION REPORT — The following field descriptions describe the information that appears on the SETL01 report if there are any discrepancies between the credits and debits calculated by the BIC ISO Interface process and the reconciliation records received from the co-network.

COMPUTED TOTALS

This section summarizes the totals computed by the BIC ISO Interface process.

DEBITS COUNT

DEBITS AMOUNT — The total number and amount of all debit transactions.

CREDITS COUNT

CREDITS AMOUNT — The total number and amount of all credit transactions.

RVSL/ADJ DEBITS COUNT

RVSL/ADJ DEBITS AMOUNT — The total number and amount of all reversal debit and adjustment transactions.

RVSL/ADJ CREDITS COUNT

RVSL/ADJ CREDITS AMOUNT — The total number and amount of all reversal credit and adjustment transactions.

ACQUIRER SETTLEMENT TOTALS

This section summarizes the acquirer settlement totals.

DEBITS COUNT

DEBITS AMOUNT — The total number and amount of all acquirer debit transactions.

CREDITS COUNT

CREDITS AMOUNT — The total number and amount of all acquirer credit transactions.

RVSL/ADJ DEBITS COUNT

RVSL/ADJ DEBITS AMOUNT — The total number and amount of all acquirer reversal debit and adjustment transactions.

RVSL/ADJ CREDITS COUNT

RVSL/ADJ CREDITS AMOUNT — The total number and amount of all acquirer reversal credit and adjustment transactions.

ISSUER SETTLEMENT TOTALS

This section summarizes the issuer settlement totals.

DEBITS COUNT

DEBITS AMOUNT — The total number and amount of all issuer debit transactions.

CREDITS COUNT

CREDITS AMOUNT — The total number and amount of all issuer credit transactions.

RVSL/ADJ DEBITS COUNT

RVSL/ADJ DEBITS AMOUNT — The total number and amount of all issuer reversal debit and adjustment transactions.

RVSL/ADJ CREDITS COUNT

RVSL/ADJ CREDITS AMOUNT — The total number and amount of all issuer reversal credit and adjustment transactions.

STAT09-BIC Transaction Detail Report

The BIC Transaction Detail Report provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the BIC ISO Interface process during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts.

This report also includes a summary of the transaction detail information contained in the report.

A sample of the STAT09-BIC Transaction Detail Report is shown on the following pages, followed by descriptions of the fields that appear on the report.

[illegible]

STAT09-BIC	TRANSACTION DETAIL REPORT FOR BIC ISO 1	DATE YY/MM/DD
		PAGE 02
ATM TRANSACTION SUMMARY		
CHECK GUAR	0	.00
CHECK VERIFY	0	.00
WITHDRAWALS	0	.00
DEPOSITS	0	.00
BALANCE INFO	0	.00
TRANSFERS	0	.00
PAYMENTS	0	.00
TOTALS	0	.00

DATE — The reporting date, in YY/MM/DD format, to which these settlement figures apply.

CARDHOLDER NUMBER — The number identifying the cardholder who initiated this transaction.

TRAN DESC — A brief description of the transaction.

PST DAT — The Interchange Log File (ILF) date assigned to this transaction according to the interchange reporting day.

B24 DAT-TIM — The date and time this transaction occurred according to the BASE24 reporting day. This date determines whether this transaction is considered to be part of today's business.

B24 RESP — The first digit indicates the entity responsible for authorizing this transaction. The next three digits indicate the response action taken for this transaction.

CARD LN — The logical network to which this cardholder belongs.

ATM FROM ACCT #/POS ACCT # — The account accessed in this transaction. This field applies to both BASE24-atm and BASE24-pos transactions.

TRAN/BAL/ORIG AMOUNT — The transaction amount if this is a BASE24-atm transaction. If the transaction is a balance inquiry, this field contains the balance amount. If this is a BASE24-pos transaction, this field contains the original amount.

TRM LN — The logical network to which this terminal belongs.

TRM ID — The identification number of this terminal.

SEQUENCE # — The sequence number the terminal assigned to this transaction.

TRACE # — An additional identification number the interchange assigned to the transaction.

SWI DAT-TIM — Date and time of this transaction according to the interchange reporting day.

SWI RESP — Response code indicating the action taken for this transaction, assigned by the interchange.

RV CDE — An identifier indicating the reversal code if this transaction was reversed. If this transaction was not reversed, this field is blank.

ATM TO ACCT #/POS APPR CODE — The account to which this transaction was made if this is a BASE24-atm transaction. If this is a BASE24-pos transaction, this field contains the approval code.

COMPLETED AMOUNT — The total amount of the completed transaction.

ATM TRM OWNER/POS RETLR — The terminal-owning institution if this is a BASE24-atm transaction. If this a BASE24-pos transaction, this field contains the name of the retailer who owns this terminal.

TRM NAME-LOCATION — The terminal name as assigned by the terminal-owning institution and its location.

TRM CITY — The city in which this terminal is located.

TRM ST — The state in which this terminal is located.

PRD — The product identifier indicating what type of product initiated this transaction (i.e., BASE24-atm or BASE24-pos).

B24 TYP — The message type the BASE24 system assigned to this transaction.

SWI TYP — The message type the interchange assigned to this transaction.

SPLIT DEP AMT 1 — The first amount involved in a split deposit transaction.

SPLIT DEP AMT 2 — The second amount involved in a split deposit transaction.

MICR — The magnetic ink character recognition (MICR) data from the check involved in this transaction.

TRANSACTION SUMMARY — A summary page of all approved and denied BASE24 transactions by transaction type. The summary is divided into sections for BASE24-atm and BASE24-pos transactions.

ACI Worldwide, Inc.

A: BIC ISO Interface Process Startup and Control

This appendix provides information on the configuration of the BIC ISO Interface process. It also describes the processing involved for initialization and network control.

Startup

The BIC ISO Interface process should be assigned a startup logic of AUTOMATIC in the XPNET system. AUTOMATIC means that the BIC ISO Interface process is started automatically when the XPNET node is initiated. This ensures that the BIC ISO Interface process is ready when it receives its first transaction.

Event Messages

Event messages are routed to certain destinations, based on the routing code that appears in the event messages. The BIC ISO Interface process routes all of its event messages with a standard routing code of zero.

The BIC Interface process generates event messages to assist network operators in performing their daily tasks. Event messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access event message documentation electronically. Event messages are contained in files on a designated HP NonStop volume and subvolume. Each file contains event messages for a specific BASE24 process. Files on the designated HP NonStop volume and subvolume are organized according to product module ID (PMID). One PMID exists for every BASE24 process. PMIDs identify a specific process in the BASE24 system (e.g., SWBICI identifies the BIC ISO Interface process). Providing event message documentation in files on the HP NonStop computer gives customers greater flexibility by allowing them to print event message documentation as needed at their location to suit their requirements.

Documentation is available in file form for the event messages generated by the BIC ISO Interface process. It is located on the HP NonStop SWxxBABI subvolume in a file named BABILOGM, where xx is the number of the current release. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

Initialization

The following steps are taken by the BIC ISO Interface process when it is first started. To initialize, the BIC ISO Interface process performs as follows:

1. Reads the name of its LCONF. The startup message from the XPNET process contains the name of the LCONF. This name allows the BIC ISO Interface process to access the LCONF for the assigns and params it needs for processing.
2. Opens the LCONF and reads the following assigns and params. If an error occurs, the BIC ISO Interface process generates an event message and abends.

Assigns	Params
APCFEMT	ATM-LN-CUTOVER
EMF	ACCT-NUM-TYP
ICF	ATM-DISCR-DATA-STM-SUPPRS
ICFE	CURRENCY CODE
ILF	ENHANCED-CRNCY-CONV
IPCFEMT	ENHANCED-EMV-RC-MAPPING
ITF	DATE-DISPLAY-TYPE
KEY6	DES-TRIPLE-SINGLE
KEYF	ILF-NOTIFY-INTERVAL
LMCF	ILF-NOTIFY-PERCENTAGE
RPT	OVRRD-INST-ID-FLDS
RPT-ERROR-FILE	POS-DISCR-DATA-PSTM-SUPPRS
SAF	POS-LN-CUTOVER
SW-RPT-OBEY-FILE	POS-B24-TO-POS-ACCT-TYP-SUPPRS
SW-RPT01-PROG-NAME	PRE-VERIFIED-PIN-PAD-CHAR
SW-RPT09-PROG-NAME	PSEUDO-3DES-CBC-MAC-TYPE

Assigns	Params
TKN	RACAL-ANSI-METHOD REVERSAL-BAL-INQ RSM-TERM-MASTER-KEY SAF-NOTIFY-INTERVAL SAF-NOTIFY-PERCENTAGE SEND-ETX SPAN-FLAG SWI-ACQ-AVS-DATA-SUPPRS SWI-ACQ-CVD2-DATA-SUPPRS SWI-ACQ-DISCR-DATA-SUPPRS SWI-ISS-AVS-DATA-SUPPRS SWI-ISS-CVD2-DATA-SUPPRS SWI-ISS-DISCR-DATA-SUPPRS SW-ILF-FILE-RETENTION SW-RPT-STARTUP UPDT-ILF-ON-LATE-RESP

3. Reads the LMCF for its modified event messages. The LMCF contains a record for each modified event message in the system. The BIC ISO Interface process opens the LMCF, retrieves its modified event messages, and closes the LMCF. Whenever the BIC ISO Interface process generates an event message, it checks the LMCF records in memory to determine whether the message requires modification.
4. Opens the the Enhanced Interchange Configuration File (ICFE). If an error occurs, the BIC ISO Interface process logs an error message and abends.
5. Using its process name, reads its own ICFE record.
6. Opens the Token File (TKN) if a TKN assign was found in the LCONF.
7. Initializes the extended memory required for this process.

8. Initializes its timer table.
9. Opens the Key File (KEYF) or Key 6 File (KEY6). If an error occurs, the BIC ISO Interface process logs an error message and abends.
10. Using its process name, reads its own KEYF or KEY6 record.
11. Builds the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICFE and KEYF, and includes all of the controls needed to process transactions.
12. Opens the External Message File (EMF), reads the appropriate records, and then closes the EMF. If an error occurs, the BIC ISO Interface process logs an error message and abends.
13. Reads the appropriate TKN records into extended memory.
14. Opens the Interchange Log Files (ILFs) for the previous day, the current day, and the next day. If an error occurs, the BIC ISO Interface process logs an error message and abends.
15. Initializes the performance counters if the NET24 Performance Monitoring subsystem is being used.
16. Opens the Store-and-Forward File. If an error occurs, the BIC ISO Interface process logs an error message and abends.
17. If the BIC ISO Interface process is not a secondary processor, the Interchange Totals File (ITF) is opened, initialized, and updated.
18. Determines the posting dates for the BASE24-atm product, the BASE24-pos product, and the co-network by comparing the current time to the BASE24-atm and BASE24-pos cutover times taken from the LCONF.
19. Updates the ICFE with the current posting dates and then closes the ICFE.
20. Initializes a security device if applicable.
21. Reads the token log records into extended memory.
22. If reports are required, starts a timer to expire when the reports are to be run. If no reports are to be generated, the BIC ISO Interface process logs an error message and abends.
23. Starts a performance timer.
24. Starts the ITF interval timer if the BIC ISO Interface process is set up to keep totals.

25. If the automatic logon option was selected, sends a logon message to the co-network.
 - a. Sends a MAC and PIN key change messages.
 - b. Starts timers for PIN and MAC key changes.
26. Closes the LCONF and waits for a message from the co-network or a local BASE24 Authorization process.
27. Closes the TKN.

Network Control Commands

This section provides a list of the BASE24 commands that can be received by the BIC ISO Interface process. Many of these commands can be entered by an operator using a network control facility and serve as a tool that operators can use to perform network management functions such as changing processing parameters, warmbooting processes, and monitoring processes.

For detailed descriptions and the syntax of all network control commands, refer to the ***BASE24 Text Command Reference Manual***.

- **ALTERATTRIBUTE.** Directs the BIC ISO Interface process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from a network control facility. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.
- **CUTOVER.** Initiates interchange cutover. The steps taken to cut over the interchange depend on whether the co-networks are equal partners or one network is the main network and the second network is the secondary network. Cutover processing and cutover message flows are described in section 5.
- **DEVICE STATE.** Process-generated commands that inform the BIC ISO Interface process of the current state of communications between the network and the co-network.
- **DEBUG.** Places the BIC ISO Interface process in the DEBUG state. The debug message places the BIC ISO Interface process into Inspect and brings up an Inspect prompt on the home terminal (the terminal on which the network was brought up). Inspect is an HP NonStop programming tool. This command should only be used to interactively debug a running process when in a test environment. Executing this command in a production environment can result in a significant impact on system performance.
- **DUMP.** Directs the BIC ISO Interface process to request a programmatic dump. A snapshot dump is taken which can then be read using RDEBUG.
- **KEY CHANGE.** Requests made by a secondary network or co-network that a PIN or MAC working key be generated by the other BIC processor. The other BIC processor can be either a main network or another co-network. If this command is entered and the BIC ISO Interface process specified is in the main or co-network whose outbound key requires changing, a new PIN or MAC working key is generated.

Operators can specify whether to change the inbound, outbound, or both keys for either PIN or MAC working keys associated with the BIC ISO Interface process. Co-networks are only responsible for generating their outbound keys. This command is used with dynamic key management (DKM).

- **KEY REPEAT.** Requests that a PIN or MAC key be repeated. This command is used with DKM and can only be initiated by the secondary network or co-network to repeat their inbound key.

- **KEY VERIFY.** Requests that a PIN or MAC key be verified. This command is used with DKM and can only be initiated by the secondary network or co-network to verify their inbound key.
- **LOGOFF.** Initiates a logoff sequence for one or more stations. It causes the BIC ISO Interface process to send a logoff message to each station indicated in the message.
- **LOGON.** Initiates a logon sequence for one or more stations. It causes the BIC ISO Interface process to send a logon message to each station indicated in the message.
- **PERFORMANCE.** Requests that the performance statistics kept in memory by the BIC ISO Interface process be sent to the log. The BIC ISO Interface process tracks performance statistics for each station assigned to it. The statistics are displayed for the BIC ISO Interface process as a whole. These statistics are gathered on an interval basis—the interval length being established in the ICFE—and statistics are available for the last five intervals.

Performance monitoring statistics commands are used to stop and start performance monitoring.

- **STATUS.** Requests the status of the BIC ISO Interface process. The requested information is sent to the current logging facility.
- **SWL.** Changes the event message logging facility from the current facility to the indicated facility.
- **TOTALS.** Requests current information regarding totals for transaction activity. The requested information is sent to current logging facility. Totals can be displayed for either the acquirer or the issuer.
- **TRACE OFF.** Stops the trace function within the BIC ISO Interface process. For a description of the trace function, please refer to the TRACE ON message description.
- **TRACE ON.** Starts a trace function within the BIC ISO Interface process. This trace function generates event messages at periodic intervals while the trace function is enabled. The event messages display the name of the procedure from which it was written or other helpful processing information. The trace function should only be enabled in a test environment. Enabling the trace function in a production environment can result in a significant impact on system performance.

- **WARMBOOT.** Warmboots the BIC ISO Interface process. Warmboot processing is similar to process initialization, except that the BIC ISO Interface process closes all open files after they have been successfully reopened. If an error is encountered, the old environment is restored.

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